TEXTBOOK · FALL 2026

Linear Algebra I and Math I Weeks 5–7

All questions and all formulas from each course file and assigned textbook sections — continuity and inverse functions, derivatives and related rates; geometry and elimination, matrix operations and decompositions, transpositions, Permutations and vector spaces — each problem with a complete step-by-step solution. A guide for preparing for the Midterm Exam is included.

Compiled for independent study and preparation for the Midterm

Tashkent Technological University · Fall 2026
Sources: course notes, lectures, ISA, practical sessions;
OpenStax Calculus 1; Strang, Introduction to Linear Algebra (4e)
and Linear Algebra and its Applications (4e)

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1 Course Map and How to Use This Document

This manual continues the official collection "Linear Algebra I and Math I — Weeks 1–4" and covers **weeks 5–7** of the fall semester 2026. It contains all questions and all formulas from each course file and assigned textbook sections — each problem with a complete step-by-step solution, as in the main collection.

Week	Math I	Linear Algebra I
5	Continuity: three conditions at a point, types of discontinuities, continuity on an interval, theorems on the limit of a composite function and on the intermediate value (§2.4 + Instructor's NOTES); inverse functions; Weierstrass's theorem	Geometry and elimination: row picture and column picture, Gaussian elimination, back substitution, elimination matrices E, permutation matrices P, connection with rank, A = LU (Strang §§2.1–2.3)
6	Derivatives: difference quotient and secants, definition via limit, tangents and velocities, differentiation rules, derivative as a function, one-sided derivatives, higher derivatives (§§3.1–3.4 + NOTES)	Matrix operations: multiplication rules and four interpretations of AB, special matrices, inverses and their derivation, elimination matrices, LU/LDU decompositions, transpositions, permutations (Strang §§2.4–2.7)
7	Related rates (§3.9): strategy and geometric models; derivatives of exponential and logarithmic functions (§3.5, §3.9); review of Chapter 3 for Midterm	Transpositions and permutations: transposition identities, symmetric matrices, six 3×3 permutations; vector spaces and subspaces, column space C(A), null space N(A) (Strang §§2.7, 3.1)

Course files covered by this manual (weeks 5–7): lectures; Required Reading and Instructor's NOTES; problem sets with solutions; in-class problems; Independent Study Assignments (ISA); EXTRA-material (solutions to week 5 classroom exercises, week 6 examples with explanations); expanded Beamer notes for week 5 Linear Algebra; Guide to MIDTERM Exam for Math I. Textbooks: OpenStax Calculus Volume 1 (chapters 2–3) and G. Strang, Introduction to Linear Algebra (4th ed.), with parallel notes from Linear Algebra and Its Applications (4th ed.).

How each week is structured. First — **formula catalog**: all definitions, theorems, and rules of the week with explanations. Then for each source file comes a "questions and complete solutions" section: problem set, in-class problems, independent study assignments (ISA), required reading, and additional material. Problems are reproduced verbatim; solutions are complete, step-by-step, with verifications.

Notation. Matrices are written row-wise: [a b; c d] — first row a b, second row c d; column vector is written as [x; y], row vector as [x y]. Powers and indices are written with ^ and _: A^{-1}, A^{T}, a_{11}, x^{2}. Derivative is denoted f′(x) or dy/dx.

Week 7 takes place immediately before the **Midterm Exam** (Monday, November 2, 2026, 2 hours, written). The exam covers chapter 2 (§§2.1–2.4) and chapter 3 (§§3.1–3.5 and 3.9, except subsection 3.9.3) of the textbook and instructor's notes; material from weeks 1–3 is considered known from Test 1. Exam rules and preparation strategy are in chapter 5.

2 Week 5 — Continuity and Inverse Functions (Math I); Geometry and Elimination (Linear Algebra I)

Week 5 of the **Math I** course (Fall 2026, lecturer G. Neofotistos) is dedicated to **continuity of functions** (continuity): section **2.4** of the textbook (OpenStax Calculus, vol. 1) and the instructor's notes "Continuity, Inverse Functions, and Weierstrass' Theorem" (Instructor's NOTES on Moodle). Week's objectives (Learning Objectives 2.4.1–2.4.5): explain the three conditions for continuity at a point; describe types of discontinuities; define continuity on an interval; state the theorem on the limit of a composite function; provide an example of application of the intermediate value theorem. To these are added inverse functions and Weierstrass's theorem — the foundation of "predictable behavior" in engineering models.

2.1 Math I: Week 5 Formula Catalog

[Source: week 5 lecture "Continuity" (§2.4 of textbook + Instructor's NOTES), files $m_{w5lecg01}$–g13]

2.1.1 Definition of continuity at a point

A function is called *continuous* if its graph can be drawn with a pencil without lifting it from the paper. Formally: a function f is continuous at the point x = a, if

$$lim\ x \to a\ f(x) = f(a)$$

This condition contains **three requirements**: (1) f(a) is defined; (2) lim x→a f(x) exists; (3) the limit equals the function value. The function is *discontinuous at a* if it is not continuous at a.

$$lim\ x \to a-\ f(x) = lim\ x \to a+\ f(x) = f(a)$$

Equivalent form: the left-hand and right-hand limits are equal to each other and equal to the function value at the point.

- Strategy for checking continuity at a point (Problem-Solving Strategy "Determining Continuity at a Point"), step 1: check if f(a) is defined. If f(a) is not defined — the function is discontinuous at a, the check is complete.
- Step 2: compute lim x→a f(x) (if necessary, first compute lim x→a− f(x) and lim x→a+ f(x)). If the limit does not exist (is not a real number) — the function is discontinuous at a.
- Step 3: compare f(a) and lim x→a f(x). If lim x→a f(x) ≠ f(a) — the function is discontinuous at a; if equal — continuous at a.
- Continuity on an interval: f is continuous on the open interval (a, b) if it is continuous at each of its points; on the closed interval [a, b] — if additionally continuous from the right at point a: lim x→a+ f(x) = f(a), and continuous from the left at point b: lim x→b− f(x) = f(b).

2.1.2 Types of Discontinuities

If f is discontinuous at point a, the discontinuity is classified as follows (Definition, §2.4):

- **Removable discontinuity**: lim x→a f(x) exists and equals a real number L, but f(a) is not defined or f(a) ≠ L — a "hole" in the graph.
- **Jump discontinuity**: lim x→a− f(x) and lim x→a+ f(x) both exist and are finite, but lim x→a− f(x) ≠ lim x→a+ f(x) — the parts of the graph do not connect.
- **Infinite discontinuity**: lim x→a− f(x) = ±∞ and/or lim x→a+ f(x) = ±∞ — a discontinuity at a vertical asymptote.
- **Oscillatory discontinuity**: the function oscillates, not approaching any single value — for example, f(x) = sin(1/x) as x → 0; lim x→0 f(x) does not exist.

Not all discontinuities neatly fit into these categories — this is a convenient classification of three typical cases.

2.1.3 Continuity of Elementary Functions, Polynomials, Rational Functions

Theorem 2.8 (continuity of polynomials and rational functions): polynomials and rational functions are continuous at each point of their domains. **Theorem 2.10:** trigonometric functions are continuous on their entire domains.

$$f + g,\ f - g,\ c \cdot f,\ f \cdot g \text{ — are continuous at point } a, \text{ if } f \text{ and } g \text{ are continuous at } a$$

$$f/g \text{ is continuous at } a, \text{ if } f \text{ and } g \text{ are continuous at } a \text{ and } g(a) \neq 0$$

Algebra of continuous functions ("very useful for engineering!"). **Important to know:** polynomial functions are continuous everywhere; rational functions — where the denominator is not zero; exponential, logarithmic, and trigonometric functions are continuous on their domains.

$$\lim_{x \to a} f(g(x)) = f\left(\lim_{x \to a} g(x)\right) = f(L), \text{ if } f \text{ is continuous at } L \text{ and } \lim_{x \to a} g(x) = L$$

Theorem 2.9 (on composite functions, Composite Function Theorem): if f is continuous at L and lim x→a g(x) = L, then the limit can be moved inside f. Corollary: the composition of continuous functions is continuous.

2.1.4 Intermediate Value Theorem (IVT)

$$f \text{ is continuous on } [a, b],\ z \text{ lies between } f(a) \text{ and } f(b) \rightarrow \text{there exists } c \in [a, b]: f(c) = z$$

$$f(a) \cdot f(b) < 0 \rightarrow \text{there exists } c \in (a, b): f(c) = 0$$

Theorem 2.11 (Intermediate Value Theorem, important for engineering): a function continuous on a closed bounded interval takes every intermediate value. Special case: if the values at the endpoints have different signs, then there is a root inside the interval. Engineering meaning: if the temperature of a continuously operating system rises from 40°C to 80°C, it must pass through every intermediate value, including 60°C. For a discontinuous model, this conclusion is not guaranteed. The theorem guarantees **existence**, but not uniqueness.

2.1.5 Weierstrass Theorem (Extreme Value Theorem)

$$f \text{ is continuous on the closed bounded interval } [a, b] \rightarrow f \text{ reaches absolute maximum and minimum on } [a, b]$$

$$\text{there exist points } x_{min},\ x_{max} \in [a, b]: f(x_{min}) \leq f(x) \leq f(x_{max}) \text{ for all } x \in [a, b]$$

Pay attention to the word **"attains"**: maximum and minimum are the actual values of the function, not just values that the function approaches. Both conditions are essential: (1) the function must be continuous; (2) the interval must be closed and bounded. The theorem guarantees existence, but does not say **where** the extrema are, if they are unique, how to compute them, or if a local extremum is global — for this, derivatives, numerical methods, and comparison of candidate values are needed.

- Practical search for absolute extrema of a function continuous on [a, b], step 1: find critical points inside (a, b), usually by solving f'(x) = 0 (derivatives are material for week 6).
- Step 2: compute f at all critical points.

- Step 3: compute f(a) and f(b).
- Step 4: compare all values — the largest is the absolute maximum, the smallest is the absolute minimum. The Weierstrass theorem guarantees that these extrema exist; differential calculus helps find them.

Engineering interpretation: the Weierstrass theorem is important not as a computational technique, but as a **guarantee of existence** of an optimum: "continuous model + closed bounded feasible set → achievable maximum and minimum". This principle underlies structural optimization, power management, control theory, circuit design, thermal analysis, material selection, and numerical optimization: before seeking an optimum, an engineer must ask whether its existence is guaranteed. Three consequences for engineering: (1) an optimal design exists; (2) there exists a worst-case operating mode — a continuous function of voltage, temperature, error, or vibration over a limited operating range achieves its maximum value; (3) numerical optimization is meaningful.

2.1.6 Inverse functions: monotonicity and domain restriction

The inverse function reverses the "input–output" relationship: if y = f(x), then x = $f^{(-1)}$(y). Defining properties:

$$f^{(-1)}(f(x)) = x \text{ and } f(f^{(-1)}(y)) = y$$

$$f^{(-1)}(x) \neq 1/f(x)$$

Attention: the notation $f^{(-1)}$(x) **does not** mean 1/f(x). For example, for f(x) = 3x + 2, solving y = 3x + 2 for x gives x = (y − 2)/3, that is, $f^{(-1)}$(y) = (y − 2)/3.

- Monotonicity on an interval: a function is **monotonic** if its values consistently move in one direction as x increases.
- increasing: $x_1 < x_2 \rightarrow f(x_1) \leq f(x_2)$; strictly increasing: $x_1 < x_2 \rightarrow f(x_1) < f(x_2)$;
- decreasing: $x_1 < x_2 \rightarrow f(x_1) \geq f(x_2)$; strictly decreasing: $x_1 < x_2 \rightarrow f(x_1) > f(x_2)$;
- strictly monotonic = strictly increasing or strictly decreasing. Every strictly monotonic function is one-to-one: two different inputs cannot give the same output.

$$f(x_1) = f(x_2) \rightarrow x_1 = x_2 \text{ (one-to-one correspondence, one-to-one; horizontal line test)}$$

A function has an inverse on its domain if and only if it is one-to-one; graphically — each horizontal line intersects the graph at most once. A continuous and strictly monotonic function on an interval is one-to-one there and therefore invertible. Example of domain restriction: f(x) = x² is not one-to-one on R, since f(−2) = f(2) = 4; on [0, +∞) it is strictly increasing and $f^{(-1)}$(x) = sqrt(x).

2.1.7 Computing the inverse function

f is continuous and strictly monotonic on interval I → f is one-to-one, and $f^{(-1)}$: f(I) → I exists and is continuous

continuity (no value gaps) + strict monotonicity (no value repetitions) → continuous inverse (stable inversion)

Recipe for computing the inverse: solve the equation y = f(x) for x; the domain of the inverse coincides with the range of f. Classic pair: f(x) = e^x is continuous and strictly increasing on R, its range is (0, +∞), so there exists a continuous inverse $f^{(-1)}$(x) = ln x, x > 0; the points of the graphs are inverses: e^0 = 1 (point (0, 1) on y = e^x) and ln 1 = 0 (point (1, 0) on y = ln x). The graph of y = $f^{(-1)}$(x) is obtained

by reflecting the graph of f over the line y = x.

2.2 Math I: Week 5 Practical Session — questions and complete solutions

[Source: Week 5 Practical Session, MATH I (Fall 2026), prepared by G. Neofotistos — files $m_{w5pracg01}$–g32; all 14 exercises with complete solutions]

Quick reference: a function f is continuous at x = a if f(a) is defined, lim x→a f(x) exists, and lim x→a f(x) = f(a). Typical types of discontinuities: removable (the limit exists but differs from f(a) or f(a) is undefined); jump (one-sided limits are finite but not equal); infinite (the function is unbounded near the point); oscillating (oscillations without approaching one value).

Exercise 1. Checking continuity. Determine if the function f(x) = {x^2 + 1, if x < 2; 5, if x = 2; 3x − 1, if x > 2} is continuous at the point x = 2.
Solution. Left-hand limit: lim x→2− f(x) = 2^2 + 1 = 5. Right-hand limit: lim x→2+ f(x) = 3·2 − 1 = 5. Value at the point: f(2) = 5. Therefore, lim x→2 f(x) = f(2) = 5, so f is continuous at the point x = 2.

Exercise 2. Removing a discontinuity. Find the value of k for which the function f(x) = {(x^2 − 9)/(x − 3), if x ≠ 3; k, if x = 3} is continuous at the point x = 3.
Solution. For x ≠ 3: f(x) = ((x − 3)(x + 3))/(x − 3) = x + 3. Therefore lim x→3 f(x) = 6. Continuity requires k = lim x→3 f(x). Therefore, k = 6.

Exercise 3. Continuous switching law. A power converter is described by the function P(t) = {4t + 1, if t < 2; a·t^2 − 1, if t ≥ 2}. Find a for which P is continuous at the point t = 2.
Solution. Left-hand limit: lim t→2− P(t) = 4·2 + 1 = 9. From the right: P(2) = 4a − 1. For continuity: 4a − 1 = 9. Thus, 4a = 10 and a = 5/2.

Exercise 4. Composition and continuity. Let f(x) = sqrt(x) and g(t) = 4 − t^2. Determine the domain of h(t) = f(g(t)) and specify where h is continuous.
Solution. Composition: h(t) = sqrt(4 − t^2). The expression under the root must be non-negative: 4 − t^2 ≥ 0, from which −2 ≤ t ≤ 2. Both functions — g and the square root — are continuous on their respective domains. Therefore, h(t) = sqrt(4 − t^2) is continuous on the interval [−2, 2].

Exercise 5. Finding the inverse. The drive displacement x is related to the input voltage V by x(V) = 0.5·V + 2, 0 ≤ V ≤ 10. (1) Find the range of x. (2) Find the inverse function V(x). (3) Explain why the inverse is continuous.
Solution. At the endpoints: x(0) = 2, x(10) = 7. Since x(V) is increasing, its range is [2, 7]. To find the inverse, solve x = 0.5·V + 2 for V: x − 2 = 0.5·V, from which V(x) = 2·(x − 2) = 2x − 4, 2 ≤ x ≤ 7. The original function is continuous and strictly increasing on [0, 10], so its inverse is continuous on [2, 7].

Exercise 6. Why domain restriction is needed. Consider f(x) = x^2. (1) Explain why f has no inverse on R. (2) Find the inverse when the domain is restricted to [0, +∞). (3) Find the inverse when restricted to (−∞, 0].
Solution. (1) On the entire real line R: f(−x) = f(x), meaning different inputs can give the same output; for example, f(−3) = f(3) = 9. Thus, f is not one-to-one on R and has no inverse. (2) On [0, +∞) the function f is strictly increasing, so $f^{(-1)}$(y)

= sqrt(y). (3) On (−∞, 0] the function is strictly decreasing: as x moves toward zero −4 < −3 < −2 < −1 < 0, but the corresponding squares 16 > 9 > 4 > 1 > 0 — larger inputs give smaller outputs (on (−∞, 0] f is strictly decreasing, while on [0, +∞) it is strictly increasing). The inverse must return a non-positive number, so $f^{(-1)}(y) = -\text{sqrt}(y)$. Both inverse functions are continuous for y ≥ 0.

Exercise 7. Thermistor calibration. Thermistor model: $R(T) = 100 \cdot e^{(-0.02 \cdot T)}$, T ≥ 0, where R is resistance in kΩ, T is temperature in degrees Celsius. (1) Show that the function has an inverse. (2) Find T(R). (3) Calculate the temperature when R = 40 kΩ.

Solution. (1) The function is continuous and strictly decreasing for T ≥ 0, so it is one-to-one and has a continuous inverse. (2) Start with $R = 100 \cdot e^{(-0.02 \cdot T)}$. Divide by 100: $R/100 = e^{(-0.02 \cdot T)}$. Take natural logarithms: $\ln(R/100) = -0.02 \cdot T$. Therefore, $T(R) = -50 \cdot \ln(R/100)$. (3) When R = 40: $T = -50 \cdot \ln(0.4) \approx 45.8$ (degrees Celsius).

Exercise 8. Application of the Intermediate Value Theorem. The temperature of a component is modeled by $T(t) = 20 + 5t - t^2$, 0 ≤ t ≤ 4. Show that the component reaches 25°C during this interval.

Solution. The function is a polynomial and therefore continuous. Evaluate it at two convenient times: T(1) = 20 + 5 − 1 = 24 and T(2) = 20 + 10 − 4 = 26. Since 24 < 25 < 26, the Intermediate Value Theorem guarantees the existence of at least one c ∈ (1, 2) for which T(c) = 25. For the exact times, solve $20 + 5t - t^2 = 25$, that is, $t^2 - 5t + 5 = 0$, from which $t = (5 - \text{sqrt}(5))/2$ or $t = (5 + \text{sqrt}(5))/2$. Both values are in [0, 4], so the component reaches 25°C twice: at $t = (5 - \text{sqrt}(5))/2 \approx 1.382$ and $t = (5 + \text{sqrt}(5))/2 \approx 3.618$. The theorem guarantees existence; solving the equation gives the actual times.

Exercise 9. Application of the Weierstrass Theorem. The power of a small device is modeled by $P(t) = -2t^2 + 12t + 5$, 0 ≤ t ≤ 5. (1) Explain why P has absolute maximum and minimum. (2) Find both extrema.

Solution. (1) The function is a polynomial, so it is continuous; its domain [0, 5] is closed and bounded. Therefore, the Weierstrass Theorem guarantees the existence of both an absolute maximum and an absolute minimum. (2) In the course materials, this part is postponed: "For part 2 (Find both extrema) wait for next week — we need derivatives!" — to find the extrema themselves, derivatives are needed (material from week 6; see the procedure for finding absolute extrema in the week 5 formula catalog).

Exercise 10. Why it's important that the interval is closed. Consider the efficiency model η(v) = v, 0 < v < 10. Does the Weierstrass theorem guarantee a maximum and minimum?

Solution. The function is continuous, but the interval (0, 10) is open, not closed. The function approaches 0 as v → 0+, but never reaches 0; it approaches 10 as v → 10−, but never reaches 10. Therefore, the function has neither an absolute minimum nor an absolute maximum on this domain. The Weierstrass theorem does not apply because the domain is not closed.

Exercise 11. Structural optimization problem. The temperature along a metal component is approximately described by the function $T(x) = 40 + 12x - 3x^2$, 0 ≤ x ≤ 5, where x is measured in meters. (1) Using the Weierstrass theorem, justify the existence of the hottest and coldest points. (2) Find their locations and temperatures.

Solution. (1) Since T(x) is a polynomial, it is continuous; the interval [0, 5] is closed and bounded. Therefore, absolute extrema exist. (2) In the course materials, this part is postponed: "For part 2 (Find both extrema) wait till next week — we need derivatives!" — derivatives are needed (week 6); use the procedure for finding absolute extrema from the formula catalog of week 5.

Exercise 12. Connecting all three ideas. A pressure sensor is modeled by the function V(p) = sqrt(p + 1), 0 ≤ p ≤ 8. (1) Show that V is continuous. (2) Explain why it has a continuous inverse. (3) Find the inverse. (4) Using the Weierstrass theorem, specify the minimum and maximum voltage of the sensor.

Solution. (1) The function p + 1 is continuous, and the square root is continuous for non-negative inputs, so V is continuous on [0, 8]. (2) It is also strictly increasing, hence one-to-one; therefore, a continuous inverse exists. (3) Start with V = sqrt(p + 1); square both sides: $V^2 = p + 1$, from which $p(V) = V^2 - 1$. (4) The voltage range is obtained at the endpoints: V(0) = 1, V(8) = 3. By the Weierstrass theorem, these extrema are achieved: $V_{min} = 1$ at p = 0 and $V_{max} = 3$ at p = 8. The inverse is defined for 1 ≤ V ≤ 3.

Exercise 13. When the theorem cannot be applied. The cost of constructing a pipe: $C(r) = \pi \cdot r^2 + 1/r$, 0 < r ≤ 5. A student claims: "The cost is continuous, so the Weierstrass theorem guarantees a maximum." Evaluate this statement.

Solution. The function is continuous for r > 0, but the domain (0, 5] is not closed, as it does not contain r = 0. As r → 0+, we have 1/r → +∞, so C(r) → +∞. Therefore, the function has no absolute maximum on (0, 5] — it becomes arbitrarily large as r approaches zero. The student's reasoning is incorrect: continuity alone is insufficient; the Weierstrass theorem requires a closed and bounded domain.

Exercise 14. Discontinuous protection system. A protection controller has an output u(T) = {T, if 0 ≤ T < 100; 0, if T = 100}. (1) Is the domain closed and bounded? (2) Is u continuous? (3) Does u achieve an absolute maximum? (4) Why is the Weierstrass theorem not applicable?

Solution. (1) The domain [0, 100] is closed and bounded. (2) However, lim T→100− u(T) = 100, while u(100) = 0; therefore, the function is discontinuous at T = 100. (3) The function takes values arbitrarily close to 100 but never equal to 100: for each T < 100, there exists a value closer to 100 that gives a larger output. Thus, u has no absolute maximum. (4) The Weierstrass theorem is not applicable because the function is not continuous on the closed interval. A useful warning: even a finite operating range does not guarantee the achievement of the worst-case value if the model contains discontinuous switches.

Main concepts/ideas to remember: continuity means lim x→a f(x) = f(a); continuous functions preserve smooth "input-output" behavior; a function must be bijective to have an inverse; a continuous strictly monotonic function on an interval has a continuous inverse; the Intermediate Value Theorem guarantees that continuous functions take every intermediate value; the Weierstrass Theorem guarantees that a continuous function on a closed bounded interval attains its absolute maximum and minimum; Weierstrass guarantees the existence of extrema — derivatives help find them.

2.3 Math I: Week 5 Independent Study Assignments — questions and complete solutions

[Source: Week 5 — Independent Study Assignments (Homework), MATH I (Fall 2026); solutions to problems 4–6 are based on NOTES and methods of week 5]

Independent Study Assignments (ISA) — homework assignments that are not submitted or graded: their purpose is to master the week's material more deeply by solving problems independently, with classmates, or with an AI assistant (to clarify concepts, not to get ready-made solutions).

Assignment 1 (optional). Interact with an AI assistant (ChatGPT, Claude, DeepSeek, Gemini, etc.) in a structured conversation: formulate the most important points of the lecture content and ask the assistant to evaluate your answer (usually — on content and clarity; you can also ask to evaluate depth and complexity or choose other criteria). If you find erroneous conclusions, discrepancies, or contradictory statements — point them out to the assistant and ask for explanations; if the contradiction cannot be resolved, consult the instructor.

How to complete. This assignment is an instruction for working with AI in the subject area; it has no single "solution." Make a list of the key ideas of week 5 (three conditions of continuity; types of discontinuities; algebra of continuous functions; theorem on composite functions; IVT; inverse functions and monotonicity; Weierstrass Theorem and its conditions), ask the assistant to evaluate the completeness and accuracy of the formulations, and then analyze each comment.

Assignment 2. Study thoroughly all problems, exercises, and examples discussed in the lecture and practical session.

How to complete. All exercises of the practical session of week 5 (exercises 1–14) are given with complete solutions in the previous section; examples from the lecture and NOTES are in the Required reading section below; solutions to in-class exercises are in the EXTRA section.

Assignment 3. From section 2.4 of the textbook (OpenStax Calculus 1), solve exercises 131–136, 138, 145, 147, 149.

Solution. Complete solutions to all listed exercises are contained in the EXTRA material of week 5 (see the "EXTRA material of week 5" section below): exercises 131, 132, 133, 134, 135, 136, 138 (classification of discontinuity points) and exercises 145, 147, 149 (finding continuity parameter for piecewise functions). Exercise 137 (H(x) = tan 2x), which was allowed to skip according to the lecture, is also analyzed there.

Assignment 4. Consider s(t) = sin(1/t) for t ≠ 0. Can we define s(0) so that s becomes continuous at t = 0?

Solution. No, we cannot. As t → 0, the argument 1/t grows without bound, and sin(1/t) oscillates infinitely between −1 and 1, not approaching any value: lim t→0 sin(1/t) does not exist (oscillating discontinuity — NOTES, section 3). Continuity at zero would require lim t→0 s(t) = s(0), but the limit does not exist for any choice of s(0), so no extension of the value at zero makes the function continuous.

Assignment 5. Consider y(t) = {t·sin(1/t), if t ≠ 0; 0, if t = 0}. Is y continuous at t = 0?

Solution. Yes, it is continuous. For $t \neq 0$, the estimate $|y(t)| = |t| \cdot |\sin(1/t)| \leq |t|$ holds (since $|\sin| \leq 1$). As $t \to 0$, both bounding functions $-|t|$ and $|t|$ approach zero, so by the squeeze theorem, $\lim_{t \to 0} t \cdot \sin(1/t) = 0 = y(0)$. All three conditions of continuity are satisfied: $y(0)$ is defined, the limit exists and equals the function value. (Note that there is no differentiability at zero: $(y(t) - y(0))/t = \sin(1/t)$, and this limit does not exist — this fact is discussed in week 6 lecture.)

Assignment 6. Designing a continuous sensor model. A sensor uses the model $S(x) = \{a \cdot x + b, \text{ if } x < 1;\ 3, \text{ if } x = 1;\ x^2 + 2, \text{ if } x > 1\}$. Find the condition on a and b for which S is continuous at x = 1. Specify two possible pairs (a, b).

Solution. At the point $x = 1$: the left-hand limit equals $a \cdot 1 + b = a + b$; the function value $S(1) = 3$; the right-hand limit equals $1^2 + 2 = 3$. Continuity requires all three values to be equal: $a + b = 3$. This is the condition on the parameters. Two possible options: $(a, b) = (1, 2)$ and $(a, b) = (2, 1)$ — in both cases $a + b = 3$, both branches meet at the point $(1, 3)$. (The method is similar to exercises 145–149 from the EXTRA material.)

2.4 Math I: Required reading for week 5 — summary and key points

[Source: Week 5 — NOTES on Continuity, Inverse Functions, and Weierstrass' Theorem (Instructor's NOTES, Moodle) and lecture "Continuity" (§2.4 of the textbook)]

The NOTES explain why mathematical models behave predictably: small changes in input lead to small changes in output, measurements can sometimes be inverted to recover unknown quantities, and engineering variables attain maximum and minimum values on finite operating ranges. Key points of the NOTES:

- **Informal meaning of continuity (1.1):** the graph of a function has no breaks, jumps, holes, or vertical asymptotes in the domain under consideration. In engineering terms: if $y = f(x)$ is continuous near $x = a$, then a sufficiently small change in the input x produces a small change in the output y.
- **Continuity at a point (1.2):** $\lim_{x \to a} f(x) = f(a)$; three requirements — $f(a)$ is defined, the limit exists, the limit equals the value. Equivalent to: $\lim_{x \to a-} f(x) = \lim_{x \to a+} f(x) = f(a)$.
- **Continuity on an interval (1.3):** on (a, b) — continuity at each point; on $[a, b]$ — additionally continuity from the right at a and from the left at b (one-sided conditions are needed because outside $[a, b]$ the function may not be defined).
- **Algebra of continuous functions (2):** if f and g are continuous at a, then $f + g$, $f - g$, $c \cdot f$, $f \cdot g$ and f/g are continuous at $g(a) \neq 0$. Polynomials are continuous everywhere; rational functions are continuous where the denominator is non-zero; exponential, logarithmic, and trigonometric functions are continuous on their domains.
- **Theorem on Composition (2):** if g is continuous at a, and f is continuous at $g(a)$, then the composition $(f \circ g)(x) = f(g(x))$ is continuous at a; consequently, $\lim_{x \to a} f(g(x)) = f(\lim_{x \to a} g(x))$ under the appropriate continuity conditions.
- **Types of Discontinuities (3):** removable, jump, infinite, oscillating — with examples (see qa below).
- **Inverse Functions (4):** $y = f(x)$ is inverted to $x = f^{(-1)}(y)$; defining properties $f^{(-1)}(f(x)) = x$ and $f(f^{(-1)}(y)) = y$; the notation $f^{(-1)}(x)$ does not mean $1/f(x)$. The inverse exists if the function is one-to-one: $f(x_1) = f(x_2) \to x_1 = x_2$ (horizontal

line test). A continuous strictly monotonic function on an interval is one-to-one and invertible.

- **Continuity of the Inverse (4.3):** if f is continuous and strictly monotonic on an interval I, then f is one-to-one, its range J = f(I) is an interval, and the inverse $f^{(-1)}$: J → I is continuous. Geometrically: the graph of $f^{(-1)}$ is the reflection of the graph of f across the line y = x; if the graph of a continuous strictly monotonic function has no discontinuities, then neither does its reflection.
- **Engineering Interpretation (4.4):** if a sensor is defined by the relationship V = f(T) with a continuous strictly increasing f, then each temperature gives exactly one voltage; each voltage in the sensor's range corresponds to exactly one temperature; calibration can be reversed: $T = f^{(-1)}(V)$; small changes in the measured voltage give small changes in the recovered temperature. Continuity alone is not sufficient for invertibility — the calibration curve must be one-to-one.
- **Intermediate Value Theorem (5):** if f is continuous on [a, b] and L lies between f(a) and f(b), then there exists at least one c ∈ [a, b]: f(c) = L. In particular, f(a)·f(b) < 0 gives c ∈ (a, b) with f(c) = 0. Existence is guaranteed, but not uniqueness. Engineering meaning: the temperature of a system continuously operating and rising from 40°C to 80°C must pass through all intermediate values, including 60°C; for a discontinuous model, this is not necessarily true.
- **Weierstrass Theorem (6):** if f is continuous on a closed bounded [a, b], then f attains its absolute maximum and minimum: there exist x_{min}, x_{max} ∈ [a, b], for which $f(x_{min}) \le f(x) \le f(x_{max})$ for all x ∈ [a, b]. The word "attains" means that the extrema are actual values of the function. Both conditions are essential (counterexamples are in qa below). In several variables: if f: R^n → R is continuous and the feasible set K is non-empty and compact (in finite-dimensional design "compact" means "closed and bounded"), then f attains its maximum and minimum on K — one of the theoretical foundations of constrained optimization.

Examples discussed in the lecture notes and in class (with solutions):

Example (NOTES, 1.1). Why is the voltage of a charging capacitor V(t) = V_s·(1 − e^(−t/(R·C))) continuous for t ≥ 0?

Solution. The function is formed by composition and combination of continuous functions: the exponential is continuous everywhere, the linear function t/(R·C) is continuous, multiplication by a constant and subtraction from a constant preserve continuity. The model predicts that the voltage changes smoothly, not abruptly.

Example (NOTES, 2). Explain the continuity of the model $T(t) = 20 + 50 \cdot e^{(-0.2 \cdot t^2)}$.

Solution. T(t) is continuous since it is formed by composition and combination of continuous functions: t^2 is continuous, $e^{(-0.2 \cdot t^2)}$ is a composition of the continuous exponential with a continuous function, adding the constant 20 preserves continuity.

Example (NOTES, 1.1; lecture — condition 1 of continuity). Determine by definition whether the function $f(x) = (x^2 - 4)/(x - 2)$ is continuous at the point x = 2.

Solution. Let's start by calculating f(2): we get 0/0 — the value is undefined. Therefore, f(x) = (x^2 − 4)/(x − 2) is discontinuous at the point 2, since f(2) is undefined (the first condition of continuity is violated; on the graph — a line with a "hole"). It is noted in NOTES: for x ≠ 2, f(x) = x + 2 and lim x→2 f(x) = 4, so the discontinuity is removable — defining f(2) = 4 makes the function continuous.

Example (lecture — condition 2 of continuity). Determine by definition whether the function f(x) = {−x^2 + 4, if x ≤ 3; 4x − 8, if x > 3} is continuous at the point x = 3.

Solution. Calculate f(3) = −(3^2) + 4 = −5 — the value is defined. Next, calculate the one-sided limits: lim x→3− f(x) = −(3^2) + 4 = −5 and lim x→3+ f(x) = 4·3 − 8 = 4. Since the one-sided limits are not equal, lim x→3 f(x) does not exist. Therefore, f(x) is not continuous at the point 3 (the second condition is violated; on the graph — a jump).

Example (lecture — condition 3 of continuity). Determine whether the function f(x) = {sin(x)/x, if x ≠ 0; 1, if x = 0} is continuous at the point x = 0.

Solution. First, f(0) = 1. Second, lim x→0 f(x) = lim x→0 sin(x)/x = 1. Third, compare: f(0) = 1 = lim x→0 f(x). All three conditions of the definition of continuity are satisfied, therefore f(x) is continuous at the point x = 0.

Example (lecture, example 2.29; theorem 2.8). For which x is the rational function f(x) = (x + 1)/(x − 5) continuous?

Solution. The rational function f(x) = (x + 1)/(x − 5) is continuous for all values of x, except x = 5 (where the denominator becomes zero).

Example (lecture). Specify the intervals of continuity of the function f(x) = (x − 1)/(x^2 + 2x).

Solution. Since f is a rational function, it is continuous at every point of its domain. The domain: the denominator x^2 + 2x = x·(x + 2) ≠ 0, that is, x ≠ 0 and x ≠ −2. Thus, f is continuous on each of the intervals (−∞, −2), (−2, 0) and (0, +∞).

Example (lecture). Specify the intervals of continuity of the function f(x) = sqrt(4 − x^2).

Solution. By the laws of limits lim x→a sqrt(4 − x^2) = sqrt(4 − a^2) for all values of a in (−2, 2). Additionally, lim x→−2+ sqrt(4 − x^2) = 0 exists and lim x→2− sqrt(4 − x^2) = 0 exists. Therefore, f(x) is continuous on the entire interval [−2, 2].

Example 2.35 (lecture; limit of a composite function). Calculate lim x→π/2 cos(x − π/2).

Solution. This function is a composition of cos x and x − π/2. Since lim x→π/2 (x − π/2) = 0 and cos x is continuous at 0, we can apply the theorem on composite functions: lim x→π/2 cos(x − π/2) = cos(lim x→π/2 (x − π/2)) = cos(0) = 1.

Example 2.36 (lecture; application of IVT). Show that f(x) = x − cos x has at least one zero.

Solution. The function f(x) = x − cos x is continuous on (−∞, +∞), hence continuous on any closed interval of the form [a, b]. If we find an interval [a, b] at the endpoints of which f(a) and f(b) have different signs, then by the Intermediate Value Theorem there exists a real number c ∈ (a, b) for which f(c) = 0. Note: f(0) = 0 − cos(0) = −1 < 0 and f(π/2) = π/2 − cos(π/2) = π/2 > 0. By IVT there exists a real number c ∈ [0, π/2] for which f(c) = 0. Therefore, f(x) = x − cos x has at least one zero.

Example 2.37 (lecture). Let f(x) be continuous on [0, 2], with f(0) > 0 and f(2) > 0. Can we conclude by the Intermediate Value Theorem that f(x) has no zeros on [0, 2]?

Solution. No. The Intermediate Value Theorem only allows us to conclude that f takes all values between f(0) and f(2); it does not allow us to conclude that f does not take other values. Counterexample: $f(x) = (x - 1)^2$ satisfies f(0) = 1 > 0, f(2) = 1 > 0, but f(1) = 0 — there is a zero on the interval.

Example 2.38 (lecture). For f(x) = 1/x we have f(−1) = −1 < 0 and f(1) = 1 > 0. Can we conclude that f(x) has a zero on the interval [−1, 1]?

Solution. No. The function is not continuous on [−1, 1] (it has an infinite discontinuity at zero), so the Intermediate Value Theorem is not applicable here.

Example (lecture; inverse of the exponential). Explain why the function $f(x) = e^x$ has a continuous inverse, and find it.

Solution. The function $f(x) = e^x$ is continuous and strictly increasing on R; its range is (0, +∞). Therefore it is bijective and has a continuous inverse: $f^{(-1)}(x) = \ln x$, x > 0. The points of the two graphs are inverses: $e^0 = 1$ means that the point (0, 1) lies on $y = e^x$, and ln 1 = 0 means that the point (1, 0) lies on y = ln x.

Example (NOTES, 4.3; inverting a sensor model). The sensor follows the relationship $V(T) = 2 + 0.01 \cdot T^2$, T ≥ 0. The function is continuous and strictly increasing for T ≥ 0 — hence it has a continuous inverse. Find T(V) and calculate the temperature at V = 11 V.

Solution. Solve for T: $V - 2 = 0.01 \cdot T^2$, from which $T^2 = 100 \cdot (V - 2)$. Since T ≥ 0, we choose the non-negative root: T(V) = 10·sqrt(V − 2), V ≥ 2. At V = 11 V: T = 10·sqrt(11 − 2) = 10·3 = 30. The measured temperature is 30°C.

Example (NOTES, 6; importance of Weierstrass conditions — open interval). Consider f(x) = x on the interval (0, 1). Why doesn't the Weierstrass Theorem guarantee extrema?

Solution. The function is continuous, but the interval is not closed. The function approaches 0 and 1 but does not achieve either value. Therefore on (0, 1) it has no absolute maximum or absolute minimum. The conditions of the theorem are sufficient: if any one is violated, the theorem no longer guarantees the result (although a specific function may still have extrema).

Example (NOTES, 6; importance of Weierstrass conditions — discontinuous function). Consider f(x) = {x, if 0 ≤ x < 1; 0, if x = 1} on [0, 1]. Why does the function have no absolute maximum?

Solution. The domain [0, 1] is closed and bounded, but f is discontinuous at x = 1. The function values approach 1, but the value f(x) = 1 is not achieved at any point. Thus there is no absolute maximum — the Weierstrass Theorem is not applicable due to the discontinuity.

Weekly multimedia material: the first half of the video presentation "Continuity and Differentiability EXPLAINED with Examples" (Ace Tutors, youtube.com/watch?v=nRv1lPqrx3k) was used in the lecture — the part concerning continuity is relevant for week 5; also presented is the Wikipedia article on Karl Weierstrass (1815–1897), "the father of modern analysis." Instructor's note: the Weierstrass Theorem from the course is the Extreme Value Theorem; it does not coincide with the Bolzano-Weierstrass Theorem, which is not covered in the course.

Final lecture assignment (In-Class activity): "If there is time left, solve as many as you can from the exercises in SECTION 2.4 EXERCISES (exercises 131–138; skip exercise 137): for each function, determine the points of discontinuity and classify the discontinuity as jump, removable, infinite, or other." Complete solutions to these exercises are provided in the next section (EXTRA-material). The thin file Week 5 Required reading (88 picture-pages without text) contains no additional content.

2.5 Math I: EXTRA material for week 5 — solutions to class exercises

[Source: Week 5 — EXTRA Material: Solutions to class exercises (Section 2.4); instructor's note: pay special attention to exercises with "removable discontinuities"]

Below are solutions to the exercises covered in class (section 2.4 of the textbook). General approach: points of discontinuity are found by checking where functions are undefined, with calculation of the corresponding limits.

Exercise 131. f(x) = 1/sqrt(x). Determine and classify points of discontinuity.

Solution. For a real-valued function, the square root requires $x \geq 0$, but the denominator cannot be zero; therefore, Dom(f) = $(0, +\infty)$. As $x \to 0+$, we have $1/\sqrt{x} \to +\infty$. Therefore, the line $x = 0$ is a vertical asymptote, and $x = 0$ is a point of infinite discontinuity. The function is continuous everywhere on its domain $(0, +\infty)$. Conclusion: $x = 0$ is an infinite discontinuity.

Exercise 132. f(x) = 2/(x^2 + 1). Determine and classify points of discontinuity.

Solution. Since $x^2 + 1 \geq 1 > 0$ for each real x, the denominator is never zero. A rational function is continuous wherever its denominator is non-zero. Therefore, f is continuous at each $x \in R$. There are no discontinuities.

Exercise 133. f(x) = x/(x^2 − x). Determine and classify points of discontinuity.

Solution. Factor the denominator: $x^2 - x = x \cdot (x - 1)$. Then $f(x) = x/(x \cdot (x - 1)) = 1/(x - 1)$ for $x \neq 0, 1$. The original function is undefined at $x = 0$ and $x = 1$. At $x = 0$: the factor x cancels, and $\lim_{x \to 0} f(x) = 1/(0 - 1) = -1$. The limit exists and is finite, but f(0) is undefined — therefore, $x = 0$ is a removable discontinuity (it can be "fixed" by additionally defining $f(0) = -1$). At $x = 1$: using the simplified expression $f(x) = 1/(x - 1)$, we get $\lim_{x \to 1-} f(x) = -\infty$ and $\lim_{x \to 1+} f(x) = +\infty$ — therefore, $x = 1$ is an infinite discontinuity. Conclusion: $x = 0$ is a removable discontinuity; $x = 1$ is an infinite discontinuity.

Exercise 134. g(t) = $t^{(-1)}$ + 1. Determine and classify points of discontinuity.

Solution. Rewrite the function as $g(t) = 1/t + 1$. It is undefined at $t = 0$. At this point, $\lim_{t \to 0-} (1/t + 1) = -\infty$, and $\lim_{t \to 0+} (1/t + 1) = +\infty$. Therefore, $t = 0$ is a vertical asymptote: a point of infinite discontinuity. Conclusion: $t = 0$ is an infinite discontinuity.

Exercise 135. f(x) = 5/(e^x − 2). Determine and classify points of discontinuity.

Solution. The denominator equals zero when $e^x - 2 = 0$, that is, $e^x = 2$, from which $x = \ln 2$. As $x \to (\ln 2)-$, we have $e^x - 2 \to 0-$, therefore $\lim_{x \to (\ln 2)-} 5/(e^x$

$- 2) = -\infty$. As $x \to (\ln 2)+$, we have $e^x - 2 \to 0+$, therefore $\lim x\to(\ln 2)+ 5/(e^x - 2) = +\infty$. Therefore, $x = \ln 2$ is a point of infinite discontinuity, and the line $x = \ln 2$ is a vertical asymptote. Conclusion: $x = \ln 2$ is an infinite discontinuity.

Exercise 136. f(x) = |x − 2|/(x − 2). Determine and classify points of discontinuity.

Solution. The function is undefined at $x = 2$. Recall the absolute value definition: $|x - 2| = \{-(x - 2)$, if $x < 2$; $x - 2$, if $x > 2\}$. One-sided limits: $\lim x\to2- f(x) = -1$ and $\lim x\to2+ f(x) = 1$. Since these limits are different, the two-sided limit does not exist. It is impossible to define a value at the point $x = 2$ to remove the discontinuity — the one-sided limits are not equal. Conclusion: $x = 2$ is a jump discontinuity.

Exercise 137. H(x) = tan(2x). Determine and classify points of discontinuity.

Solution. Recall: $\tan(2x) = \sin(2x)/\cos(2x)$. The function is undefined where $\cos(2x) = 0$. This occurs when $2x = \pi/2 + k\cdot\pi$, $k \in Z$; dividing by 2, we get $x = \pi/4 + k\cdot\pi/2$. At each of these points, the tangent becomes unbounded and has a vertical asymptote. Therefore, $x = \pi/4 + k\cdot\pi/2$, $k \in Z$ are points of infinite discontinuities. Some of these points: ..., $-3\pi/4$, $-\pi/4$, $\pi/4$, $3\pi/4$, ...

Exercise 138. f(t) = (t + 3)/(t² + 5t + 6). Determine and classify points of discontinuity.

Solution. Factor the denominator: $t^2 + 5t + 6 = (t + 2)(t + 3)$. Then $f(t) = (t + 3)/((t + 2)(t + 3)) = 1/(t + 2)$ for $t \neq -3, -2$. The original function is undefined at $t = -3$ and $t = -2$. At $t = -3$: the factor $(t + 3)$ cancels, and $\lim t\to-3 f(t) = 1/(-3 + 2) = -1$. The limit exists, but the original function is undefined at this point — thus, $t = -3$ is a removable discontinuity; the "hole" can be eliminated by defining $f(-3) = -1$. At $t = -2$: the remaining denominator becomes zero — $\lim t\to-2- f(t) = -\infty$ and $\lim t\to-2+ f(t) = +\infty$. Conclusion: $t = -3$ is a removable discontinuity; $t = -2$ is an infinite discontinuity.

Exercise
131
132
133
134
135
136
137
138

Second series of problems (note from instructor: exercise 146 can be skipped). General method: for piecewise functions, individual formulas are continuous on their intervals, so only the point of formula change needs to be checked.

Exercise 145. f(x) = {3x + 2, if x < k; 2x − 3, if k ≤ x ≤ 8}. Find k for which f is continuous.

Solution. The only possible discontinuity is at $x = k$. Left-hand limit: $\lim x\to k- f(x) = 3k + 2$. The value and right-hand limit from the second branch: $f(k) = 2k - 3$ and $\lim x\to k+ f(x) = 2k - 3$. For continuity: $3k + 2 = 2k - 3$. Solving, we get $k =$

−5. Verification: 3·(−5) + 2 = −13 and 2·(−5) − 3 = −13 — both branches meet at the point (−5, −13).

Exercise 146. f(θ) = {sin θ, if 0 ≤ θ < π/2; cos(θ + k), if π/2 ≤ θ ≤ π}. Find k for which f is continuous.

Solution. The only possible discontinuity is at θ = π/2. Left-hand limit: lim θ→(π/2)− sin θ = 1. The value from the second branch: f(π/2) = cos(π/2 + k). Continuity requires cos(π/2 + k) = 1. Cosine equals 1 when its argument is an integer multiple of 2π: π/2 + k = 2π·n, n ∈ Z. Hence k = 2π·n − π/2, n ∈ Z (for example, k = −π/2 when n = 0).

Exercise 147. f(x) = {(x^2 + 3x + 2)/(x + 2), if x ≠ −2; k, if x = −2}. Find k for which f is continuous at x = −2.

Solution. Factor the numerator: x^2 + 3x + 2 = (x + 1)(x + 2). For x ≠ −2: f(x) = ((x + 1)(x + 2))/(x + 2) = x + 1. Therefore lim x→−2 f(x) = lim x→−2 (x + 1) = −1. For continuity, the function value at the point must equal this limit: k = −1. This extension fills the removable "hole" in the original rational expression.

Exercise 148. f(x) = {$e^{(k·x)}$, if 0 ≤ x ≤ 4; x + 3, if 4 ≤ x ≤ 8}. Find k for which f is continuous.

Solution. The branches meet at x = 4. From the first branch: f(4) = $e^{(4k)}$. From the second branch: f(4) = 4 + 3 = 7. For correctness and continuity at x = 4, these values must coincide: $e^{(4k)}$ = 7. Take natural logarithms: 4k = ln 7. Therefore k = ln(7)/4. Numerically: k ≈ 1.94591/4 ≈ 0.4865.

Exercise 149. f(x) = {sqrt(k·x), if 0 ≤ x ≤ 3; x + 1, if 3 < x ≤ 10}. Find k for which f is continuous.

Solution. The possible discontinuity is at x = 3. From the first branch: f(3) = sqrt(3k). Right-hand limit from the second branch: lim x→3+ (x + 1) = 4. Continuity requires sqrt(3k) = 4. Squaring both sides, we get 3k = 16, from which k = 16/3. This value is positive, so sqrt(k·x) is real for each x ∈ [0, 3]. Verification: f(3) = sqrt(3 · 16/3) = sqrt(16) = 4 — matches the right-hand limit.

Exercise
145
146
147
148
149

Week 5 of **Linear Algebra I** (TT 2026) connects the geometry of the equation Ax = b with the algorithm for solving it—Gaussian elimination. Sources: Strang, Introduction to Linear Algebra, 4th ed., §§2.1–2.3; Linear Algebra and Its Applications, 4th ed., §§1.2–1.4; Beamer lecture notes "Geometry and Elimination" (80 minutes lecture, 90 minutes problems). Weekly plan: review of rank and inverses (10 min), row and column pictures (20 min), elimination and back substitution (25 min), elimination matrices E and permutation matrices P (20 min), summary (5 min). Objectives: to see Ax = b as the intersection of lines and planes (row picture) and as a linear combination of columns (column picture); to perform elimination with multipliers and back substitution; to represent row operations with matrices E, cancel them with matrices E^{-1}, and

swap equations with matrices P. Notation: matrices are written row-wise — [a b; c d] (first row a b, second c d); column vector [x; y]; augmented matrix [A | b].

2.6 Linear Algebra I, week 5: formula catalog

[Source: Beamer lecture notes week 5 "Geometry and Elimination" (Linear Algebra I, TT 2026): Strang, Introduction to Linear Algebra, 4th ed., §§2.1–2.3; Linear Algebra and Its Applications, §§1.2–1.4]

2.6.1 Connection with rank: n pivots mean invertibility

Review: A = [1 2; 2 4] has rank 1 and no inverse; Ax = [3; 6] has infinitely many solutions, while Ax = [3; 7] has none. Key fact: for an n × n square matrix, having **n pivots** is equivalent to invertibility—then Ax = b has a unique solution for every b; the number of pivots found by elimination is the rank. Both algorithms act on [A | b]: **Gauss–Jordan** reduces the matrix to reduced echelon form; **Gaussian elimination** stops at upper triangular U and completes with back substitution.

$$A = [1\ 2; 2\ 4]\text{: rank } A = 1,\ A^{-1} \text{ does not exist; } Ax = [3; 6] \text{ — infinitely many solutions, } Ax = [3; 7] \text{ — no solution}$$

2.6.2 Row picture: rows as lines and planes

$$2x - y = 1,\ x + y = 5;\ [2\ -1; 1\ 1]\cdot[x; y] = [1; 5]$$

Each row is one geometric constraint: in 2D lines, in 3D planes; the solution is their common point. The example lines intersect at (x, y) = (2, 3); three planes in 3D:

$$[2\ 1\ 1; 4\ -6\ 0; -2\ 7\ 2]\cdot[u; v; w] = [5; -2; 9] \text{ — planes intersect at } (u, v, w) = (1, 1, 2)$$

2.6.3 Column picture: Ax as a linear combination of columns

$$x\cdot[2; 1] + y\cdot[-1; 1] = [1; 5]\text{; check: } 2\cdot[2; 1] + 3\cdot[-1; 1] = [1; 5]$$

Strang emphasizes: **Ax is a linear combination of the columns of A**—we choose coefficients so that $x_1 \cdot a_1 + \ldots + x_n \cdot a_n = b$. Three-dimensional version (weights (1, 1, 2)):

$$u\cdot[2; 4; -2] + v\cdot[1; -6; 7] + w\cdot[1; 0; 2] = [5; -2; 9]$$

Both pictures are the same equation Ax = b: rows are intersections (equations), columns are linear combinations (vectors).

2.6.4 Three possible outcomes

- **One solution**—independent constraints.
- **Infinitely many solutions:** x + y = 2 and 2x + 2y = 4—the second equation is twice the first, the lines coincide.
- **No solutions:** x + y = 2 and 2x + 2y = 5—the left sides are proportional, the right sides are not, the lines are parallel.

$$\text{elimination reveals outcome: } [0\ 0\ \ldots\ 0 \mid 0] \to 0 = 0 \text{ — infinitely many solutions; } [0\ 0\ \ldots\ 0 \mid c],\ c \neq 0, \to 0 = c \text{ — no solutions}$$

2.6.5 Pivot, multiplier, and forward elimination

$$\lambda_{ij} = a_{ij}/a_{jj},\ \text{operation } R_i \leftarrow R_i - \lambda_{ij}\cdot R_j$$

To eliminate element a_{ij} below a nonzero pivot a_{jj} (the multiplier is denoted λ_{ij} here), subtract row j multiplied by λ_{ij} from row i. Running example: [2 1 1 | 5; 4

$-6\ 0 \mid -2;\ -2\ 7\ 2 \mid 9]$; pivot 2, multipliers $\lambda_{21} = 4/2 = 2$ and $\lambda_{31} = -2/2 = -1$:

$$R_2 \leftarrow R_2 - 2R_1,\ R_3 \leftarrow R_3 + R_1\text{: } [2\ 1\ 1 \mid 5;\ 0\ -8\ -2 \mid -12;\ 0\ 8\ 3 \mid 14]$$

Second pivot -8, multiplier $\lambda_{32} = 8/(-8) = -1$, operation $R_3 \leftarrow R_3 + R_2$. Triangular system and result of forward elimination:

$$[2\ 1\ 1 \mid 5;\ 0\ -8\ -2 \mid -12;\ 0\ 0\ 1 \mid 2],\ U = [2\ 1\ 1;\ 0\ -8\ -2;\ 0\ 0\ 1],\ \text{pivot elements } 2, -8, 1$$

2.6.6 Back substitution

$$w = 2;\ -8v - 2w = -12 \rightarrow v = 1;\ 2u + v + w = 5 \rightarrow u = 1;\ x = [1;\ 1;\ 2]$$

The triangular system is solved from bottom up: the last equation contains only w, the second to last contains v and w. The answer is verified by substitution into the original equation: $[2\ 1\ 1;\ 4\ -6\ 0;\ -2\ 7\ 2]\cdot[1;\ 1;\ 2] = [5;\ -2;\ 9]$. A row operation is reversible (add $2R_1$ back), so elimination preserves solutions; but the operation is applied to both left and right sides, otherwise the equations are not equivalent.

2.6.7 Elimination matrices E

$$E_{21} = [1\ 0\ 0;\ -2\ 1\ 0;\ 0\ 0\ 1],\ E_{21}\cdot A = [2\ 1\ 1;\ 0\ -8\ -2;\ -2\ 7\ 2],\ E_{21}\cdot b = [5;\ -12;\ 9]$$

Left multiplication performs a row operation, and the right side is transformed by the same matrix: $Ax = b$ becomes $(E_{21}A)x = E_{21}b$. Full pass to U (the rightmost multiplier acts first—order of multiplication matters):

$$E_{31} = [1\ 0\ 0;\ 0\ 1\ 0;\ 1\ 0\ 1],\ E_{32} = [1\ 0\ 0;\ 0\ 1\ 0;\ 0\ 1\ 1],\ E_{32}\cdot E_{31}\cdot E_{21}\cdot A = U$$

Undo step: subtraction is replaced by addition—for $E = [1\ 0;\ -3\ 1]$ the inverse is $E^{-1} = [1\ 0;\ 3\ 1]$ and $E^{-1}\cdot E = I$.

2.6.8 Zero pivot: permutation matrices P

$$[0\ 2 \mid 4;\ 3\ 1 \mid 7] \rightarrow [3\ 1 \mid 7;\ 0\ 2 \mid 4]\ \text{(row exchange), then } y = 2,\ x = 5/3$$

A zero pivot does not necessarily mean singularity: if there is a nonzero element below, rows are swapped. The swap is performed by a permutation matrix, and it also transforms the right side:

$$P = [0\ 1;\ 1\ 0],\ P\cdot[0\ 2;\ 3\ 1] = [3\ 1;\ 0\ 2],\ P\cdot[4;\ 7] = [7;\ 4],\ \text{equivalent system } PAx = Pb$$

Properties of P: one 1 in each row and column, rest zeros; $P^{-1} = P^T$; for a single swap $P^2 = I$; swapping rows 1 and 3 — $P_{13} = [0\ 0\ 1;\ 0\ 1\ 0;\ 1\ 0\ 0]$; left multiplication changes equations, not unknowns. Elimination failures: **temporary** — zero pivot with nonzero element below (swap saves); **permanent** — no available pivot in column (singularity: no solution or infinitely many).

2.6.9 Looking ahead: A = LU and PA = LU

$$\text{without row exchanges } A = LU\ (U \text{ — upper triangular, } L \text{ stores elimination factors}); \text{ with exchanges } PA = LU$$

Week summary: rows — intersections, columns — combinations; elimination — zeros below pivots plus back substitution; E — row replacement, P — row swap. LU decomposition ($A = LU$, $PA = LU$) is built in week 6.

2.7 Linear Algebra I, week 5: in-class problems — questions and complete solutions

[Source: Week 5 Class Problems (Projection), problems 1–15 (la$_{w5classg01}$–g11 and text layer pp. 23–31)]

Exercise 1. Row picture and column picture. Solve the system [2 −1; 1 1]·[x; y] = [1; 5] and describe both geometric pictures.

Solution. Add equations: 3x = 6, x = 2, then y = 5 − 2 = 3. Lines 2x − y = 1 and x + y = 5 intersect at (2, 3); by columns: 2·[2; 1] + 3·[−1; 1] = [4 − 3; 2 + 3] = [1; 5]. Answer: x = 2, y = 3.

Exercise 2. Three outcomes. Classify the system [1 1; 2 2]·[x; y] = [2; c] for c = 4 and c = 5.

Solution. $R_2 \leftarrow R_2 - 2R_1$ gives row [0 0 | c − 4]. When c = 4 this is 0 = 0: x + y = 2 remains, infinitely many solutions — [2 − t; t]. When c = 5 this is 0 = 1 — contradiction, no solution.

Exercise 3. Find the multiplier. Eliminate below the first pivot in the augmented matrix [2 3 | 7; 8 13 | 29] and find both pivots.

Solution. Multiplier $\lambda_{21} = 8/2 = 4$. $R_2 \leftarrow R_2 - 4R_1$: [8, 13 | 29] − 4·[2, 3 | 7] = [0, 1 | 1]. Result: [2 3 | 7; 0 1 | 1]; pivots are 2 and 1.

Exercise 4. Back substitution. Solve the system [2 1 −1; 0 3 2; 0 0 4]·[x; y; z] = [3; 7; 8].

Solution. Bottom up: 4z = 8, z = 2; 3y + 2·2 = 7, y = 1; 2x + 1 − 2 = 3, x = 2. Answer: x = [2; 1; 2]. Check: 2·2 + 1 − 2 = 3, 3 + 4 = 7, 8 = 8 — correct.

Exercise 5. Full elimination. Solve the system [1 1 1 | 6; 2 3 1 | 10; 1 2 3 | 14] by elimination.

Solution. $R_2 \leftarrow R_2 - 2R_1 \rightarrow$ [0 1 −1 | −2]; $R_3 \leftarrow R_3 - R_1 \rightarrow$ [0 1 2 | 8]; $R_3 \leftarrow R_3 - R_2 \rightarrow$ [0 0 3 | 10], so 3z = 10. Back substitution: z = 10/3; y − z = −2 → y = 4/3; x + y + z = 6 → x = 4/3. Answer: [4/3; 4/3; 10/3]. Check: rows give 6, 10, 14 — correct.

Exercise 6. Identify elementary matrix. Find E that performs $R_3 \leftarrow R_3 - 4R_1$ on a 3 × 3 matrix.

Solution. E is identity matrix with −4 in position (3, 1): E = [1 0 0; 0 1 0; −4 0 1]; row 3 of EA equals (row 3) − 4·(row 1), rows 1–2 unchanged. Inverse operation is addition: E^{-1} = [1 0 0; 0 1 0; 4 0 1] with +4 in position (3, 1).

Exercise 7. EA multiplication. Let E = [1 0; −2 1] and A = [3 1; 6 5]. Compute EA and explain the operation performed.

Solution. Row 1 of product: [3 1]; row 2: −2·[3 1] + [6 5] = [0 3]. So EA = [3 1; 0 3]: E subtracts twice row 1 from row 2 — elimination with multiplier 6/3 = 2.

Exercise 8. Transformed right side. For A = [3 1; 6 5] and b = [4; 17] use E = [1 0; −2 1] to solve the system.

Solution. Eb = [4; −8 + 17] = [4; 9]; together with EA = [3 1; 0 3] (exercise 7) we get 3y = 9, y = 3 and 3x + 3 = 4, x = 1/3. Check in original system: 3·(1/3) + 3 = 4, 6·(1/3) + 5·3 = 17 — correct.

Exercise 9. Zero pivot. Solve the system [0 2; 3 1]·[x; y] = [4; 7] by row swap; find P.

Solution. Zero in position (1, 1) is temporary failure: P = [0 1; 1 0], PA = [3 1; 0 2], Pb = [7; 4]. Then 2y = 4, y = 2; 3x + y = 7, x = 5/3. Check: 2·2 = 4, 3·(5/3) + 2 = 7 — correct; zero pivot didn't mean singularity.

Exercise 10. Three-row permutation. Find P that swaps rows 1 and 3 of matrix A = [1 2 0; 0 1 3; 4 0 2]; compute PA.

Solution. P = [0 0 1; 0 1 0; 1 0 0] — rows of identity matrix in order 3, 2, 1; row 1 of PA equals row 3 of matrix A and so on: PA = [4 0 2; 0 1 3; 1 2 0]. Swap 1 ↔ 3 is involution: $P^{-1} = P$ and $P^2 = I$ (consistent with $P^{-1} = P^T$).

Exercise 11. Singular but consistent. Solve the system [1 2 1; 2 4 2; 0 1 1]·[x; y; z] = [3; 6; 2].

Solution. Row 2 with the right-hand side (6 = 2·3) is twice row 1, so $R_2 \leftarrow R_2 - 2R_1$ gives 0 = 0. We are left with x + 2y + z = 3 and y + z = 2; free variable z = t: y = 2 − t, x + 2(2 − t) + t = 3 → x = t − 1. Answer: [t − 1; 2 − t; t] — infinitely many solutions. Verification for t = 1: [0; 1; 1] gives 3, 6, 2 — correct.

Exercise 12. Degenerate and inconsistent. Replace the right-hand side in the previous system with b = [3; 7; 2]. Is the system consistent?

Solution. Now 7 ≠ 2·3. $R_2 \leftarrow R_2 - 2R_1$ gives the row [0 0 0 | 7 − 6] = [0 0 0 | 1], that is, 0 = 1 — a contradiction. The system is inconsistent: the right-hand side violates the necessary condition $b_2 = 2b_1$.

Exercise 13. Column interpretation of consistency. For A = [1 2; 2 4], which vectors b = [b_1; b_2] admit a solution Ax = b?

Solution. The second row is twice the first: after $R_2 \leftarrow R_2 - 2R_1$ we have [0 0 | $b_2 - 2b_1$], so consistency requires $b_2 = 2b_1$. The columns [1; 2] and [2; 4] = 2·[1; 2] are multiples of [1; 2], so Ax always lies on the line $b_2 = 2b_1$. Answer: b = [b_1; $2b_1$] — b must lie on the line through the columns of A.

Exercise 14. Order of matrix operations. Let E = [1 0; −2 1] and P = [0 1; 1 0]. Compare EP and PE.

Solution. EP: row 1 = [0 1], row 2 = −2·[0 1] + [1 0] = [1, −2], i.e. EP = [0 1; 1 −2]. PE: row 1 = 1·[−2 1] = [−2 1], row 2 = [1 0], i.e. PE = [−2 1; 1 0]. The products are different: row operations, in general, do not commute.

Exercise 15. Challenge: construct a system. Build a 3 × 3 system requiring row exchange, with three pivots and solution x = [1; 2; 3].

Solution. A zero in position (1, 1) with a nonzero element below: A = [0 1 0; 1 0 0; 0 0 1]; set the right-hand side by the column picture: b = A·[1; 2; 3] = [2; 1; 3]. Swapping rows 1 and 2 (P = [0 1 0; 1 0 0; 0 0 1]) turns A into the identity matrix: pivots 1, 1, 1, solution x = [1; 2; 3]. Verification: A·[1; 2; 3] = [2; 1; 3] — correct.

2.8 Linear Algebra I, week 5: practical session — questions and complete solutions

[Source: Week 5 Practical Session problems with solutions (Week5 TA Problems Projection), problems 1–15; matrices restored from the text layer]

Exercise 1. System of two lines. Solve the system [1 2; 3 1]·[x; y] = [7; 8].

Solution. $R_2 \leftarrow R_2 - 3R_1$: (1 − 6)y = 8 − 21, that is, −5y = −13, y = 13/5; then x = 7 − 2·(13/5) = 9/5. Answer: (9/5, 13/5). Verification: 9/5 + 26/5 = 7, 27/5 + 13/5 = 8 — correct.

Exercise 2. Parallel or coincident? Classify the system [2 4; 1 2]·[x; y] = [6; c] for c = 3 and c = 4.

Solution. Row 1 is twice row 2 on the left; consistency requires the same on the right: 6 = 2c, that is, c = 3. For c = 3, we are left with one equation x + 2y = 3 — infinitely many solutions [3 − 2t; t] (the lines coincide). For c = 4, doubling the second equation gives 2x + 4y = 8 ≠ 6 — no solutions (the lines are parallel).

Exercise 3. One elimination step. Eliminate below the first pivot in the matrix [3 2; 9 8].

Solution. The multiplier $\lambda_{21} = 9/3 = 3$; $R_2 \leftarrow R_2 - 3R_1$: [9 − 9, 8 − 6] = [0, 2]. Result: [3 2; 0 2]; pivots 3 and 2.

Exercise 4. Triangular system. Solve [1 2 1; 0 2 −2; 0 0 3]·[x; y; z] = [8; 0; 6].

Solution. From bottom up: 3z = 6, z = 2; 2y − 2·2 = 0, y = 2; x + 4 + 2 = 8, x = 2. Answer: [2; 2; 2]. Verification: 2 + 4 + 2 = 8, 4 − 4 = 0, 3·2 = 6 — correct.

Exercise 5. Complete system with three unknowns. Solve the system [1 1 0 | 3; 2 3 1 | 9; 1 0 1 | 4].

Solution. $R_2 \leftarrow R_2 - 2R_1 \rightarrow$ [0 1 1 | 3]; $R_3 \leftarrow R_3 - R_1 \rightarrow$ [0 −1 1 | 1]; $R_3 \leftarrow R_3 + R_2 \rightarrow$ [0 0 2 | 4], that is, 2z = 4, z = 2. Backwards: y + z = 3 → y = 1; x + y = 3 → x = 2. Answer: [2; 1; 2]. Verification: 2 + 1 = 3, 4 + 3 + 2 = 9, 2 + 2 = 4 — correct.

Exercise 6. Construct E_{32}. Write the 3 × 3 matrix for the operation $R_3 \leftarrow R_3 + 5R_2$.

Solution. E_{32} is the identity matrix with 5 in position (3, 2): E_{32} = [1 0 0; 0 1 0; 0 5 1]; rows 1–2 are unchanged, and row 3 of the product equals (row 3) + 5·(row 2).

Exercise 7. Reverse the operation. Find the inverse matrix for E = [1 0 0; 0 1 0; 0 −3 1].

Solution. E implements $R_3 \leftarrow R_3 - 3R_2$ (subtracting three times row 2); the inverse operation is addition: E^{-1} = [1 0 0; 0 1 0; 0 3 1] (the sign changed). Verification: $E^{-1}\cdot E = I$.

Exercise 8. Transform the right-hand side. Apply E = [1 0 0; −2 1 0; 0 0 1] to b = [3; 5; 7].

Solution. Eb = [3; 5 − 2·3; 7] = [3; −1; 7]: the second component becomes 5 − 2·3 = −1. E eliminates x from the second equation with multiplier 2 — the right-hand side is transformed by the same operation.

Exercise 9. Row exchange matrix. Find P that swaps rows 2 and 3 of a 3 × 3 matrix.

Solution. P = [1 0 0; 0 0 1; 0 1 0] is the identity matrix with rows 2 and 3 swapped. A double swap returns the original order: $P^2 = I$, so $P^{-1} = P$ (consistent with $P^{-1} = P^T$); left multiplication swaps equations, not unknowns.

Exercise 10. System requiring exchange. Solve the system [0 1; 2 3]·[x; y] = [4; 14].

Solution. The pivot in position (1, 1) is zero — swap rows: 2x + 3y = 14 and y = 4. Then 2x + 12 = 14, x = 1. Answer: (1, 4). Check: 0·1 + 4 = 4, 2 + 12 = 14 — correct.

Exercise 11. Free variable. Solve the system [1 1 1; 0 1 1; 0 0 0]·[x; y; z] = [4; 3; 0].

Solution. The third row is 0 = 0, giving no constraints; there are two pivots and three unknowns. Let z = t: y + t = 3 → y = 3 − t; x + (3 − t) + t = 4 → x = 1. Answer: [1; 3 − t; t]. Check at t = 0: [1; 3; 0] gives 4, 3, 0 — correct.

Exercise 12. Inconsistent row. Explain why the system [1 1 0 | 2; 1 1 3 | 0; 0 0 0 | 1] has no solutions.

Solution. The last row reads as 0·x + 0·y + 0·z = 1, i.e., 0 = 1 — an impossible equality: a zero left-hand side cannot give a nonzero right-hand side for any x, y, z. The system is inconsistent.

Exercise 13. Compare EP and PE. For E = [1 0; 4 1] and P = [0 1; 1 0] compute both products.

Solution. EP: row 1 = [0 1], row 2 = 4·[0 1] + [1 0] = [1, 4], i.e., EP = [0 1; 1 4]. PE: row 1 = 1·[4 1] = [4 1], row 2 = [1 0], i.e., PE = [4 1; 1 0]. The products are different: "first elimination, then exchange" and "first exchange, then elimination" give different matrices.

Exercise 14. Column space test. Can the system [1 3; 2 6]·[x; y] = [2; 5] be solved?

Solution. Both columns have the second component twice the first: [1; 2] and [3; 6] = 3·[1; 2]; any combination x·[1; 2] + y·[3; 6] has the form [s; 2s]. For b = [2; 5] we have 5 ≠ 2·2 = 4 — b is not in the column space. Answer: no (doubling the first equation gives 2x + 6y = 4 ≠ 5 — the lines are parallel).

Exercise 15. Challenge: construct a permutation. Construct P that maps rows (R_1, R_2, R_3) to (R_3, R_1, R_2).

Solution. New row 1 is old R_3, new 2 is old R_1, new 3 is old R_2, so the rows of P are the rows of the identity matrix in order 3, 1, 2: P = [0 0 1; 1 0 0; 0 1 0]. Check on rows [1 2 3; 4 5 6; 7 8 9]: P·A = [7 8 9; 1 2 3; 4 5 6]. Left multiplication permutes rows; P^3 = I.

2.9 Linear Algebra I, week 5: ISA exercises — questions and complete solutions

[Source: Week 5 Independent Study Assignments, practices 1–15; no solutions file provided — complete solutions with verification are given below]

Practice 1. Row picture and column picture. Solve the system [2 −1; 1 1]·[x; y] = [1; 5]. Describe both geometric pictures. Show each elimination step and verify the answer.

Solution. Add the equations: 3x = 6, x = 2, y = 3. Check: 2·2 − 3 = 1, 2 + 3 = 5. The lines 2x − y = 1 and x + y = 5 intersect at (2, 3); in columns: 2·[2; 1] + 3·[−1; 1] = [1; 5]. Answer: x = 2, y = 3.

Practice 2. Three outcomes. Classify the system [1 1; 2 2]·[x; y] = [2; c] for c = 4 and c = 5. Show each elimination step and verify the answer.

Solution. $R_2 \leftarrow R_2 - 2R_1$ gives [0 0 | c − 4]. For c = 4: 0 = 0, solutions [2 − t; t] (check: (2 − t) + t = 2, 2(2 − t) + 2t = 4). For c = 5: 0 = 1 — no solutions.

Practice 3. Find the multiplier. Eliminate below the first pivot in [2 3 | 7; 8 13 | 29]; find both pivots. Show each step and verify the answer.

Solution. λ_{21} = 8/2 = 4; $R_2 \leftarrow R_2 - 4R_1$: [8, 13 | 29] − 4·[2, 3 | 7] = [0, 1 | 1]. Result: [2 3 | 7; 0 1 | 1], pivots are 2 and 1. Check by invertibility: [0, 1 | 1] + 4·[2, 3 | 7] = [8, 13 | 29].

Practice 4. Back substitution. Solve [2 1 −1; 0 3 2; 0 0 4]·[x; y; z] = [3; 7; 8]. Show each step and verify the answer.

Solution. From bottom up: z = 8/4 = 2; 3y + 2·2 = 7, y = 1; 2x + 1 − 2 = 3, x = 2. Verification: [2 1 −1; 0 3 2; 0 0 4]·[2; 1; 2] = [3; 7; 8]. Answer: x = [2; 1; 2].

Practice 5. Complete elimination. Solve the system [1 1 1 | 6; 2 3 1 | 10; 1 2 3 | 14] by elimination. Show each step and verify the answer.

Solution. $R_2 \leftarrow R_2 - 2R_1$ and $R_3 \leftarrow R_3 - R_1$ give [1 1 1 | 6; 0 1 −1 | −2; 0 1 2 | 8]; then $R_3 \leftarrow R_3 - R_2$ gives [0 0 3 | 10]. Backwards: z = 10/3, y = z − 2 = 4/3, x = 6 − y − z = 4/3. Verification: the rows give 6, 10, 14 — correct. Answer: [4/3; 4/3; 10/3].

Practice 6. Identify the elementary matrix. Find E that implements $R_3 \leftarrow R_3 - 4R_1$ on a 3 × 3 matrix. Show each step and verify the answer.

Solution. E = [1 0 0; 0 1 0; −4 0 1] — the identity with −4 in position (3, 1); the third row of EA is (row 3) − 4·(row 1). Verification on [1 1 1; 2 3 1; 1 2 3]: row 3 is [1 2 3] − 4·[1 1 1] = [−3 −2 −1]. Inverse: E^{-1} = [1 0 0; 0 1 0; 4 0 1].

Practice 7. Multiplication EA. Let E = [1 0; −2 1], A = [3 1; 6 5]. Compute EA and explain the operation. Show each step and verify the answer.

Solution. Row 1: [3 1]; row 2: [6 5] − 2·[3 1] = [0 3]; EA = [3 1; 0 3] — subtracting twice row 1 from row 2 (multiplier 2). Verification: E^{-1} = [1 0; 2 1] returns A.

Practice 8. Transformed right-hand side. For A = [3 1; 6 5] and b = [4; 17] use E = [1 0; −2 1] to solve the system. Show each step and verify the answer.

Solution. Eb = [4; −2·4 + 17] = [4; 9]; EA = [3 1; 0 3]; then 3y = 9 → y = 3 and 3x + 3 = 4 → x = 1/3. Verification in the original system: 3·(1/3) + 3 = 4, 6·(1/3) + 5·3 = 17 — correct.

Practice 9. Zero pivot element. Solve [0 2; 3 1]·[x; y] = [4; 7] by row exchange; find P. Show each step and verify the answer.

Solution. P = [0 1; 1 0]; PA = [3 1; 0 2], Pb = [7; 4]; 2y = 4 → y = 2, 3x + 2 = 7 → x = 5/3. Verification: 0·(5/3) + 2·2 = 4, 3·(5/3) + 2 = 7 — correct.

Practice 10. Permutation of three rows. Find P that swaps rows 1 and 3 of matrix A = [1 2 0; 0 1 3; 4 0 2]; compute PA. Show each step and verify the answer.

Solution. P = [0 0 1; 0 1 0; 1 0 0]; row 1 of PA = row 3 of A and row 3 of PA = row 1 of A, so PA = [4 0 2; 0 1 3; 1 2 0]. Verification: P^2 = I, i.e., P·PA = A; P^{-1} = P = P^T.

Practice 11. Degenerate but consistent. Solve [1 2 1; 2 4 2; 0 1 1]·[x; y; z] = [3; 6; 2]. Show each step and verify the answer.

Solution. $R_2 \leftarrow R_2 - 2R_1$ gives [0 0 0 | 6 − 2·3] = [0 0 0 | 0] — a row disappeared, the system is consistent but degenerate. Remaining: x + 2y + z = 3 and y + z = 2; z = t: y = 2 − t, x = t − 1. Answer: [t − 1; 2 − t; t]. Verification at t = 2: [1; 0; 2] gives 3, 6, 2 — correct.

Practice 12. Degenerate and inconsistent. In the previous system, replace the right-hand side with b = [3; 7; 2]. Is the system consistent? Show each step and verify the answer.

Solution. $R_2 \leftarrow R_2 - 2R_1$ gives [0 0 0 | 7 − 2·3] = [0 0 0 | 1], i.e., 0 = 1 — contradiction. The system is inconsistent: the necessary condition $b_2 = 2b_1$ is violated (7 ≠ 6).

Practice 13. Column interpretation of consistency. For A = [1 2; 2 4], which vectors b = [b_1; b_2] admit a solution to Ax = b? Show each step and verify the answer.

Solution. $R_2 \leftarrow R_2 - 2R_1$ leaves [1 2 | b_1; 0 0 | $b_2 - 2b_1$]: consistency requires $b_2 = 2b_1$. The columns [1; 2] and [2; 4] = 2·[1; 2] are multiples of [1; 2], so Ax always lies on the line $b_2 = 2b_1$. Answer: b = [b_1; $2b_1$]. Verification: [4; 8] is solved (x = 4, y = 0), [4; 9] is not.

Practice 14. Order of matrix operations. Let E = [1 0; −2 1] and P = [0 1; 1 0]. Compare EP and PE. Show each step and verify the answer.

Solution. EP: rows [0 1] and [1, −2], i.e., EP = [0 1; 1 −2]. PE: rows [−2 1] and [1 0], i.e., PE = [−2 1; 1 0]. The products are different — row operations do not commute: in EP elimination acts on the permuted rows, in PE on the original ones.

Practice 15. Challenge: construct a system. Build a 3 × 3 system that requires row exchange, with three pivots and solution x = [1; 2; 3]. Show each step and verify the answer.

Solution. A = [0 1 0; 1 0 0; 0 0 1] — zero in position (1, 1), with a one below; b = A·[1; 2; 3] = [2; 1; 3]. Swapping rows 1 and 2 (P = [0 1 0; 1 0 0; 0 0 1]) gives the identity matrix: pivots 1, 1, 1, solution x = [1; 2; 3]. Verification: A·[1; 2; 3] = [2; 1; 3]; without exchange, the first column is not eliminated.

Week 6 — Derivatives (Math I); Matrix operations, inverses and decompositions (Linear Algebra I)

Week 6 of **Math I** (Fall 2026, lecturer G. Neofotistos) is dedicated to **derivatives**: sections **3.1–3.4** of the textbook (OpenStax Calculus, vol. 1) — definition of the derivative, derivative as a function, differentiation rules, derivatives as rates of change — plus the instructor's NOTES on differentiability, one-sided derivatives, kinks, cusps, and vertical tangents. Week's objectives (Learning Objectives 3.1–3.4): tangent line and its slope; derivative as the limit of the difference quotient and its calculation at a point; rate as rate of change (average and instantaneous) and estimating the derivative from a table; derivative function and its graph; connection between differentiability and continuity and conditions for non-existence of the derivative; higher-order derivatives; constant, power, sum/difference, constant multiple, product, and quotient rules; increment formula; applications of derivatives to motion, population, and marginal quantities. Lecture note: the derivative is *local sensitivity* or *rate of change*.

3.1 Math I: Week 6 Formula Catalog

[Source: Week 6 lecture "Derivatives" (§§3.1–3.4 of OpenStax Calculus, vol. 1) + Instructor's NOTES; files $m_{w6lecg01}$–$g17$, $m_{w6rrg01}$–$g28$]

3.1.1 Difference quotient, secant line, and tangent line

The slope of the *secant* line to the graph y = f(x) through the points (a, f(a)) and (x, f(x)) is given by the **difference quotient**; replacing x with a + h gives the form with increment h:

$$m_{sec} = (f(x) - f(a))/(x - a) \quad (3.1)$$

$$m_{sec} = (f(a + h) - f(a))/h \quad (3.2)$$

When the second point approaches the first (x → a or h → 0), the secant lines approach the limiting line — the **tangent line**: the line through (a, f(a)) with

slope

$$m_{tan} = lim\ x \to a\ (f(x) - f(a))/(x - a)\ (3.3)$$

$$m_{tan} = lim\ h \to 0\ (f(a + h) - f(a))/h\ (3.4)$$

— provided the limit exists. Both forms are interchangeable; the choice depends on computational convenience. Near the point of tangency, the graph is *locally linear*: on a narrow interval, the graph of f(x) = sqrt(x) and its tangent at (1, 1) practically coincide.

3.1.2 Derivative at a point and derivative function

The limit of the form (3.3)–(3.4) appears in physics (velocity, acceleration), economics (marginal functions), biology (growth rates) — so it is given the name **derivative**; the process of finding it is **differentiation**.

$$f'(a) = lim\ x \to a\ (f(x) - f(a))/(x - a)\ (3.5)$$

$$f'(a) = lim\ h \to 0\ (f(a + h) - f(a))/h\ (3.6)$$

$$f'(x) = lim\ h \to 0\ (f(x + h) - f(x))/h\ (3.9)$$

A function is **differentiable at a point a** if f′(a) exists; on an open set S — if f′ exists at every point of S. Formula (3.9) defines the **derivative function** f′; the **instantaneous rate of change** of f at point a is f′(a). Notations: f′(x), y′, dy/dx, d/dx(f(x)), $dy/dx|_{x=a}$; Leibniz notation dy/dx = lim Δx→0 Δy/Δx. Derivatives of elementary functions by definition: $d/dx(x^2) = 2x$; d/dx(sqrt(x)) = 1/(2·sqrt(x)); $d/dx(1/x) = -1/x^2$; lim t→0 (sin t)/t = 1; from a table of values: for x^2, the ratios $(x^2 - 9)/(x - 3)$ as x → 3 give 5.9, 5.99, 5.999, … → f′(3) = 6 (example 3.4).

3.1.3 Velocity and rates of change

If s(t) is the position of an object on the coordinate axis, then the average velocity on [a, t] is the difference quotient (3.7), and the **instantaneous velocity** at moment a is the derivative of position (3.8):

$$v_{ave} = (s(t) - s(a))/(t - a)\ (3.7)$$

$$v(a) = s'(a) = lim\ t \to a\ (s(t) - s(a))/(t - a)\ (3.8)$$

№	Quantity	Definition	Interpretation	SI Unit
1	Velocity	v = dr/dt	Rate of change of position	m/s
2	Speed	v = ds/dt = |dr/dt|	Rate of change of distance traveled s	m/s
3	Acceleration	a = dv/dt = d^2 r/dt^2	Rate of change of velocity	m/s^2
4	Jerk	j = da/dt	Rate of change of acceleration; important for vibrations and smooth motion	m/s^3

Table 1. Quick lecture reference: derivatives in mechanics

3.1.4 Differentiation rules

Finding derivatives by definition is time-consuming; rules allow bypassing limit calculation. Functions f(x) = c and $g(x) = x^n$ are building blocks of polynomials and rational functions.

$$d/dx(c) = 0\ (constant\ rule,\ theorem\ 3.2)$$

$$d/dx(x^n) = n \cdot x^{n-1} \text{ (power rule, theorem 3.3)}$$

$$d/dx(x^k) = k \cdot x^{k-1} \text{ (extended power rule, theorem 3.7; } k \text{ — integer)}$$

$$(f + g)' = f' + g';\ (f - g)' = f' - g';\ (k \cdot f)' = k \cdot f' \text{ (theorem 3.4)}$$

$$(f \cdot g)' = f' \cdot g + g' \cdot f \text{ (product rule, theorem 3.5)}$$

$$(f/g)' = (f' \cdot g - g' \cdot f)/g^2 \text{ (quotient rule, theorem 3.6)}$$

$$(f \cdot g \cdot h)' = f' \cdot g \cdot h + f \cdot g' \cdot h + f \cdot g \cdot h' \text{ (extended product rule, example 3.29)}$$

- Constant — horizontal line: d/dx(8) = 0. Power rule: the exponent becomes the coefficient, the power is reduced by 1 ($d/dx(x^{10}) = 10x^9$); does not apply to functions like 3^x.
- The derivative of a product is NOT equal to the product of derivatives (for x·x it would be 1, but actually 2x); the derivative of a quotient is NOT equal to the quotient of derivatives.
- Example 3.21: $d/dx(2x^5 + 7) = 10x^4$ (order: sum → constant multiple → power/constant). Example 3.28: k(x) = 3h(x) + x^2·g(x) → k′(x) = 3h′(x) + 2x·g(x) + x^2·g′(x): rules are applied in the reverse order of the function evaluation.
- Technique: rewrite fractions as powers — $d/dx(6/x^2) = d/dx(6x^{-2}) = -12/x^3$ (example 3.27), this is simpler than the quotient rule.

3.1.5 One-sided derivatives; differentiability and continuity

$$f'_-(a) = \lim_{h \to 0-} (f(a + h) - f(a))/h;\ f'_+(a) = \lim_{h \to 0+} (f(a + h) - f(a))/h$$

$$f'(a) \text{ exists} \iff f'_-(a) = f'_+(a) \text{ and both are finite (at an interior point of the domain)}$$

$$f \text{ differentiable at } a \Rightarrow f \text{ continuous at } a \text{ (theorem 3.1)}$$

Proof idea for theorem 3.1: f(a + h) − f(a) = h·(f(a + h) − f(a))/h → 0·f′(a) = 0, so f(a + h) → f(a). **The converse implication is false.** Typical reasons for the absence of a derivative:

- discontinuity: a function not continuous at a point cannot be differentiable (contrapositive of theorem 3.1);
- kink/corner: two different finite one-sided slopes — |x| at zero: f′_−(0) = −1, f′_+(0) = 1;
- cusp: infinite slopes of opposite signs — $|x|^{2/3}$: −∞ from the left, +∞ from the right;
- vertical tangent: identical infinite slopes — $x^{1/3}$: $(x^{1/3} - 0)/x = 1/x^{2/3} \to +\infty$ from both sides; informally written as "f′(0) = +∞", but there is no finite derivative;
- oscillation: for f(x) = {x·sin(1/x), x ≠ 0; 0, x = 0} the difference quotient at zero equals sin(1/x) — the limit does not exist (fig. 3.16);
- procedure (NOTES): first check continuity (if none, no differentiability), then compare one-sided difference quotients — need the same finite limit.

3.1.6 Higher derivatives

$$f''(x),\ f'''(x),\ f^{(4)}(x),\ \ldots,\ f^{(n)}(x);\ d^2y/dx^2,\ d^3y/dx^3,\ \ldots,\ d^ny/dx^n$$

The derivative is itself a function, so it can be differentiated again: this gives the **second derivative**, then the third, and so on — **higher derivatives** (d^2y/dx^2 is the compact form of d/dx(dy/dx)). The derivative of position is velocity, the

derivative of velocity is acceleration; each differentiation of a polynomial reduces its degree by 1, until a constant is obtained, whose derivative is 0.

3.1.7 Derivatives as rates of change (§3.4)

$$f(a + h) \approx f(a) + f'(a)\cdot h \text{ (increment formula, 3.10)}$$

Amount of change on [a, a + h] is f(a + h) − f(a); **average rate of change** is (f(a + h) − f(a))/h; instantaneous rate is f′(a). For small h, the increment formula estimates a nearby unknown value from a known value and the rate of change (example 3.33: f(3) = 2, f′(3) = 5 → f(3.2) ≈ 2 + 0.2·5 = 3).

$$v(t) = s'(t) \text{ — speed; } |v(t)| \text{ — scalar speed (speed); } a(t) = v'(t) = s''(t) \text{ — acceleration}$$

$$P'(t) \text{ — population growth rate; } MC(x) = C'(x),\ MR(x) = R'(x),\ MP(x) = P'(x) = R'(x) - C'(x)$$

Marginal quantities in economics are derivatives of cost C(x), revenue R(x), and profit P(x) = R(x) − C(x); substituting h = 1 gives MC(x) ≈ C(x + 1) − C(x): the derivative estimates the change from producing one additional unit. Lecture note: chain rule, implicit and trigonometric differentiation not required this week; "differentiability" means a finite derivative.

3.2 Math I: Week 6 Practical Session — questions and full solutions

[Source: Week 6 Practical Session exercises with solutions, MATH I (Fall 2026), prepared by G. Neofotistos — files $m_{w6pracg01-g24}$; all 12 exercises with full solutions]

Exercise 1. Smooth motion on a track. The track is described by the function f(x) = {(1/10)x^2 + bx + c, if x < −10; −(1/4)x + 5/2, if x ≥ −10}. Is the car's motion on the track smooth? Discuss in terms of continuity and differentiability.

Solution. Smoothness means that at the junction of the branches at x = −10 there is neither a discontinuity nor a kink: the function must be both continuous and differentiable there. Continuity: lim x→−10− f(x) = 10 − 10b + c must equal f(−10) = −(1/4)(−10) + 5/2 = 5, that is, 10 − 10b + c = 5. Differentiability: the slope of the parabola at x = −10 is −2 + b and must match the slope of the line −1/4, that is, b − 2 = −1/4. For arbitrary b and c the conditions are not met: violation of the first causes a jump at the junction, the second — a kink; the motion is not smooth. Both conditions are satisfied only when b = 7/4, c = 10b − 5 = 25/2 (full derivation — in exercise 2) — only then is the motion on the track smooth.

Exercise 2. Piecewise function, continuous and differentiable (example 3.14). The track for a toy car starts with a parabola and transitions to a straight line: f(x) = {(1/10)x^2 + bx + c, if x < −10; −(1/4)x + 5/2, if x ≥ −10} (in inches). For smooth motion f must be continuous and differentiable at −10. Find b and c.

Solution. Continuity at x = −10: lim x→−10− f(x) = (1/10)·100 − 10b + c = 10 − 10b + c = f(−10) = 5, whence c = 10b − 5. Differentiability: f′(−10) = lim x→−10 (f(x) − f(−10))/(x + 10) must exist; we compute the one-sided limits and equate them. From the left (substituting c = 10b − 5): lim x→−10− [(1/10)x^2 + bx + 10b − 10]/(x + 10); multiplying numerator and denominator by 10: lim (x^2 + 10bx + 100b − 100)/(10(x + 10)); grouping x^2 − 100 + 10b(x + 10) = (x + 10)(x − 10 + 10b) gives lim (x − 10 + 10b)/10 = (−20 + 10b)/10 = b − 2. From the right: lim x→−10+ [−(1/4)x + 5/2 − 5]/(x + 10) = lim [−(1/4)x − 5/2]/(x + 10) = lim −(x + 10)/(4(x + 10)) = −1/4. Equating: b − 2 = −1/4, whence b = 7/4 and c = 10·(7/4)

$- 5 = 25/2$.

Exercise 3. Reciprocal function. Let f(x) = 1/x, x ≠ 0. (a) Find f′(x) by definition of the limit. (b) Find the tangent line at x = 2. (c) Is the slope of the tangent line positive or negative there? Explain geometrically.

Solution. (a) For $x \neq 0$: $f'(x) = \lim_{h\to 0} (1/(x + h) - 1/x)/h = \lim_{h\to 0} [(x - (x + h))/(x(x + h))]/h = \lim_{h\to 0} (-1)/(x(x + h)) = -1/x^2$. (b) Point of tangency: $f(2) = 1/2$; slope $f'(2) = -1/4$. Tangent line: $y - 1/2 = -(1/4)(x - 2)$, that is, $y = 1 - x/4$. (c) The slope is negative ($-1/4 < 0$): the graph of $1/x$ near $x = 2$ is decreasing — as x increases, the function values decrease.

Exercise 4. Absolute value. Let f(x) = |x|. (a) Show that f is continuous at zero. (b) Using difference quotients, find f′₋(0) and f′₊(0). (c) Explain why f′(0) does not exist. (d) Sketch the graph and indicate the corner point.

Solution. (a) $\lim_{x\to 0} |x| = 0 = f(0)$ — f is continuous at zero. (b) Difference quotient: $(f(h) - f(0))/(h - 0) = |h|/h = -1$ for $h < 0$ and 1 for $h > 0$; therefore, $f'_-(0) = -1$, $f'_+(0) = 1$. (c) The one-sided derivatives are different ($-1 \neq 1$) — the two-sided limit does not exist, $f'(0)$ does not exist. (d) The graph is a V-shaped broken line; a corner (kink) at the vertex (0, 0): slope −1 to the left, +1 to the right.

Exercise 5. Branch Junction. Consider f(x) = {x^2, if x ≤ 1; 2x − 1, if x > 1}. (a) Is f continuous at x = 1? (b) Calculate the left-hand and right-hand derivatives at x = 1. (c) Is f differentiable there? (d) Find the tangent at the junction point, if it exists.

Solution. (a) $f(1) = 1$; $\lim_{x\to 1-} f(x) = 1$; $\lim_{x\to 1+} f(x) = 2\cdot 1 - 1 = 1$ — all three quantities are equal, f is continuous at 1. (b) $f'_-(1) = (x^2)'|_{x=1} = 2$; $f'_+(1) = (2x - 1)'|_{x=1} = 2$. (c) The one-sided derivatives exist, are finite, and are equal: f is differentiable at 1, $f'(1) = 2$. (d) Tangent through (1, 1) with slope 2: $y = 2x - 1$. Note: the right branch itself is this tangent, so the junction is smooth.

Exercise 6. True or false? With justification. (a) If f is differentiable at a, then f is continuous at a. (b) If f is continuous at a, then f is differentiable at a. (c) If f′(a) = 0, then f must be a constant. (d) If both one-sided derivatives exist and are finite, then f′(a) exists. (e) A function can have a vertical tangent at a point where it is continuous. (f) A corner can occur where two smooth pieces of the graph meet.

Solution. (a) True: Theorem 3.1 "differentiability implies continuity". (b) False: counterexample — |x| at zero: continuous, but $f'(0)$ does not exist. (c) False: counterexample — x^2: $f'(0) = 0$, but the function is not constant. (d) False: the one-sided derivatives must also coincide; for |x| at zero, both are finite (−1 and 1), but $f'(0)$ does not exist. (e) True: $x^{1/3}$ is continuous at zero and has a vertical tangent there. (f) True: two smooth pieces can meet with different slopes — like the rays $y = -x$ and $y = x$ for the function |x|.

Exercise 7. Four cases at zero. For each function, determine whether it is continuous and differentiable at zero; indicate a corner, cusp, or vertical tangent: (a) f(x) = x^2; (b) g(x) = |x|; (c) p(x) = $|x|^{2/3}$; (d) q(x) = $x^{1/3}$.

Solution. (a) x^2 is continuous; $(h^2 - 0)/h = h \to 0$, therefore $f'(0) = 0$ — horizontal tangent. (b) |x| is continuous; $f'_-(0) = -1$, $f'_+(0) = 1$ — kink, derivative does not exist. (c) $|x|^{2/3}$ is continuous; $|h|^{2/3}/h \to -\infty$ from the left and $+\infty$ from the right — cusp: the graph is infinitely steep from both sides with opposite signs; no finite

derivative. (d) $x^{1/3}$ is continuous; $(h^{1/3} - 0)/h = 1/h^{2/3} \to +\infty$ from both sides — vertical tangent $x = 0$; no (finite) derivative. Conclusion: all four functions are continuous at zero, derivative exists only for x^2.

Exercise 8. The function $x\cdot|x|$. Let $f(x) = x\cdot|x|$. (a) Write f piecewise. (b) Find $f'(0)$ by definition. (c) Find $f'(x)$ for $x \neq 0$. (d) Is f' continuous at zero? (e) Does $f''(0)$ exist? Explain using one-sided limits.

Solution. (a) $f(x) = \{-x^2$, if $x < 0$; x^2, if $x \geq 0\}$. (b) $f'(0) = \lim_{h\to 0} (h\cdot|h| - 0)/h = \lim_{h\to 0} |h| = 0$. (c) For $x \neq 0$: $f'(x) = -2x$ for $x < 0$ and $2x$ for $x > 0$; together with the value at zero $f'(x) = \{-2x, x < 0; 0, x = 0; 2x, x > 0\} = 2|x|$. (d) $2|x| \to 0 = f'(0)$ as $x \to 0$ — f' is continuous at zero. (e) $(f'(h) - f'(0))/h = 2|h|/h \to -2$ from the left and $+2$ from the right; the one-sided limits are different, therefore $f''(0)$ does not exist (the graph of $f' = 2|x|$ has a kink at zero).

Exercise 9. Removable "hole". Consider $f(x) = (x^2 - 4)/(x - 2)$, $x \neq 2$. (a) Simplify the expression while preserving the domain restriction. (b) Find $f'(x)$ on its domain. (c) Does the original function have a derivative at point 2? (d) Define an extension at point 2 that will be differentiable.

Solution. (a) For $x \neq 2$: $f(x) = ((x - 2)(x + 2))/(x - 2) = x + 2$ — point $x = 2$ is still excluded. (b) $f'(x) = 1$ for $x \neq 2$. (c) The original function is not defined at point 2, therefore it has no derivative at point 2. (d) Extend $\tilde{f}(2) = 4 = \lim_{x\to 2} f(x)$; then $\tilde{f}(x) = x + 2$ for all x and $\tilde{f}'(2) = 1$ — the extension removed the "hole" and made the function differentiable.

Exercise 10. Correcting common mistakes. A student writes: (a) $d/dx(x^2\cdot x^3) = (2x)(3x^2) = 6x^3$; (b) $d/dx((x^2 + 1)/(x + 1)) = 2x/1$; (c) $d/dx(7x^3 + 5) = 21x^2 + 5$. For each statement, indicate the error, give the correct derivative, and name the corresponding rule.

Solution. (a) Error: the derivative of a product is not the product of derivatives. Correctly (product rule): $d/dx(x^2\cdot x^3) = (2x)\cdot x^3 + x^2\cdot(3x^2) = 5x^4$ (check: $x^2\cdot x^3 = x^5$, and $d/dx(x^5) = 5x^4$). (b) Error: the derivative of a quotient is not the quotient of derivatives. Correctly (quotient rule): $d/dx((x^2 + 1)/(x + 1)) = (2x(x + 1) - (x^2 + 1))/(x + 1)^2 = (x^2 + 2x - 1)/(x + 1)^2$, $x \neq -1$. (c) Error: the derivative of a constant is zero. Correctly: $d/dx(7x^3 + 5) = 21x^2$.

Exercise 11. Higher derivatives of $1/x$. Let $f(x) = 1/x$, $x \neq 0$. (a) Find $f'(x)$, $f''(x)$, $f'''(x)$. (b) Describe the pattern in signs and coefficients. (c) Propose a formula for $f^{(n)}(x)$.

Solution. (a) $f'(x) = -1/x^2$; $f''(x) = 2/x^3$; $f'''(x) = -6/x^4$. (b) Signs alternate $(-, +, -, \ldots)$, absolute values of coefficients are factorials ($1 = 1!$, $2 = 2!$, $6 = 3!$), the exponent of the denominator increases by 1. (c) $f^{(n)}(x) = (-1)^n\cdot n!/x^{n+1}$, $x \neq 0$. Check for $n = 3$: $(-1)^3\cdot 3!/x^4 = -6/x^4$ — matches.

Exercise 12. Heating experiment. During the experiment $T(t) = 20 + 8t - t^2$, $0 \leq t \leq 4$ (T in degrees Celsius, t in minutes). (a) Find $T'(t)$. (b) Interpret $T'(1)$ and $T'(3)$. (c) Find the average heating rate on $[1, 3]$. (d) Using $T(2)$ and $T'(2)$, estimate $T(2.1)$. (e) Calculate the exact $T(2.1)$ and the estimation error.

Solution. (a) $T'(t) = 8 - 2t$. (b) $T'(1) = 6$ °C/min, $T'(3) = 2$ °C/min: the component is heating at both moments, but slower at $t = 3$. (c) $T(1) = 27$, $T(3) = 35$; average rate $(35 - 27)/(3 - 1) = 4$ °C/min. (d) $T(2) = 32$, $T'(2) = 4$; using the increment formula $T(2.1) \approx 32 + 4\cdot 0.1 = 32.4$ °C. (e) Exactly: $T(2.1) = 20 + 16.8 - 4.41 = 32.39$ °C; the tangent overestimates by 0.01 °C.

Main ideas of the lesson: smoothness of a piecewise function's joint requires both continuity and matching one-sided slopes; continuity does not guarantee differentiability (kink |x|, cusp $|x|^{2/3}$, vertical tangent $x^{1/3}$); derivatives of products and quotients are not replaced by products/quotients of derivatives; extension can remove a "hole" and restore differentiability.

3.3 Math I: Week 6 Independent Study Assignments — questions and complete solutions

[Source: Week 6 — Independent Study Assignments (Homework), MATH I (Fall 2026); no solutions in file — complete solutions prepared for this manual]

Independent Study Assignments (ISA) — homework assignments that are not submitted or graded: their purpose is to master the week's material more deeply by solving problems independently, with classmates, or with an AI assistant (to clarify concepts, not to obtain ready-made solutions).

Assignment 1 (optional). Interact with an AI assistant (ChatGPT, Claude, DeepSeek, Gemini, etc.) in a structured conversation: formulate the most important points from the lecture content and ask the assistant to evaluate your answer (for content and clarity; also for depth and complexity if applicable). If you find erroneous conclusions, discrepancies, or contradictions — point them out and request explanations; if the contradiction is not resolved, consult the instructor.

How to complete. This is an instructional assignment; there is no single "solution." Create a list of key ideas from week 6 (difference quotient and tangent; derivative as a limit; differentiation rules; one-sided derivatives and the criterion for the existence of f′(a); the "differentiability ⇒ continuity" theorem and its violations — corner, cusp, vertical tangent, oscillation; higher derivatives; increment formula; velocity/acceleration/jerk; marginal quantities), ask the assistant to evaluate the completeness and accuracy of your formulations, and discuss each remark.

Assignment 2. Thoroughly study all problems, exercises, and examples discussed in the lecture and practical session.

How to complete. All 12 exercises from the practical session — with complete solutions in the previous section; lecture exercises (In-Class activity № 1–10) and examples from Required reading — in the next section; examples 3.34–3.35 (Assignment 3) — below.

Assignment 3. Study examples 3.34 and 3.35 from the textbook (OpenStax Calculus 1).

Solution. Example 3.34 (instantaneous and average velocity). A ball falls from a height of 64 feet: $s(t) = -16t^2 + 64$. (a) The ground is reached when $s(t) = 0$: $t = 2$ s. The instantaneous velocity at impact: $v(t) = s'(t) = -32t$, so $v(2) = -64$ ft/s (the "minus" sign indicates downward motion; there is a typo in the transcript file "v(t) = −64", corrected). (b) Average velocity during the fall: $v_{ave} = (s(2) - s(0))/(2 - 0) = -32$ ft/s. Example 3.35 (relationship between v(t) and a(t)). A particle: $s(t) = t^3 - 4t + 2$; $v(t) = 3t^2 - 4$, $a(t) = 6t$; at $t = 1$: $v(1) = -1$, $a(1) = 6$. (a) Since $v(1) < 0$, the particle is moving from right to left. (b) $v(1) < 0$ and $a(1) > 0$ — opposite directions: |v(t)| is decreasing, the particle is slowing down.

Assignment 4. Solve the following problems (they are from the textbook, pages 196, 210, 211, 212, 232, …).

Solution. The list of problems is provided in the assignment file as images (the text layer did not preserve them; the list is cut off at "...232, ..."). The specified pages correspond to checkpoint questions and examples from the textbook: checkpoint 3.4 (p. 196), 3.8 (p. 210), 3.9 and 3.10 (pp. 211–212) and example 3.34 (p. 232, discussed in assignment 3). Solutions: Checkpoint 3.4 (a stone falls from a height of 64 feet, $s(t) = -16t^2 + 64$, $0 \le t \le 2$; instantaneous velocity after 1 s by equation 3.5): $s(1) = 48$; $v(1) = \lim t\to1\ (-16t^2 + 16)/(t - 1) = \lim t\to1\ -16(t - 1)(t + 1)/(t - 1) = -32$ ft/s. Checkpoint 3.8 (find a and b such that $f(x) = \{ax + b, \text{if } x < 3; x^2, \text{if } x \ge 3\}$ is continuous and differentiable at $x = 3$): continuity gives $3a + b = f(3) = 9$, differentiability gives equality of slopes $a = (x^2)'|_{x=3} = 6$; hence $a = 6$, $b = -9$ (check: $3\cdot6 - 9 = 9 = 3^2$, slopes $6 = 6$). Checkpoint 3.9 ($f(x) = x^2$, find $f''(x)$): $f'(x) = 2x$, $f''(x) = 2$. Checkpoint 3.10 ($s(t) = t^3$, find $a(t)$): $v(t) = 3t^2$, $a(t) = 6t$.

Assignment 5. Consider $f(x) = \{2x + 1, \text{if } x < 0; x^2 + 1, \text{if } x \ge 0\}$. (a) Check continuity at zero. (b) Find $f'_-(0)$ and $f'_+(0)$. (c) Determine if f is differentiable at zero. (d) Describe the graph near the junction point.

Solution. (a) $f(0) = 0^2 + 1 = 1$; $\lim x\to0-\ f(x) = 2\cdot0 + 1 = 1$; $\lim x\to0+\ f(x) = 0^2 + 1 = 1$ — the limit exists and equals the value: f is continuous at zero. (b) The slope of the left branch $2x + 1$ at zero: $f'_-(0) = 2$; the slope of the right branch $x^2 + 1$: $f'_+(0) = (x^2 + 1)'|_{x=0} = 0$. (c) $f'_-(0) = 2 \ne 0 = f'_+(0)$: f is not differentiable at zero. (d) On the left — a line with slope 2, on the right — a parabola with a horizontal tangent at (0, 1); the branches continuously join at (0, 1), but with different slopes — there is a kink at the connection point.

Problem 6. True or false? Justify your answer. (a) If f is differentiable at a, then f is continuous at a. (b) If f is continuous at a, then f is differentiable at a. (c) If $f'(a) = 0$, then f must be a constant function. (d) If both one-sided derivatives exist and are finite, then $f'(a)$ exists. (e) A function can have a vertical tangent at a point where it is continuous.

Solution. (a) True — Theorem 3.1. (b) False — $|x|$ at zero: continuous, but $f'(0)$ does not exist. (c) False — x^2: $f'(0) = 0$, but the function is not constant. (d) False — one-sided derivatives must also be equal: for $|x|$ at zero, both are finite (-1 and 1), but $f'(0)$ does not exist. (e) True — $x^{1/3}$ is continuous at zero and has a vertical tangent there ($1/h^{2/3} \to +\infty$ from both sides), so no finite derivative exists.

Optional assignment (for racing enthusiasts). Study the student project "Formula One Grandstands" (p. 226 of the textbook): the track section is modeled by the function $f(x) = x^3 + 3x^2 + x$, and the front corner of the grandstand is planned at point $(-1.9, 2.8)$. (1) Drivers most often lose control when entering the turn — where the slope of the tangent is 1; find the coordinates (x, y) of this point on the turn. (2) Find the equation of the tangent at this point. (3) Find the x-coordinate of the point where the tangent intersects the line $y = 2.8$: is this point safely to the right of the grandstand or are spectators in danger? (4) Suppose a driver loses control earlier — at point $(-2.5, 0.625)$; what is the slope of the tangent there? (5) Are spectators safe in this scenario? (6) Keep the current grandstand design or move the grandstands?

Solution. (1) $f'(x) = 3x^2 + 6x + 1$; $f'(x) = 1$ gives $3x(x + 2) = 0$, so $x = 0$ or $x = -2$. The point at the turn entrance is $x = -2$: $f(-2) = -8 + 12 - 2 = 2$, point $(-2, 2)$. (2) Tangent through $(-2, 2)$ with slope 1: $y = x + 4$. (3) Intersection with $y = 2.8$: $x = -1.2$. Point $(-1.2, 2.8)$ is to the right of the front corner of the grandstand ($x = -1.9$) — the car's trajectory passes to the right of the grandstand, spectators

are safe. (4) f′(−2.5) = 18.75 − 15 + 1 = 4.75. (5) Tangent: y = 0.625 + 4.75·(x + 2.5); at y = 2.8: x + 2.5 = 2.175/4.75 ≈ 0.458, x ≈ −2.04 — left of the grandstand corner: a steep tangent passes through the grandstand area, spectators are in danger. (6) The design is unsafe: grandstands need to be moved further from the track or relocated.

3.4 Math I: Required reading for week 6 — summary and key points

[Source: Week 6 Required Reading (Chapter 3, §§3.1–3.4 of OpenStax Calculus, vol. 1; files $m_{w6rrg01}$–g28) + Instructor's NOTES "Continuity, Differentiability, Left- and Right-sided derivatives, kink points, cusps, vertical tangents"]

The NOTES summary complements §§3.1–3.2 of the textbook: *continuity* means that function values connect without a break; *differentiability* means that the graph has a well-defined finite slope of the tangent. A function can be continuous but not differentiable at a corner point or with a vertical tangent. Key points:

- Continuity at a: lim x→a f(x) = f(a) — three conditions: f(a) is defined, the limit exists, and they are equal. Differentiability at a: lim h→0 (f(a + h) − f(a))/h exists as a finite number; the difference quotient is the slope of the secant, approaching the slope of the tangent.
- Main relationship: differentiability at a implies continuity at a; the converse is false (|x| at zero: continuous, but the sharp angle prevents differentiability). Why: f(a + h) − f(a) = h·(f(a + h) − f(a))/h → 0·f′(a) = 0, so f(a + h) → f(a).
- One-sided derivatives f′_−(a), f′_+(a); at an interior point f′(a) exists ⟺ they are equal and finite. Example: for x^2 at zero (f(h) − f(0))/h = h → 0 from both sides — f′(0) = 0, the tangent is horizontal.
- Kink: two different finite slopes (|x|: −1 and 1); engineering interpretation — a position graph where velocity jumps from one finite value to another. Cusp: infinite slopes of opposite signs ($|x|^{2/3}$: −∞ and +∞). Vertical tangent: identical infinite slopes ($x^{1/3}$: +∞ from both sides, tangent x = 0); although informally written as "f′(0) = +∞", no derivative exists as a finite number.
- Procedure: first check continuity (no continuity — no differentiability), then compare one-sided difference quotients. Other failures are also possible: slopes of secants may oscillate without limit (x·sin(1/x)).

Function	Left slope	Right slope	Continuous at 0?	Differentiable at 0?	Behavior
x^2	0	0	yes	yes	horizontal tangent
|x|	−1	1	yes	no	kink
$\|x\|^{2/3}$	−∞	+∞	yes	no	cusp
$x^{1/3}$	+∞	+∞	yes	no	vertical tangent

Table 2. Four main cases of behavior at zero (Instructor's NOTES)

Examples from Required reading (not included in the formula catalog and practical session), with solutions:

Example 3.8 (RR; rate of change of speed). Hennessey Venom GT accelerates from 0 to 60 mph in 3.05 s, to 100 mph in 5.88 s, to 200 mph in 14.51 s, and to 229.9 mph in 19.96 s. What happens to the acceleration as it approaches 229.9 mph: does it increase, decrease, or remain constant?

Solution. Convert speeds to ft/s: 60 mph = 88; 100 ≈ 146.67; 200 ≈ 293.33; 229.9 ≈ 337.19. Average accelerations (v(t) − 337.19)/(t − 19.96) on intervals [t,

19.96]: 16.89; 14.74; 13.53; 8.05 ft/s per second at t = 0; 3.05; 5.88; 14.51 respectively. They are decreasing — the acceleration decreases as it approaches 229.9 mph.

Example 3.10 (RR; rate of change of profit). A firm sells x game consoles at price p = −0.01x + 400 dollars; costs C(x) = 100x + 10000 dollars. Find the rate of change of profit when producing 10000 games. Should production be increased?

Solution. Revenue R(x) = x·p = $-0.01x^2 + 400x$; profit P(x) = R − C = $-0.01x^2 + 300x - 10000$. Value: P(10000) = 1 990 000 dollars (in transcript there is a typo "1900000", corrected). By definition (3.5): $P'(10000) = \lim_{x\to 10000} (P(x) - P(10000))/(x - 10000) = \lim_{x\to 10000} (-0.01x^2 + 300x - 2\,000\,000)/(x - 10000)$; the numerator factors as $-0.01(x - 10000)(x - 20000)$, so the limit equals $-0.01\cdot(10000 - 20000) = 100$. Thus, $P'(10000) = 100 > 0$ and $P(10000) > 0$ — profit is increasing, the firm should increase production.

Example 3.13 (lecture and RR; sketch of derivative). From the graph of f(x) with a local maximum at (3, 6) and a local minimum at (−2, 2), sketch the graph of f′(x).

Solution. The function increases on (−2, 3) and decreases on (−∞, −2) and (3, +∞). Therefore, $f'(x) > 0$ on (−2, 3), $f'(x) < 0$ outside this interval, and at points of horizontal tangents $f'(-2) = 0$ and $f'(3) = 0$. The derivative graph lies above the x-axis on (−2, 3), crosses it at points −2 and 3, and lies below outside.

Example 3.24 (RR; product of binomials). For $j(x) = (x^2 + 2)(3x^3 - 5x)$, find j′(x) using the product rule and verify the result by expanding the product.

Solution. $f(x) = x^2 + 2$ ($f' = 2x$), $g(x) = 3x^3 - 5x$ ($g' = 9x^2 - 5$). $j'(x) = 2x(3x^3 - 5x) + (9x^2 - 5)(x^2 + 2) = (6x^4 - 10x^2) + (9x^4 + 13x^2 - 10) = 15x^4 + 3x^2 - 10$. Verification: $j(x) = 3x^5 + x^3 - 10x \to j'(x) = 15x^4 + 3x^2 - 10$ — matches.

Example 3.25 (RR; quotient rule). Find the derivative of $k(x) = 5x^2/(4x + 3)$.

Solution. $f(x) = 5x^2$ ($f' = 10x$), $g(x) = 4x + 3$ ($g' = 4$). $k'(x) = (10x(4x + 3) - 4\cdot 5x^2)/(4x + 3)^2 = (20x^2 + 30x)/(4x + 3)^2$.

Example 3.31 (RR and lecture In-Class activity; horizontal tangents). Determine the values of x at which $f(x) = x^3 - 7x^2 + 8x + 1$ has a horizontal tangent.

Solution. Horizontal tangent — where $f'(x) = 0$. $f'(x) = 3x^2 - 14x + 8 = (3x - 2)(x - 4)$; solutions: x = 2/3 and x = 4 — tangents are horizontal there.

Example 3.36 (RR; position and velocity). Position of a particle: $s(t) = t^3 - 9t^2 + 24t + 4$, $t \geq 0$. (a) Find v(t). (b) When is the particle at rest? (c) Where is it moving left and where right? (d) Sketch the path on the coordinate axis.

Solution. (a) $v(t) = 3t^2 - 18t + 24$. (b) At rest when v(t) = 0: $3(t - 2)(t - 4) = 0$, i.e., t = 2 and t = 4. (c) $v(t) > 0$ on $[0, 2) \cup (4, +\infty)$ — moving left to right; $v(t) < 0$ on (2, 4) — moving right to left. (d) Key positions: s(0) = 4, s(2) = 24, s(4) = 20: particle starts at point 4, moves right to 24 (t = 2), then left to 20 (t = 4), then right again.

Example 3.37 (RR; population estimation). A city's population triples every 5 years; currently 10,000. What is the approximate population in 2

years?

Solution. Let P(t) be the population (in thousands) after t years: P(0) = 10, P(5) = 30. Growth rate estimate: P′(0) ≈ (30 − 10)/5 = 4 (thousands per year). Using the increment formula: P(2) ≈ P(0) + 2·P′(0) ≈ 10 + 8 = 18 — approximately 18,000 people.

Example 3.38 (RR; marginal revenue). Price for BBQ lunches p(x) = 9 − 0.03x, 0 ≤ x ≤ 300; revenue R(x) = x·p(x) = $-0.03x^2 + 9x$. Estimate using marginal revenue the revenue from selling the 101st lunch and compare with the actual value.

Solution. MR(x) = R′(x) = −0.06x + 9; MR(100) = 3 — revenue from the 101st lunch is approximately 3 dollars. Actually: R(101) − R(100) = 602.97 − 600 = 2.97 dollars. Marginal revenue is a sufficiently accurate and simple to calculate estimate.

Example (lecture, In-Class activity; oscillation of slopes). Investigate f(x) = {x·sin(1/x), if x ≠ 0; 0, if x = 0} at zero: is it continuous? is it differentiable?

Solution. Continuity: |f(x)| = |x|·|sin(1/x)| ≤ |x| → 0, therefore by the squeeze theorem lim x→0 f(x) = 0 = f(0). Differentiability: f′(0) = lim x→0 (x·sin(1/x))/(x) = lim x→0 sin(1/x) — the limit does not exist: the slopes of the secants constantly change direction (Fig. 3.16). The function is continuous at zero but not differentiable.

In-Class exercises, discussed in the lecture with solutions (lecture instruction: show reasoning and intermediate calculations; where the derivative definition is requested — do not use rules; in engineering applications, specify units):

Exercise 1 (In-Class activity). Derivative at a point. For $f(x) = x^2 + 3x$, using the definition f′(a) = lim h→0 (f(a + h) − f(a))/h, find f′(2) and explain the geometric meaning.

Solution. f(2) = 10; $f(2 + h) = 10 + 7h + h^2$. Then $f′(2) = \lim_{h→0} (7h + h^2)/h = \lim_{h→0} (7 + h) = 7$. Geometric meaning: the tangent at the point (2, 10) has slope 7.

Exercise 2 (In-Class activity). Derivative as a function. For $f(x) = 3x^2 − 2x + 4$, using the definition: (a) find f′(x); (b) the slope of the tangent at x = 1; (c) the equation of this tangent.

Solution. (a) $f′(x) = \lim_{h→0} [3(x + h)^2 − 2(x + h) + 4 − (3x^2 − 2x + 4)]/h = \lim_{h→0} (6xh + 3h^2 − 2h)/h = 6x − 2$. (b) f′(1) = 4. (c) f(1) = 5; tangent: y − 5 = 4(x − 1), that is, y = 4x + 1.

Exercise 3 (In-Class activity). Differentiating Polynomials. Find the derivatives: (a) $f(x) = 5x^4 − 3x^3 + 2x − 7$; (b) $g(x) = x^5/5 − 3x^2/2 + 8$; (c) $p(x) = 7 − 4x + x^6$.

Solution. (a) $f′(x) = 20x^3 − 9x^2 + 2$. (b) $g′(x) = x^4 − 3x$. (c) $p′(x) = 6x^5 − 4$.

Exercise 4 (In-Class activity). Powers. Rewrite using powers and differentiate; specify the domain of the derivative: (a) $f(x) = 3/x^2$; (b) g(x) = 2·sqrt(x) − 1/x; (c) $p(x) = x^{1/3} + x^2$.

Solution. (a) $f(x) = 3x^{-2} → f′(x) = -6x^{-3} = -6/x^3$, x ≠ 0. (b) $g(x) = 2x^{1/2} − x^{-1} → g′(x) = 1/\text{sqrt}(x) + 1/x^2$, x > 0. (c) $p′(x) = 1/(3x^{2/3}) + 2x$, x ≠ 0: p itself is defined at zero, but there is no finite derivative there (vertical tangent).

Exercise 5 (In-Class activity). The Quotient Rule. Let $f(x) = (x^2 + 1)/(x + 2)$, $x \neq -2$. (a) Find $f'(x)$ using the quotient rule. (b) Find $f'(0)$. (c) Find the tangent at $x = 0$. (d) Why is there no derivative at $x = -2$?

Solution. (a) $f'(x) = [2x(x + 2) - (x^2 + 1)]/(x + 2)^2 = (x^2 + 4x - 1)/(x + 2)^2$. (b) $f'(0) = -1/4$. (c) $f(0) = 1/2$; tangent $y = 1/2 - x/4$. (d) At $x = -2$ the function is not defined (denominator is zero), so there is no derivative there.

Exercise 6 (In-Class activity). Higher Derivatives. For $f(x) = 2x^4 - 3x^3 + x^2 - 5x + 6$ find $f'(x)$, $f''(x)$, $f'''(x)$, $f^{(4)}(x)$, $f^{(5)}(x)$ and explain the meaning of the fifth derivative.

Solution. $f'(x) = 8x^3 - 9x^2 + 2x - 5$; $f''(x) = 24x^2 - 18x + 2$; $f'''(x) = 48x - 18$; $f^{(4)}(x) = 48$; $f^{(5)}(x) = 0$. Each differentiation reduces the degree by 1; after four steps we get the constant 48, and the derivative of a constant is zero.

Exercise 7 (In-Class activity). Two Methods. Differentiate $f(x) = \text{sqrt}(x)\cdot(x^2 + 4)$, $x > 0$: (a) using the product rule; (b) by expanding into powers; (c) show equivalence.

Solution. (a) $f'(x) = (x^2 + 4)/(2\cdot\text{sqrt}(x)) + 2x\cdot\text{sqrt}(x)$; since $2x\cdot\text{sqrt}(x) = 4x^2/(2\cdot\text{sqrt}(x))$, we get $f'(x) = (5x^2 + 4)/(2\cdot\text{sqrt}(x))$. (b) $f(x) = x^{5/2} + 4x^{1/2} \to f'(x) = (5/2)x^{3/2} + 2x^{-1/2}$. (c) Bringing to denominator $2\cdot\text{sqrt}(x)$: $(5/2)x^{3/2} = 5x^2/(2\cdot\text{sqrt}(x))$ and $2x^{-1/2} = 4/(2\cdot\text{sqrt}(x))$; in sum again $(5x^2 + 4)/(2\cdot\text{sqrt}(x))$.

Exercise 9 (In-Class activity). Cart Motion. A cart moves according to the law $s(t) = t^4 - 4t^3 + 6t^2$ (meters), $0 \le t \le 3$ s. (a) Find the velocity $v(t)$. (b) The acceleration $a(t)$. (c) The jerk $j(t)$. (d) Calculate all three at $t = 1$ s with units.

Solution. (a) $v(t) = 4t^3 - 12t^2 + 12t$. (b) $a(t) = 12t^2 - 24t + 12$. (c) $j(t) = 24t - 24$. (d) $v(1) = 4$ m/s; $a(1) = 0$ m/s^2; $j(1) = 0$ m/s^3.

Exercise 10 (In-Class activity). Small Vehicle. Position $s(t) = t^3 - 6t^2 + 9t$, $0 \le t \le 4$ (s in meters, t in seconds). (a) Average velocity on [0, 2]. (b) Instantaneous velocity at $t = 2$. (c) All moments when the vehicle is at rest.

Solution. (a) $s(0) = 0$, $s(2) = 8 - 24 + 18 = 2$; $v_{ave} = 2/2 = 1$ m/s. (b) $v(t) = 3t^2 - 12t + 9 = 3(t - 1)(t - 3)$; $v(2) = -3$ m/s (the "minus" sign indicates motion in the negative direction). (c) Rest occurs when $v(t) = 0$: $t = 1$ s and $t = 3$ s.

Lecture Exercise 8 ("correcting typical mistakes") literally coincides with Exercise 10 of the practical session — the solution is given above. Example 3.22 (tangent to $f(x) = x^2 - 4x + 6$ at $x = 1$: $f(1) = 3$, $f'(x) = 2x - 4$, tangent $y = -2x + 5$) and Example 3.23 (for $j = f\cdot g$ with $f(2) = 3$, $f'(2) = -4$, $g(2) = 1$, $g'(2) = 6$: $j'(2) = f'(2)g(2) + g'(2)f(2) = -4 + 18 = 14$) are solved using the same techniques.

Week's multimedia material: video presentation "Continuity and Differentiability EXPLAINED with Examples" (Ace Tutors, youtube.com/watch?v=nRv1lPqrx3k; the part about differentiability relates to week 6), derivative videos (youtube.com/watch?v=N2PpRnFqnqY, youtube.com/watch?v=wc7HHA6vIUw) and a short video about velocity (youtube.com/shorts/UOTmbY2ejkg). The student project "Formula One Grandstands" (textbook p. 226) is discussed in the optional ISA assignment above.

Week 6 of the **Linear Algebra I** course (Tashkent University of Technology, Fall 2026) covers sections **2.4–2.7** of G. Strang, Introduction to Linear Algebra, 4th ed. (pp. 68–120); supplementary reading — Lay, Linear Algebra and Its Applications, 4th ed., §§1.4–1.6. Topics: rules of matrix multiplication and four

interpretations of the product AB; algebraic identities and noncommutativity; special matrices; elimination matrices; definition of inverse, 2×2 formula and Gauss-Jordan method; LU and LDU decompositions and solving systems through factors; transposition, symmetric matrices, LDL^T; permutations and PA = LU; applications. Matrices are written row-wise: [a b; c d] — first row a b, second row c d; column vector — [x; y], row vector — [x y].

3.5 Linear Algebra I, week 6: formula catalog

[Source: Week 6 Lecture presentation (Strang, Introduction to Linear Algebra, 4th ed., §§2.4–2.7; companion: Lay, §§1.4–1.6)]

3.5.1 Review of week 5: two pictures of the system and elimination

Row picture: equations are intersecting lines or planes. Column picture: Ax = $x_1 \cdot a_1 + \ldots + x_n \cdot a_n = b$. Example: [2 1; 1 1]·[x; y] = [5; 3] gives (x, y) = (2, 1); both pictures describe the same system. Elimination: with pivot $a_{kk} \neq 0$, the multiplier $m_{ik} = a_{ik}/a_{kk}$ is taken and $R_i \leftarrow R_i - m_{ik} \cdot R_k$ is performed; thus [1 2 1; 2 5 3; 1 3 3] → [1 2 1; 0 1 1; 0 1 2] → [1 2 1; 0 1 1; 0 0 1], then back substitution solves the triangular system. Degeneracy: [1 2; 2 4]·[x; y] = [3; 6] has infinitely many solutions, while changing 6 to 7 gives the contradiction 0 = 1. For square matrices, a full set of pivots means a unique solution for any right-hand side.

3.5.2 Multiplication rules and four interpretations of AB

$$A\ (m\times n),\ B\ (n\times p) \to AB\ (m\times p),\ (AB)_{ij} = \sum_k a_{ik} \cdot b_{kj}$$

Example: [1 2 0; −1 3 4]·[2 1; 0 −1; 3 2] = [2 −1; 10 4]. Four views of AB: (1) element-wise — row i of matrix A is multiplied by column j of matrix B; (2) column-wise — AB = [Ab_1 ... Ab_p]; (3) row-wise — each row of AB is a combination of rows of B; (4) outer products — AB = $\sum_k a_k \cdot r_k$ (column k of matrix A times row k of matrix B).

$$(AB)C = A(BC),\ A(B + C) = AB + AC,\ (A + B)C = AC + BC,\ AI = IA = A$$

$$\text{generally speaking } AB \neq BA;\ (A + B)^2 = A^2 + AB + BA + B^2 \neq A^2 + 2AB + B^2$$

An explicit example of non-commutativity: A = [1 1; 0 1], B = [1 0; 1 1] give AB = [2 1; 1 1], but BA = [1 1; 1 2] — horizontal and vertical shears depend on the order of application.

3.5.3 Special matrices and elimination matrices

The identity matrix I leaves vectors unchanged; diagonal D = diag(d_1, ..., d_n) scales coordinates. Upper triangular matrices have zeros below the diagonal, lower triangular matrices have zeros above it; triangular systems are solved by forward/backward substitution. The elimination operation is itself a matrix: $R_2 \leftarrow R_2 - 3R_1$ corresponds to E = [1 0; −3 1], and E·[2 1; 6 5] = [2 1; 0 2]; the transformed system is EAx = Eb, and E^{-1} = [1 0; 3 1] restores the original rows.

3.5.4 The inverse matrix: definition, criteria, rules

$$AA^{-1} = A^{-1}A = I \to Ax = b \text{ has solution } x = A^{-1}b \text{ for any } b$$

The inverse reverses the linear transformation. If Ax = 0 for nonzero x, the inverse does not exist. For a square A, the following are equivalent: the inverse exists ⇔ all pivots are nonzero (after possible exchanges) ⇔ columns are independent ⇔ Ax = 0 only when x = 0 ⇔ Ax = b has a unique solution for any b.

$$(AB)^{-1} = B^{-1}A^{-1},\ (A^T)^{-1} = (A^{-1})^T,\ (A^{-1})^{-1} = A,\ (A^k)^{-1} = (A^{-1})^k$$

The order of factors in the inverse of a product is reversed: $(AB)(B^{-1}A^{-1}) = I$.

3.5.5 The inverse of 2×2 and the Gauss-Jordan method

$$[a\ b;\ c\ d]\cdot[d\ -b;\ -c\ a] = (ad - bc)\cdot I \rightarrow [a\ b;\ c\ d]^{-1} = (1/(ad - bc))\cdot[d\ -b;\ -c\ a] \text{ when } ad - bc \neq 0$$

Example: $[2\ 1;\ 5\ 3]^{-1} = [3\ -1;\ -5\ 2]$, since det = 1. Gauss-Jordan method: $[A \mid I] \rightarrow [I \mid A^{-1}]$ using the same row operations on both halves: $[1\ 2\ 1\ 0;\ 3\ 7\ 0\ 1] \rightarrow [1\ 2\ 1\ 0;\ 0\ 1\ -3\ 1] \rightarrow [1\ 0\ 7\ -2;\ 0\ 1\ -3\ 1]$, that is $[1\ 2;\ 3\ 7]^{-1} = [7\ -2;\ -3\ 1]$. The columns of the inverse: if $A^{-1} = [x_1 \ldots x_n]$, then $Ax_j = e_j$ — finding the inverse means solving n systems; for a single system Ax = b, triangular elimination is usually preferable to explicit computation of A^{-1}.

3.5.6 LU decomposition and solving systems through factors

A = LU: elimination factors — below diagonal of unit lower triangular L, elimination result — upper triangular U

$$[2\ 1;\ 6\ 5] = [1\ 0;\ 3\ 1]\cdot[2\ 1;\ 0\ 2] \text{ (verify by multiplication)}$$

$$E_k \cdots E_{2E1A} = U \rightarrow A = E_1^{-1}E_2^{-1}\cdots E_k^{-1}U = LU$$

The inverse of "subtract a multiple of a row" is "add that multiple back"; the order of factors is crucial (the right operation acts first). 3×3 example: $A = [2\ 1\ 1;\ 4\ 5\ 4;\ -2\ 2\ 3]$ with multipliers $m_{21} = 2$, $m_{31} = -1$, $m_{32} = 1$ gives $L = [1\ 0\ 0;\ 2\ 1\ 0;\ -1\ 1\ 1]$ and $U = [2\ 1\ 1;\ 0\ 3\ 2;\ 0\ 0\ 2]$ (verify LU = A row by row). Solving the system: first Ly = b by forward substitution, then Ux = y by backward substitution; for $L = [1\ 0;\ 3\ 1]$, $U = [2\ 1;\ 0\ 2]$, $b = [5;\ 19]$ we get $y = [5;\ 4]$ and $x = [3/2;\ 2]$. By decomposing A once, we can solve systems with many right-hand sides.

3.5.7 LDU decomposition, pivots and determinant

U = D·(D⁻¹U), D contains reference elements of U; A = L·D·(D⁻¹U) with unit diagonals for L and D⁻¹U

$$[2\ 1;\ 6\ 5] = [1\ 0;\ 3\ 1]\cdot[2\ 0;\ 0\ 2]\cdot[1\ 1/2;\ 0\ 1]$$

Pivots and determinant: for $A = [a\ b;\ c\ d]$ with $a \neq 0$, the pivots are a and $d - cb/a$, and their product is $ad - bc = \det A$. Without row exchanges, the determinant equals the product of pivots; the normalized LDU is unique when it exists with nonzero pivots.

3.5.8 Permutations, PA = LU and partial pivoting

Why exchange rows: $A = [0\ 2;\ 3\ 4]$ is invertible, but its first pivot is zero; exchanging rows gives $PA = [3\ 4;\ 0\ 2]$ — a zero pivot may indicate a poor row order, not singularity. Permutation matrices permute rows of I; there is exactly one 1 in each row and each column. $P = [0\ 1;\ 1\ 0]$ exchanges rows: PA — rows of A, AP — columns of A; there are n! permutations of n rows. Example: $P = [0\ 1\ 0;\ 0\ 0\ 1;\ 1\ 0\ 0]$ maps $(x_1, x_2, x_3)^T$ to $(x_2, x_3, x_1)^T$ and preserves vector lengths and scalar products.

$$P^{-1} = P^T \text{ (since } P^T\cdot P = I\text{); with exchanges } PA = LU$$

Example: $A = [0\ 2;\ 3\ 4]$, $P = [0\ 1;\ 1\ 0]$, L = I, $U = [3\ 4;\ 0\ 2]$; with exchanges, it's important to correctly track the multipliers in L. Partial pivoting: $[10^{-8}\ 1;\ 1\ 1]$ is nonsingular, but the tiny first pivot gives a multiplier of 10^8 — exchanging rows avoids this; the rule is to take the largest in magnitude element of the active column (below and including the pivot).

3.5.9 Transposition, symmetric matrices, LDL^T

$$(A^T)_{ij} = a_{ji},\ (A + B)^T = A^T + B^T,\ (AB)^T = B^T \cdot A^T$$

Example: [1 2 3; 4 5 6]^T = [1 4; 2 5; 3 6]. A symmetric matrix satisfies $A^T = A$. If a symmetric A has an LDU decomposition without exchanges, its normalized upper factor equals L^T: A = LDL^T; for example [2 2; 2 5] = [1 0; 1 1]·[2 0; 0 3]·[1 1; 0 1].

3.5.10 Applications: differences, sums, tridiagonal systems

The difference matrix A = [1 0 0; −1 1 0; 0 −1 1] gives Ax = $(x_1, x_2 - x_1, x_3 - x_2)^T$; its inverse A^{-1} = [1 0 0; 1 1 0; 1 1 1] gives cumulative sums — the discrete analog of differentiation and integration. The tridiagonal A = [2 −1 0; −1 2 −1; 0 −1 2] describes nearest neighbor connections (finite difference equations): its factors preserve the narrow band, and solving via LU uses the structure, avoiding the dense inverse matrix.

3.5.11 Week's Summary

- Matrix multiplication has several equivalent interpretations, but is non-commutative.
- The inverse exists only for non-singular square matrices.
- Elimination is encoded in A = LU = LDU.
- Row exchanges are encoded by the matrix P: PA = LU.
- Transpositions and permutations prepare for the vector spaces of week 7.

3.6 Linear Algebra I, week 6: In-class problems — questions and full solutions

[Source: Week 6 Problems in class (Solved) — 20 problems with solutions, from 2×2 to advanced decompositions]

Exercise 1. 2×2 inverse. Find A^{-1} for A = [2 1; 2 3]. Verify by multiplication.

Solution. det A = 2·3 − 1·2 = 4 ≠ 0. By the 2×2 formula: A^{-1} = (1/4)·[3 −1; −2 2] = [3/4 −1/4; −1/2 1/2]. Verification: A·A^{-1} = [2 1; 2 3]·[3/4 −1/4; −1/2 1/2] = [3/2 − 1/2, −1/2 + 1/2; 3/2 − 3/2, −1/2 + 3/2] = I.

Exercise 2. 2×2 system solution. Solve Ax = b, where A = [3 2; 1 4], b = [8; 6].

Solution. System: 3x + 2y = 8, x + 4y = 6. Multiplying the second equation by 3 and subtracting the first, we get 10y = 10, that is, y = 1, x = 6 − 4 = 2. Thus, x = [2; 1]. Verification: A·x = [3·2 + 2; 2 + 4] = [8; 6] = b.

Exercise 3. 2×2 inverse. Find A^{-1} for A = [4 1; 2 5]. Verify by multiplication.

Solution. det A = 4·5 − 1·2 = 18 ≠ 0. A^{-1} = (1/18)·[5 −1; −2 4] = [5/18 −1/18; −1/9 2/9]. Verification: A·A^{-1} = I — for example, element (1,1) equals 4·(5/18) − 1/9 = 20/18 − 2/18 = 1, element (2,2) equals −2/18 + 20/18 = 1, off-diagonal elements are zeros.

Exercise 4. Solving a 2×2 system. Solve Ax = b, where A = [2 2; 1 6], b = [4; 7].

Solution. System: 2x + 2y = 4 (that is, x + y = 2), x + 6y = 7. Subtracting: 5y = 5, y = 1, x = 1. Thus, x = [1; 1]. Verification: A·x = [2 + 2; 1 + 6] = [4; 7] = b.

Exercise 5. 2×2 decomposition. Find A = LU = LDU for A = [2 1; 2 4].
Solution. Pivot element 2, multiplier $m_{21} = 2/2 = 1$: $R_2 \leftarrow R_2 - R_1$ gives U = [2 1; 0 3]. We write the multiplier in L: L = [1 0; 1 1], A = LU (verification: [2 1] + [0 3] = [2 4]). We take the pivot elements 2 and 3 to D = [2 0; 0 3]; normalizing rows of U by dividing by pivot elements, we get $D^{-1}U$ = [1 1/2; 0 1]. Thus, A = LU = $L \cdot D \cdot D^{-1}U$.

Exercise 6. 2×2 decomposition. Find A = LU = LDU for A = [3 1; 3 5].
Solution. $m_{21} = 3/3 = 1$: $R_2 \leftarrow R_2 - R_1$ gives U = [3 1; 0 4]. L = [1 0; 1 1], A = LU (verification: [3 1] + [0 4] = [3 5]). D = [3 0; 0 4], $D^{-1}U$ = [1 1/3; 0 1], and A = $L \cdot D \cdot D^{-1}U$.

Exercise 7. 2×2 decomposition. Find A = LU = LDU for A = [1 1; 4 3].
Solution. $m_{21} = 4$: $R_2 \leftarrow R_2 - 4R_1$ gives U = [1 1; 0 −1]. L = [1 0; 4 1], A = LU (verification: 4·[1 1] + [0 −1] = [4 3]). The second pivot element is negative — this is acceptable: D = [1 0; 0 −1], $D^{-1}U$ = [1 1; 0 1], and A = $L \cdot D \cdot D^{-1}U$.

Exercise 8. 2×2 decomposition. Find A = LU = LDU for A = [2 1; 5 4].
Solution. $m_{21} = 5/2$: $R_2 \leftarrow R_2 - (5/2)R_1$ gives U = [2 1; 0 3/2] (4 − 5/2 = 3/2). L = [1 0; 5/2 1], A = LU (verification: (5/2)·[2 1] + [0 3/2] = [5 4]). D = [2 0; 0 3/2], $D^{-1}U$ = [1 1/2; 0 1], and A = $L \cdot D \cdot D^{-1}U$.

Exercise 9. 3×3 elimination. Solve Ax = b by Gaussian elimination for A = [1 3 1; 2 5 2; 1 4 3], b = [6; 10; 6].
Solution. Augmented matrix [1 3 1 6; 2 5 2 10; 1 4 3 6]. Step 1 ($m_{21} = 2$, $m_{31} = 1$): $R_2 - 2R_1$ = (0, −1, 0 | −2); $R_3 - R_1$ = (0, 1, 2 | 0). Step 2 ($m_{32} = -1$): $R_3 + R_2$ = (0, 0, 2 | −2). Triangular system: x + 3y + z = 6, −y = −2, 2z = −2. Back substitution: z = −1, y = 2, x = 6 − 6 + 1 = 1. Thus, x = [1; 2; −1]. Verification: A·x = [1 + 6 − 1; 2 + 10 − 2; 1 + 8 − 3] = [6; 10; 6] = b.

Exercise 10. 3×3 elimination. Solve Ax = b by Gaussian elimination for A = [1 1 1; 2 3 2; 1 2 4], b = [2; 6; 1].
Solution. The augmented matrix is [1 1 1 2; 2 3 2 6; 1 2 4 1]. Step 1 ($m_{21} = 2$, $m_{31} = 1$): $R_2 - 2R_1$ = (0, 1, 0 | 2); $R_3 - R_1$ = (0, 1, 3 | −1). Step 2 ($m_{32} = 1$): $R_3 - R_2$ = (0, 0, 3 | −3). Triangular system: x + y + z = 2, y = 2, 3z = −3. Back substitution: z = −1, y = 2, x = 2 − 2 + 1 = 1. Thus, x = [1; 2; −1]. Verification: A·x = [1 + 2 − 1; 2 + 6 − 2; 1 + 4 − 4] = [2; 6; 1] = b.

Exercise 11. 3×3 Elimination (degenerate case). Solve Ax = b by Gaussian elimination for A = [1 2 1; 2 4 2; 1 3 5], b = [4; 8; 2].
Solution. The augmented matrix is [1 2 1 4; 2 4 2 8; 1 3 5 2]. Step 1 ($m_{21} = 2$, $m_{31} = 1$): $R_2 - 2R_1$ = (0, 0, 0 | 0) — the second equation is exactly twice the first; $R_3 - R_1$ = (0, 1, 4 | −2). After swapping rows 2 and 3: [1 2 1 4; 0 1 4 −2; 0 0 0 0] — two pivots for three unknowns, one free variable. Let z = t: y = −2 − 4t (from y + 4z = −2), x = 4 − 2y − z = 8 + 7t. General solution x = [8 + 7t; −2 − 4t; t] = [8; −2; 0] + t·[7; −4; 1], t ∈ R — infinitely many solutions; the solution given in the file x = [1; 2; −1] is a particular solution at t = −1. Verification: A·[1; 2; −1] = [1 + 4 − 1; 2 + 8 − 2; 1 + 6 − 5] = [4; 8; 2] = b. Geometrically: the first two equations define one plane, its intersection with the third is a line.

Exercise 12. 3×3 Elimination. Solve Ax = b by Gaussian elimination for A = [1 3 1; 2 5 2; 1 4 4], b = [6; 10; 5].
Solution. The augmented matrix is [1 3 1 6; 2 5 2 10; 1 4 4 5]. Step 1 ($m_{21} = 2$, $m_{31} = 1$): $R_2 - 2R_1$ = (0, −1, 0 | −2); $R_3 - R_1$ = (0, 1, 3 | −1). Step 2 ($m_{32} = -1$):

$R_3 + R_2 = (0, 0, 3 \mid -3)$. Triangular system: $x + 3y + z = 6$, $-y = -2$, $3z = -3$. Back substitution: $z = -1$, $y = 2$, $x = 6 - 6 + 1 = 1$. Thus, $x = [1; 2; -1]$. Verification: $A \cdot x = [1 + 6 - 1; 2 + 10 - 2; 1 + 8 - 4] = [6; 10; 5] = b$.

Exercise 13. LU and LDU 3×3. Find LU and LDU for A = [2 1 1; 2 3 2; −2 1 1].

Solution. Multipliers $m_{21} = 1$, $m_{31} = -1$: $R_2 - R_1 = (0, 2, 1)$; $R_3 + R_1 = (0, 2, 2)$. Second pivot is 2, $m_{32} = 1$: $R_3 - R_2 = (0, 0, 1)$. Thus, $L = [1\ 0\ 0; 1\ 1\ 0; -1\ 1\ 1]$, $U = [2\ 1\ 1; 0\ 2\ 1; 0\ 0\ 1]$; verification: $LU = A$ (third row: $-[2\ 1\ 1] + [0\ 2\ 1] + [0\ 0\ 1] = [-2\ 1\ 1]$). $D = [2\ 0\ 0; 0\ 2\ 0; 0\ 0\ 1]$, normalized upper multiplier $D^{-1}U = [1\ 1/2\ 1/2; 0\ 1\ 1/2; 0\ 0\ 1]$; $A = LU = L \cdot D \cdot D^{-1}U$.

Exercise 14. LU and LDU 3×3. Find LU and LDU for A = [2 1 1; 4 5 3; −2 5 3].

Solution. $m_{21} = 2$, $m_{31} = -1$: $R_2 - 2R_1 = (0, 3, 1)$; $R_3 + R_1 = (0, 6, 4)$. Second pivot is 3, $m_{32} = 2$: $R_3 - 2R_2 = (0, 0, 2)$. $L = [1\ 0\ 0; 2\ 1\ 0; -1\ 2\ 1]$, $U = [2\ 1\ 1; 0\ 3\ 1; 0\ 0\ 2]$; verification: third row of LU is $-[2\ 1\ 1] + 2 \cdot [0\ 3\ 1] + [0\ 0\ 2] = [-2\ 5\ 3]$. $D = [2\ 0\ 0; 0\ 3\ 0; 0\ 0\ 2]$, $D^{-1}U = [1\ 1/2\ 1/2; 0\ 1\ 1/3; 0\ 0\ 1]$; $A = LU = L \cdot D \cdot D^{-1}U$.

Exercise 15. LU and LDU 3×3. Find LU and LDU for A = [2 1 1; 2 5 2; −2 11 5].

Solution. $m_{21} = 1$, $m_{31} = -1$: $R_2 - R_1 = (0, 4, 1)$; $R_3 + R_1 = (0, 12, 6)$. Second pivot is 4, $m_{32} = 3$: $R_3 - 3R_2 = (0, 0, 3)$. $L = [1\ 0\ 0; 1\ 1\ 0; -1\ 3\ 1]$, $U = [2\ 1\ 1; 0\ 4\ 1; 0\ 0\ 3]$; verification: $-[2\ 1\ 1] + 3 \cdot [0\ 4\ 1] + [0\ 0\ 3] = [-2\ 11\ 5]$. $D = [2\ 0\ 0; 0\ 4\ 0; 0\ 0\ 3]$, $D^{-1}U = [1\ 1/2\ 1/2; 0\ 1\ 1/4; 0\ 0\ 1]$; $A = LU = L \cdot D \cdot D^{-1}U$.

Exercise 16. LU and LDU 3×3. Find LU and LDU for A = [2 1 1; 4 4 3; −2 1 1].

Solution. $m_{21} = 2$, $m_{31} = -1$: $R_2 - 2R_1 = (0, 2, 1)$; $R_3 + R_1 = (0, 2, 2)$. Second pivot is 2, $m_{32} = 1$: $R_3 - R_2 = (0, 0, 1)$. $L = [1\ 0\ 0; 2\ 1\ 0; -1\ 1\ 1]$, $U = [2\ 1\ 1; 0\ 2\ 1; 0\ 0\ 1]$; verification: second row is $2 \cdot [2\ 1\ 1] + [0\ 2\ 1] = [4\ 4\ 3]$, third is $-[2\ 1\ 1] + [0\ 2\ 1] + [0\ 0\ 1] = [-2\ 1\ 1]$. $D = [2\ 0\ 0; 0\ 2\ 0; 0\ 0\ 1]$, $D^{-1}U = [1\ 1/2\ 1/2; 0\ 1\ 1/2; 0\ 0\ 1]$; $A = LU = L \cdot D \cdot D^{-1}U$.

Exercise 17. Row Permutation and PLU. Find the row permutation P and factors satisfying PA = LU, for A = [0 2 1; 1 1 0; 2 3 2].

Solution. The first diagonal element is zero — swap rows 1 and 2: $P = [0\ 1\ 0; 1\ 0\ 0; 0\ 0\ 1]$, $PA = [1\ 1\ 0; 0\ 2\ 1; 2\ 3\ 2]$. Elimination: element (2, 1) is already zero ($m_{21} = 0$); $m_{31} = 2$: $R_3 - 2R_1 = (2, 3, 2) - 2 \cdot (1, 1, 0) = (0, 1, 2)$; then $m_{32} = 1/2$: $R_3 - (1/2)R_2 = (0, 1, 2) - (0, 1, 1/2) = (0, 0, 3/2)$. Thus, $L = [1\ 0\ 0; 0\ 1\ 0; 2\ 1/2\ 1]$, $U = [1\ 1\ 0; 0\ 2\ 1; 0\ 0\ 3/2]$. Check: $LU = [1\ 1\ 0; 0\ 2\ 1; 2\ 3\ 2] = PA$ (third row: $2 \cdot (1, 1, 0) + (1/2) \cdot (0, 2, 1) + (0, 0, 3/2) = (2, 3, 2)$). Thus, $PA = LU$: P reorders the equations, giving a suitable pivot; L contains the multipliers 2 and 1/2 with their original signs, although the operations subtract them.

Exercise 18. Transpose and inverse. For A = [1 2 0; 0 1 1; 2 0 1] compute A^T and verify $(A^T)^{-1} = (A^{-1})^T$.

Solution. Transpose: $A^T = [1\ 0\ 2; 2\ 1\ 0; 0\ 1\ 1]$. $\det A = 1 \cdot (1 \cdot 1 - 1 \cdot 0) - 2 \cdot (0 \cdot 1 - 1 \cdot 2) + 0 = 1 + 4 = 5$. By the adjugate matrix $(A^T)^{-1} = (1/5) \cdot [1\ 2\ -2; -2\ 1\ 4; 2\ -1\ 1]$. Independently compute $A^{-1} = (1/5) \cdot [1\ -2\ 2; 2\ 1\ -1; -2\ 4\ 1]$; then $(A^{-1})^T = (1/5) \cdot [1\ 2\ -2; -2\ 1\ 4; 2\ -1\ 1] = (A^T)^{-1}$. The equality holds; check: $A^T \cdot (A^T)^{-1} = I$.

Exercise 19. Symmetric LDL^T. Factor the symmetric A = [2 2 0; 2 5 1; 0 1 3] as LDL^T.

Solution. Pivot element 2; $m_{21} = 2/2 = 1$ ($m_{31} = 0$): $R_2 - R_1 = (0, 3, 1)$. Second pivot element 3; $m_{32} = 1/3$: $R_3 - (1/3)R_2 = (0, 1, 3) - (0, 1, 1/3) = (0, 0, 8/3)$. For symmetric A without exchanges, the normalized upper multiplier is L^T, that is, $A = LDL^T$, where L = [1 0 0; 1 1 0; 0 1/3 1], D = [2 0 0; 0 3 0; 0 0 8/3]. Check: L·D = [2 0 0; 2 3 0; 0 1 8/3], and (LD)·L^T = [2 2 0; 2 5 1; 0 1 3] = A.

Exercise 20. Challenge problem: tridiagonal model. Solve Ax = b using LU for A = [2 −1 0; −1 2 −1; 0 −1 2], b = [2; 0; 2]. Interpret the nearest neighbor structure.

Solution. Factorization: $m_{21} = -1/2$, so $R_2 \leftarrow R_2 + (1/2)R_1$: (−1, 2, −1) + (1, −1/2, 0) = (0, 3/2, −1); $m_{32} = -2/3$, so $R_3 \leftarrow R_3 + (2/3)R_2$: (0, −1, 2) + (0, 1, −2/3) = (0, 0, 4/3). Thus, L = [1 0 0; −1/2 1 0; 0 −2/3 1], U = [2 −1 0; 0 3/2 −1; 0 0 4/3]. Forward substitution Ly = b: $y_1 = 2$; $-(1/2)\cdot 2 + y_2 = 0$, so $y_2 = 1$; $-(2/3)\cdot 1 + y_3 = 2$, so $y_3 = 8/3$. Back substitution Ux = y: $(4/3)x_3 = 8/3$, $x_3 = 2$; $(3/2)x_2 - 2 = 1$, $x_2 = 2$; $2x_1 - 2 = 2$, $x_1 = 2$. Thus, x = [2; 2; 2]. Check: A·x = [4 − 2; −2 + 4 − 2; −2 + 4] = [2; 0; 2] = b. The matrix is tridiagonal — nonzero elements only on the main diagonal and adjacent positions: each unknown is connected only to its nearest neighbors.

3.7 Linear Algebra I, Week 6: Practical Session — Questions and Full Solutions

[Source: Week 6 Practical Session exercises with Solutions — 20 problems; Selected detailed solutions — detailed versions of problems 9–12, 17, 20]

Exercise 1. 2×2 inverse. Find A^{-1} for A = [3 2; 1 6]. Verify by multiplication.

Solution. det A = 3·6 − 2·1 = 16 ≠ 0. $A^{-1} = (1/16)\cdot$[6 −2; −1 3] = [3/8 −1/8; −1/16 3/16]. Check: $A\cdot A^{-1} = I$ — element (1,1) is 9/8 − 2/16 = 1, element (2,2) is −1/8 + 18/16 = 1, off-diagonal elements are zero.

Exercise 2. 2×2 system solution. Solve Ax = b, where A = [4 1; 2 3], b = [14; 12].

Solution. System: 4x + y = 14, 2x + 3y = 12. Multiplying the first equation by 3 and subtracting the second, we get 10x = 30, that is, x = 3, y = 14 − 12 = 2. Thus, x = [3; 2]. Check: A·x = [12 + 2; 6 + 6] = [14; 12] = b.

Exercise 3. 2×2 inverse. Find A^{-1} for A = [2 2; 1 4]. Verify by multiplication.

Solution. det A = 2·4 − 2·1 = 6 ≠ 0. $A^{-1} = (1/6)\cdot$[4 −2; −1 2] = [2/3 −1/3; −1/6 1/3]. Check: $A\cdot A^{-1} = I$ — element (1,1) is 4/3 − 1/3 = 1, element (2,2) is −1/3 + 4/3 = 1, off-diagonal elements are zero.

Exercise 4. 2×2 system solution. Solve Ax = b, where A = [3 1; 2 5], b = [8; 14].

Solution. System: 3x + y = 8, 2x + 5y = 14. Multiplying the first equation by 5 and subtracting the second, we get 13x = 26, that is, x = 2, y = 8 − 6 = 2. Thus, x = [2; 2]. Check: A·x = [6 + 2; 4 + 10] = [8; 14] = b.

Exercise 5. 2×2 factorization. Find A = LU = LDU for A = [3 1; 5 5].

Solution. $m_{21} = 5/3$: $R_2 \leftarrow R_2 - (5/3)R_1$ gives U = [3 1; 0 10/3] (5 − 5 = 10/3). L = [1 0; 5/3 1], A = LU (check: (5/3)·[3 1] + [0 10/3] = [5 5]). D = [3 0; 0 10/3], $D^{-1}U$ = [1 1/3; 0 1], and A = L·D·D^{-1}U.

Exercise 6. 2×2 factorization. Find A = LU = LDU for A = [1 1; 2 3].

Solution. $m_{21} = 2$: $R_2 \leftarrow R_2 - 2R_1$ gives U = [1 1; 0 1]. L = [1 0; 2 1], A = LU (check: 2·[1 1] + [0 1] = [2 3]). Both pivots are 1, so D = I and $D^{-1}U = U$: A = [1 0; 2 1]·[1 0; 0 1]·[1 1; 0 1].

Exercise 7. 2×2 Factorization. Find A = LU = LDU for A = [2 1; 3 4].

Solution. $m_{21} = 3/2$: $R_2 \leftarrow R_2 - (3/2)R_1$ gives U = [2 1; 0 5/2] (4 − 3/2 = 5/2). L = [1 0; 3/2 1], A = LU (check: (3/2)·[2 1] + [0 5/2] = [3 4]). D = [2 0; 0 5/2], $D^{-1}U$ = [1 1/2; 0 1], and A = L·D·D^{-1}U.

Exercise 8. 2×2 Factorization. Find A = LU = LDU for A = [3 1; 4 5].

Solution. $m_{21} = 4/3$: $R_2 \leftarrow R_2 - (4/3)R_1$ gives U = [3 1; 0 11/3] (5 − 4/3 = 11/3). L = [1 0; 4/3 1], A = LU (check: (4/3)·[3 1] + [0 11/3] = [4 5]). D = [3 0; 0 11/3], $D^{-1}U$ = [1 1/3; 0 1], and A = L·D·D^{-1}U.

Exercise 9. 3×3 Elimination. Solve Ax = b by Gaussian elimination for A = [1 1 1; 2 3 2; 1 2 4], b = [2; 6; 1].

Solution. System: x + y + z = 2, 2x + 3y + 2z = 6, x + 2y + 4z = 1; augmented matrix [1 1 1 2; 2 3 2 6; 1 2 4 1]. Eliminate x ($m_{21} = 2$, $m_{31} = 1$): new second row (2, 3, 2 | 6) − 2·(1, 1, 1 | 2) = (0, 1, 0 | 2); new third row (1, 2, 4 | 1) − (1, 1, 1 | 2) = (0, 1, 3 | −1). Eliminate y ($m_{32} = 1$): (0, 1, 3 | −1) − (0, 1, 0 | 2) = (0, 0, 3 | −3). Triangular system: x + y + z = 2, y = 2, 3z = −3. Back substitution: z = −1, y = 2, x = 2 − 2 + 1 = 1. Thus, x = [1; 2; −1]. Check: A·x = [1 + 2 − 1; 2 + 6 − 2; 1 + 4 − 4] = [2; 6; 1] = b.

Exercise 10. 3×3 Elimination (infinitely many solutions). Solve Ax = b by Gaussian elimination for A = [1 2 1; 2 4 2; 1 3 5], b = [4; 8; 2].

Solution. System: x + 2y + z = 4, 2x + 4y + 2z = 8, x + 3y + 5z = 2; augmented matrix [1 2 1 4; 2 4 2 8; 1 3 5 2]. Eliminate x ($m_{21} = 2$, $m_{31} = 1$): second row gives (0, 0, 0 | 0) — equation 0 = 0 imposes no restrictions (second equation is exactly twice the first); third row: (0, 1, 4 | −2). Swapping rows 2 and 3, we get echelon form [1 2 1 4; 0 1 4 −2; 0 0 0 0] — two pivots for three unknowns. Let z = t, t ∈ R: from second row y = −2 − 4t; from first x = 4 − 2y − z = 8 + 7t. All solutions: x = [8 + 7t; −2 − 4t; t] = [8; −2; 0] + t·[7; −4; 1], t ∈ R. Check: (8 + 7t) + 2(−2 − 4t) + t = 4 and (8 + 7t) + 3(−2 − 4t) + 5t = 2; second equation is automatically satisfied as twice the first. Geometrically: first two equations describe the same plane, and its intersection with the third is a line (solution family). Short answer file x = [1; 2; −1] — particular solution at t = −1: A·[1; 2; −1] = [4; 8; 2] = b.

Exercise 11. 3×3 Elimination. Solve Ax = b by Gaussian elimination for A = [1 3 1; 2 5 2; 1 4 4], b = [6; 10; 5].

Solution. System: x + 3y + z = 6, 2x + 5y + 2z = 10, x + 4y + 4z = 5; augmented matrix [1 3 1 6; 2 5 2 10; 1 4 4 5]. Eliminate x ($m_{21} = 2$, $m_{31} = 1$): second row (0, −1, 0 | −2); third row (0, 1, 3 | −1). Second pivot −1; to eliminate the element below it, add row 2 to row 3 ($m_{32} = -1$): (0, 1, 3 | −1) + (0, −1, 0 | −2) = (0, 0, 3 | −3). Triangular system: x + 3y + z = 6, −y = −2, 3z = −3. Back substitution: z = −1, y = 2, x = 6 − 6 + 1 = 1. Thus, x = [1; 2; −1]. Check: A·x = [1 + 6 − 1; 2 + 10 − 2; 1 + 8 − 4] = [6; 10; 5] = b.

Exercise 12. 3×3 Elimination. Solve Ax = b by Gaussian elimination for A = [1 1 1; 2 3 2; 1 2 3], b = [2; 6; 2].

Solution. System: x + y + z = 2, 2x + 3y + 2z = 6, x + 2y + 3z = 2; augmented matrix [1 1 1 2; 2 3 2 6; 1 2 3 2]. Eliminate x ($m_{21} = 2$, $m_{31} = 1$): second row (0, 1,

0 | 2); third row (0, 1, 2 | 0). Eliminate y ($m_{32} = 1$): (0, 1, 2 | 0) − (0, 1, 0 | 2) = (0, 0, 2 | −2). Triangular system: x + y + z = 2, y = 2, 2z = −2. Back substitution: z = −1, y = 2, x = 2 − 2 + 1 = 1. Thus, x = [1; 2; −1]. Check: A·x = [1 + 2 − 1; 2 + 6 − 2; 1 + 4 − 3] = [2; 6; 2] = b.

Exercise 13. LU and LDU 3×3. Find LU and LDU for A = [2 1 1; 4 5 3; −2 5 3].

Solution. $m_{21} = 2$, $m_{31} = -1$: $R_2 - 2R_1 = (0, 3, 1)$; $R_3 + R_1 = (0, 6, 4)$. Second pivot element 3, $m_{32} = 2$: $R_3 - 2R_2 = (0, 0, 2)$. L = [1 0 0; 2 1 0; −1 2 1], U = [2 1 1; 0 3 1; 0 0 2]; check: third row of LU is −[2 1 1] + 2·[0 3 1] + [0 0 2] = [−2 5 3]. D = [2 0 0; 0 3 0; 0 0 2], $D^{-1}U$ = [1 1/2 1/2; 0 1 1/3; 0 0 1]; A = LU = L·D·D^{-1}U.

Exercise 14. LU and LDU 3×3. Find LU and LDU for A = [2 1 1; 2 5 2; −2 11 5].

Solution. $m_{21} = 1$, $m_{31} = -1$: $R_2 - R_1 = (0, 4, 1)$; $R_3 + R_1 = (0, 12, 6)$. Second pivot element 4, $m_{32} = 3$: $R_3 - 3R_2 = (0, 0, 3)$. L = [1 0 0; 1 1 0; −1 3 1], U = [2 1 1; 0 4 1; 0 0 3]; check: −[2 1 1] + 3·[0 4 1] + [0 0 3] = [−2 11 5]. D = [2 0 0; 0 4 0; 0 0 3], $D^{-1}U$ = [1 1/2 1/2; 0 1 1/4; 0 0 1]; A = LU = L·D·D^{-1}U.

Exercise 15. LU and LDU 3×3. Find LU and LDU for A = [2 1 1; 4 4 3; −2 1 1].

Solution. $m_{21} = 2$, $m_{31} = -1$: $R_2 - 2R_1 = (0, 2, 1)$; $R_3 + R_1 = (0, 2, 2)$. Second pivot element 2, $m_{32} = 1$: $R_3 - R_2 = (0, 0, 1)$. L = [1 0 0; 2 1 0; −1 1 1], U = [2 1 1; 0 2 1; 0 0 1]; check: 2·[2 1 1] + [0 2 1] = [4 4 3] and −[2 1 1] + [0 2 1] + [0 0 1] = [−2 1 1]. D = [2 0 0; 0 2 0; 0 0 1], $D^{-1}U$ = [1 1/2 1/2; 0 1 1/2; 0 0 1]; A = LU = L·D·D^{-1}U.

Exercise 16. LU and LDU 3×3. Find LU and LDU for A = [2 1 1; 2 4 2; −2 5 3].

Solution. $m_{21} = 1$, $m_{31} = -1$: $R_2 - R_1 = (0, 3, 1)$; $R_3 + R_1 = (0, 6, 4)$. Second pivot element 3, $m_{32} = 2$: $R_3 - 2R_2 = (0, 0, 2)$. L = [1 0 0; 1 1 0; −1 2 1], U = [2 1 1; 0 3 1; 0 0 2]; check: third row of LU is −[2 1 1] + 2·[0 3 1] + [0 0 2] = [−2 5 3]. D = [2 0 0; 0 3 0; 0 0 2], $D^{-1}U$ = [1 1/2 1/2; 0 1 1/3; 0 0 1]; A = LU = L·D·D^{-1}U.

Exercise 17. Row Permutations and PLU. Find the row permutation P and factors satisfying PA = LU for A = [0 2 1; 1 1 0; 2 3 2].

Solution. (1) First diagonal element 0 — cannot use it as pivot; we swap rows 1 and 2. The corresponding permutation matrix (same swap in I): P = [0 1 0; 1 0 0; 0 0 1]; multiplying A by P on the left swaps rows: PA = [1 1 0; 0 2 1; 2 3 2]. (2) Eliminate below first pivot element 1: element (2, 1) is already zero ($m_{21} = 0$); $m_{31} = 2/1 = 2$, and $R_3 \leftarrow R_3 - 2R_1$: (2, 3, 2) − 2·(1, 1, 0) = (0, 1, 2). (3) Second pivot element 2; $m_{32} = 1/2$, and $R_3 \leftarrow R_3 - (1/2)R_2$: (0, 1, 2) − (0, 1, 1/2) = (0, 0, 3/2); U = [1 1 0; 0 2 1; 0 0 3/2]. (4) Build L from multipliers: L = [1 0 0; m_{21} 1 0; m_{31} m_{32} 1] = [1 0 0; 0 1 0; 2 1/2 1]. (5) Check: LU = [1 1 0; 0 2 1; 2 3 2] = PA — third row is 2·(1, 1, 0) + (1/2)·(0, 2, 1) + (0, 0, 3/2) = (2, 3, 2). Thus PA = LU: P reorders equations to provide a suitable pivot, and elimination completes the factorization; L contains the multipliers 2 and 1/2 with their original signs, although the operations subtract them.

Exercise 18. Transpose and Inverse. For A = [1 2 0; 0 1 1; 2 0 1] compute A^T and verify $(A^T)^{-1} = (A^{-1})^T$.

Solution. A^T = [1 0 2; 2 1 0; 0 1 1]. det A = 1·(1·1 − 1·0) − 2·(0·1 − 1·2) + 0·(0·0 − 1·2) = 5. From the adjugate matrix $(A^T)^{-1}$ = (1/5)·[1 2 −2; −2 1 4; 2 −1 1]. Independently: A^{-1} = (1/5)·[1 −2 2; 2 1 −1; −2 4 1], from which $(A^{-1})^T$ = (1/5)·[1 2

$-2; -2\ 1\ 4; 2\ -1\ 1] = (A^T)^{-1}$. The equality holds; check: $A^T \cdot (A^T)^{-1} = I$.

Exercise 19. Symmetric LDL^T. Factor the symmetric A = [2 2 0; 2 5 1; 0 1 3] as LDL^T.

Solution. Pivot element 2; $m_{21} = 1$ ($m_{31} = 0$): $R_2 - R_1 = (0, 3, 1)$. Second pivot element 3; $m_{32} = 1/3$: $R_3 - (1/3)R_2 = (0, 0, 8/3)$. For symmetric A without exchanges $A = LDL^T$, where L = [1 0 0; 1 1 0; 0 1/3 1], D = [2 0 0; 0 3 0; 0 0 8/3]. Check: L·D = [2 0 0; 2 3 0; 0 1 8/3], and $(LD)\cdot L^T$ = [2 2 0; 2 5 1; 0 1 3] = A.

Exercise 20. Challenge problem: tridiagonal model. Solve Ax = b using LU for A = [2 −1 0; −1 2 −1; 0 −1 2], b = [3; 0; 2]. Interpret the nearest neighbor structure.

Solution. (1) Decomposition: $m_{21} = (-1)/2 = -1/2$, so $R_2 \leftarrow R_2 + (1/2)R_1$: $(-1, 2, -1) + (1, -1/2, 0) = (0, 3/2, -1)$; $m_{32} = (-1)/(3/2) = -2/3$, so $R_3 \leftarrow R_3 + (2/3)R_2$: $(0, -1, 2) + (0, 1, -2/3) = (0, 0, 4/3)$. Thus, L = [1 0 0; −1/2 1 0; 0 −2/3 1], U = [2 −1 0; 0 3/2 −1; 0 0 4/3]; the multipliers appear in L with their original signs, including minuses. (2) Forward substitution Ly = b: $y_1 = 3$; $-(1/2)\cdot 3 + y_2 = 0$, $y_2 = 3/2$; $-(2/3)\cdot(3/2) + y_3 = 2$, $y_3 = 3$. (3) Back substitution Ux = y: $(4/3)x_3 = 3$, $x_3 = 9/4$; $(3/2)x_2 - 9/4 = 3/2$, $x_2 = (15/4)\cdot(2/3) = 5/2$; $2x_1 - 5/2 = 3$, $x_1 = 11/4$. Thus, x = [11/4; 5/2; 9/4]. Verification: $2\cdot(11/4) - 5/2 = 3$; $-(11/4) + 2\cdot(5/2) - 9/4 = 0$; $-(5/2) + 2\cdot(9/4) = 2$. (4) Nearest neighbor structure: the matrix is tridiagonal (nonzero elements only on the main diagonal and immediately above and below it); x_2 interacts with both x_1 and x_3, there is no direct x_1–x_3 connection ($a_{13} = a_{31} = 0$). Such a structure arises in heat conduction models along a rod, spring chains, and one-dimensional finite difference equations (A is the standard discrete negative second derivative up to a grid step factor); elimination preserves the narrow structure (L and U are bidiagonal), so the system is solved cheaply. (5) LDU: D = [2 0 0; 0 3/2 0; 0 0 4/3] and $D^{-1}U$ = [1 −1/2 0; 0 1 −2/3; 0 0 1], so $A = L\cdot D\cdot D^{-1}U$.

3.8 Linear Algebra I, Week 6: ISA exercises — questions and full solutions

[Source: Week 6 Independent Study Assignment — 40 assigned problems (from basic 2×2 to challenge problems); instructions: show elimination steps, verify products, and justify existence of inverse or decomposition. Solutions verified in Python]

Exercise 1. 2×2 Inverse. Find A^{-1} for A = [2 2; 1 6]. Verify by multiplication.

Solution. $\det A = 2\cdot 6 - 2\cdot 1 = 10 \neq 0$. $A^{-1} = (1/10)\cdot[6\ -2; -1\ 2] = [3/5\ -1/5; -1/10\ 1/5]$. Verification: $A\cdot A^{-1} = I$ — element (1,1) is $6/5 - 1/5 = 1$, element (2,2) is $-1/5 + 6/5 = 1$, off-diagonal elements are zeros.

Exercise 2. 2×2 System Solution. Solve Ax = b, where A = [3 1; 2 3], b = [8; 10].

Solution. System: $3x + y = 8$, $2x + 3y = 10$. Multiplying the first equation by 3 and subtracting the second, we get $7x = 14$, so $x = 2$, $y = (8 - 6)/1 = 2$. Thus, x = [2; 2]. Verification: $A\cdot x = [6 + 2; 4 + 6] = [8; 10] = b$.

Exercise 3. 2×2 Inverse. Find A^{-1} for A = [4 2; 1 4]. Verify by multiplication.

Solution. $\det A = 4\cdot 4 - 2\cdot 1 = 14 \neq 0$. $A^{-1} = (1/14)\cdot[4\ -2; -1\ 4] = [2/7\ -1/7; -1/14\ 2/7]$. Verification: $A\cdot A^{-1} = I$ — element (1,1) is $8/7 - 1/7 = 1$, element (2,2) is $-1/7 + 8/7 = 1$, off-diagonal elements are zeros.

Exercise 4. 2×2 System Solution. Solve Ax = b, where A = [2 1; 2 5], b = [4; 12].

Solution. System: 2x + y = 4, 2x + 5y = 12. Subtracting the first equation from the second: 4y = 8, y = 2; then 2x = 4 − 2, x = 1. Thus, x = [1; 2]. Verification: A·x = [2 + 2; 2 + 10] = [4; 12] = b.

Exercise 5. 2×2 Decomposition. Find A = LU = LDU for A = [2 1; 5 4].

Solution. m_{21} = 5/2: $R_2 \leftarrow R_2 - (5/2)R_1$ gives U = [2 1; 0 3/2]. L = [1 0; 5/2 1], A = LU (verification: (5/2)·[2 1] + [0 3/2] = [5 4]). D = [2 0; 0 3/2], D^{-1}U = [1 1/2; 0 1], and A = L·D·D^{-1}U.

Exercise 6. 2×2 Decomposition. Find A = LU = LDU for A = [3 1; 2 5].

Solution. m_{21} = 2/3: $R_2 \leftarrow R_2 - (2/3)R_1$ gives U = [3 1; 0 13/3] (5 − 2/3 = 13/3). L = [1 0; 2/3 1], A = LU (verification: (2/3)·[3 1] + [0 13/3] = [2 5]). D = [3 0; 0 13/3], D^{-1}U = [1 1/3; 0 1], and A = L·D·D^{-1}U.

Exercise 7. 2×2 Decomposition. Find A = LU = LDU for A = [1 1; 3 5].

Solution. m_{21} = 3: $R_2 \leftarrow R_2 - 3R_1$ gives U = [1 1; 0 2]. L = [1 0; 3 1], A = LU (verification: 3·[1 1] + [0 2] = [3 5]). D = [1 0; 0 2], D^{-1}U = [1 1; 0 1], and A = L·D·D^{-1}U.

Exercise 8. 2×2 Decomposition. Find A = LU = LDU for A = [2 1; 4 4].

Solution. m_{21} = 2: $R_2 \leftarrow R_2 - 2R_1$ gives U = [2 1; 0 2]. L = [1 0; 2 1], A = LU (verification: 2·[2 1] + [0 2] = [4 4]). D = [2 0; 0 2], D^{-1}U = [1 1/2; 0 1], and A = L·D·D^{-1}U.

Exercise 9. 3×3 Elimination. Solve Ax = b by Gaussian elimination for A = [1 3 1; 2 5 2; 1 4 4], b = [6; 10; 5].

Solution. The augmented matrix is [1 3 1 6; 2 5 2 10; 1 4 4 5]. Step 1 (m_{21} = 2, m_{31} = 1): $R_2 - 2R_1$ = (0, −1, 0 | −2); $R_3 - R_1$ = (0, 1, 3 | −1). Step 2 (m_{32} = −1): $R_3 + R_2$ = (0, 0, 3 | −3). The triangular system: x + 3y + z = 6, −y = −2, 3z = −3; hence z = −1, y = 2, x = 1. Thus, x = [1; 2; −1]. Verification: A·x = [1 + 6 − 1; 2 + 10 − 2; 1 + 8 − 4] = [6; 10; 5] = b.

Exercise 10. 3×3 Elimination. Solve Ax = b by Gaussian elimination for A = [1 1 1; 2 3 2; 1 2 3], b = [2; 6; 2].

Solution. The augmented matrix is [1 1 1 2; 2 3 2 6; 1 2 3 2]. Step 1 (m_{21} = 2, m_{31} = 1): $R_2 - 2R_1$ = (0, 1, 0 | 2); $R_3 - R_1$ = (0, 1, 2 | 0). Step 2 (m_{32} = 1): $R_3 - R_2$ = (0, 0, 2 | −2). The triangular system: x + y + z = 2, y = 2, 2z = −2; hence z = −1, y = 2, x = 1. Thus, x = [1; 2; −1]. Verification: A·x = [1 + 2 − 1; 2 + 6 − 2; 1 + 4 − 3] = [2; 6; 2] = b.

Exercise 11. 3×3 Elimination (infinitely many solutions). Solve Ax = b by Gaussian elimination for A = [1 2 1; 2 4 2; 1 3 6], b = [4; 8; 1].

Solution. The augmented matrix is [1 2 1 4; 2 4 2 8; 1 3 6 1]. Step 1 (m_{21} = 2, m_{31} = 1): $R_2 - 2R_1$ = (0, 0, 0 | 0) — the second equation is twice the first and adds no constraints; $R_3 - R_1$ = (0, 1, 5 | −3). After swapping rows 2 and 3: [1 2 1 4; 0 1 5 −3; 0 0 0 0] — two pivots for three unknowns. Free variable z = t: y = −3 − 5t, x = 4 − 2y − z = 10 + 9t. Infinitely many solutions: x = [10 + 9t; −3 − 5t; t], t ∈ R; particular solution x = [1; 2; −1] corresponds to t = −1. Verification: A·[1; 2; −1] = [1 + 4 − 1; 2 + 8 − 2; 1 + 6 − 6] = [4; 8; 1] = b.

Exercise 12. 3×3 Elimination. Solve Ax = b by Gaussian elimination for A = [1 3 1; 2 5 2; 1 4 3], b = [6; 10; 6].

Solution. The augmented matrix is [1 3 1 6; 2 5 2 10; 1 4 3 6]. Step 1 (m_{21} = 2, m_{31} = 1): $R_2 - 2R_1$ = (0, −1, 0 | −2); $R_3 - R_1$ = (0, 1, 2 | 0). Step 2 (m_{32} = −1): $R_3 + R_2$ = (0, 0, 2 | −2). Back substitution: z = −1, y = 2, x = 6 − 6 + 1 = 1. Thus, x = [1; 2; −1]. Verification: A·x = [1 + 6 − 1; 2 + 10 − 2; 1 + 8 − 3] = [6; 10; 6] = b.

Exercise 13. LU and LDU 3×3. Compute LU and LDU for A = [2 1 1; 4 4 3; −2 1 1].

Solution. m_{21} = 2, m_{31} = −1: $R_2 - 2R_1$ = (0, 2, 1); $R_3 + R_1$ = (0, 2, 2). Second pivot 2, m_{32} = 1: $R_3 - R_2$ = (0, 0, 1). L = [1 0 0; 2 1 0; −1 1 1], U = [2 1 1; 0 2 1; 0 0 1]; verification: LU = A (third row: −[2 1 1] + [0 2 1] + [0 0 1] = [−2 1 1]). D = [2 0 0; 0 2 0; 0 0 1], $D^{-1}U$ = [1 1/2 1/2; 0 1 1/2; 0 0 1]; A = LU = L·D·D^{-1}U.

Exercise 14. LU and LDU 3×3. Compute LU and LDU for A = [2 1 1; 2 4 2; −2 5 3].

Solution. m_{21} = 1, m_{31} = −1: $R_2 - R_1$ = (0, 3, 1); $R_3 + R_1$ = (0, 6, 4). Second pivot 3, m_{32} = 2: $R_3 - 2R_2$ = (0, 0, 2). L = [1 0 0; 1 1 0; −1 2 1], U = [2 1 1; 0 3 1; 0 0 2]; verification: third row of LU is −[2 1 1] + 2·[0 3 1] + [0 0 2] = [−2 5 3]. D = [2 0 0; 0 3 0; 0 0 2], $D^{-1}U$ = [1 1/2 1/2; 0 1 1/3; 0 0 1]; A = LU = L·D·D^{-1}U.

Exercise 15. LU and LDU 3×3. Compute LU and LDU for A = [2 1 1; 4 6 3; −2 11 5].

Solution. m_{21} = 2, m_{31} = −1: $R_2 - 2R_1$ = (0, 4, 1); $R_3 + R_1$ = (0, 12, 6). Second pivot 4, m_{32} = 3: $R_3 - 3R_2$ = (0, 0, 3). L = [1 0 0; 2 1 0; −1 3 1], U = [2 1 1; 0 4 1; 0 0 3]; verification: −[2 1 1] + 3·[0 4 1] + [0 0 3] = [−2 11 5]. D = [2 0 0; 0 4 0; 0 0 3], $D^{-1}U$ = [1 1/2 1/2; 0 1 1/4; 0 0 1]; A = LU = L·D·D^{-1}U.

Exercise 16. LU and LDU 3×3. Compute LU and LDU for A = [2 1 1; 2 3 2; −2 1 1].

Solution. m_{21} = 1, m_{31} = −1: $R_2 - R_1$ = (0, 2, 1); $R_3 + R_1$ = (0, 2, 2). Second pivot 2, m_{32} = 1: $R_3 - R_2$ = (0, 0, 1). L = [1 0 0; 1 1 0; −1 1 1], U = [2 1 1; 0 2 1; 0 0 1]; verification: LU = A. D = [2 0 0; 0 2 0; 0 0 1], $D^{-1}U$ = [1 1/2 1/2; 0 1 1/2; 0 0 1]; A = LU = L·D·D^{-1}U.

Exercise 17. Row Permutation and PLU. Find a row permutation P and factors satisfying PA = LU, for A = [0 2 1; 1 1 0; 2 3 2].

Solution. Zero first pivot — swap rows 1 and 2: P = [0 1 0; 1 0 0; 0 0 1], PA = [1 1 0; 0 2 1; 2 3 2]. Elimination: m_{31} = 2 ($R_3 - 2R_1$ = (0, 1, 2)); m_{32} = 1/2 ($R_3 - (1/2)R_2$ = (0, 0, 3/2)). L = [1 0 0; 0 1 0; 2 1/2 1], U = [1 1 0; 0 2 1; 0 0 3/2]; verification: LU = [1 1 0; 0 2 1; 2 3 2] = PA. Thus, PA = LU.

Exercise 18. Transpose and Inverse. For A = [1 2 0; 0 1 1; 2 0 1] compute A^T and verify $(A^T)^{-1} = (A^{-1})^T$.

Solution. A^T = [1 0 2; 2 1 0; 0 1 1]; det A = 1·(1 − 0) − 2·(0 − 2) + 0 = 5. $(A^T)^{-1}$ = (1/5)·[1 2 −2; −2 1 4; 2 −1 1]. Independently: A^{-1} = (1/5)·[1 −2 2; 2 1 −1; −2 4 1], from which $(A^{-1})^T$ = (1/5)·[1 2 −2; −2 1 4; 2 −1 1] = $(A^T)^{-1}$. The equality holds; verification: $A^T \cdot (A^T)^{-1}$ = I.

Exercise 19. Symmetric LDL^T. Factor the symmetric A = [2 2 0; 2 5 1; 0 1 3] as LDL^T.

Solution. The multipliers are m_{21} = 1, m_{31} = 0, m32 = 1/3. L = [1 0 0; 1 1 0; 0 1/3 1], D = [2 0 0; 0 3 0; 0 0 8/3]; for a symmetric matrix without pivoting, A = LDL^T. Verification: L·D·L^T = [2 2 0; 2 5 1; 0 1 3] = A.

Exercise 20. Challenge problem: tridiagonal model. Solve Ax = b using LU for A = [2 −1 0; −1 2 −1; 0 −1 2], b = [2; 0; 2]. Interpret the nearest neighbor structure.

Solution. LU: $m_{21} = -1/2$ and $m_{32} = -2/3$, so L = [1 0 0; −1/2 1 0; 0 −2/3 1], U = [2 −1 0; 0 3/2 −1; 0 0 4/3]. Forward substitution Ly = b: $y_1 = 2$, $y_2 = 1$, $y_3 = 8/3$. Back substitution Ux = y: $x_3 = 2$, $x_2 = 2$, $x_1 = 2$. Thus, x = [2; 2; 2]. Verification: A·x = [4 − 2; −2 + 4 − 2; −2 + 4] = [2; 0; 2] = b. The matrix is tridiagonal: connections only between nearest neighbors (x_2 with x_1 and x_3; no direct connection x_1–x_3, $a_{13} = a_{31} = 0$); L and U are bidiagonal — the narrow structure is preserved.

Exercise 21. 2×2 inverse. Find A^{-1} for A = [2 1; 2 3]. Verify by multiplication.

Solution. det A = 2·3 − 1·2 = 4 ≠ 0. $A^{-1} = (1/4)\cdot[3\ -1;\ -2\ 2] = [3/4\ -1/4;\ -1/2\ 1/2]$. Verification: $A\cdot A^{-1} = [2\ 1;\ 2\ 3]\cdot[3/4\ -1/4;\ -1/2\ 1/2] = [1\ 0;\ 0\ 1] = I$.

Exercise 22. Solving a 2×2 system. Solve Ax = b, where A = [3 2; 1 4], b = [8; 6].

Solution. System: 3x + 2y = 8, x + 4y = 6. From the second equation x = 6 − 4y; substitution: 3(6 − 4y) + 2y = 8, that is 18 − 10y = 8, y = 1, x = 2. Thus, x = [2; 1]. Verification: A·x = [6 + 2; 2 + 4] = [8; 6] = b.

Exercise 23. 2×2 inverse. Find A^{-1} for A = [4 1; 2 5]. Verify by multiplication.

Solution. det A = 4·5 − 1·2 = 18 ≠ 0. $A^{-1} = (1/18)\cdot[5\ -1;\ -2\ 4] = [5/18\ -1/18;\ -1/9\ 2/9]$. Verification: $A\cdot A^{-1} = I$ — element (1,1) equals 20/18 − 2/18 = 1, element (2,2) equals −2/18 + 20/18 = 1, off-diagonal elements are zeros.

Exercise 24. Solving a 2×2 system. Solve Ax = b, where A = [2 2; 1 6], b = [4; 7].

Solution. System: 2x + 2y = 4 (that is, x + y = 2), x + 6y = 7. Subtracting: 5y = 5, y = 1, x = 1. Thus, x = [1; 1]. Verification: A·x = [2 + 2; 1 + 6] = [4; 7] = b.

Exercise 25. 2×2 Factorization. Find A = LU = LDU for A = [2 1; 2 4].

Solution. $m_{21} = 1$: $R_2 \leftarrow R_2 - R_1$ gives U = [2 1; 0 3]. L = [1 0; 1 1], A = LU (verification: [2 1] + [0 3] = [2 4]). D = [2 0; 0 3], $D^{-1}U = [1\ 1/2;\ 0\ 1]$, and $A = L\cdot D\cdot D^{-1}U$.

Exercise 26. 2×2 Factorization. Find A = LU = LDU for A = [3 1; 3 5].

Solution. $m_{21} = 1$: $R_2 \leftarrow R_2 - R_1$ gives U = [3 1; 0 4]. L = [1 0; 1 1], A = LU (verification: [3 1] + [0 4] = [3 5]). D = [3 0; 0 4], $D^{-1}U = [1\ 1/3;\ 0\ 1]$, and $A = L\cdot D\cdot D^{-1}U$.

Exercise 27. 2×2 Factorization. Find A = LU = LDU for A = [1 1; 4 3].

Solution. $m_{21} = 4$: $R_2 \leftarrow R_2 - 4R_1$ gives U = [1 1; 0 −1]. L = [1 0; 4 1], A = LU (verification: 4·[1 1] + [0 −1] = [4 3]). D = [1 0; 0 −1] (negative pivot is acceptable), $D^{-1}U = [1\ 1;\ 0\ 1]$, and $A = L\cdot D\cdot D^{-1}U$.

Exercise 28. 2×2 Factorization. Find A = LU = LDU for A = [2 1; 5 4].

Solution. $m_{21} = 5/2$: $R_2 \leftarrow R_2 - (5/2)R_1$ gives U = [2 1; 0 3/2]. L = [1 0; 5/2 1], A = LU (verification: (5/2)·[2 1] + [0 3/2] = [5 4]). D = [2 0; 0 3/2], $D^{-1}U = [1\ 1/2;\ 0\ 1]$, and $A = L\cdot D\cdot D^{-1}U$.

Exercise 29. 3×3 Elimination. Solve Ax = b by Gaussian elimination for A = [1 3 1; 2 5 2; 1 4 3], b = [6; 10; 6].

Solution. Augmented matrix [1 3 1 6; 2 5 2 10; 1 4 3 6]. Step 1 (m_{21} = 2, m_{31} = 1): $R_2 - 2R_1$ = (0, −1, 0 | −2); $R_3 - R_1$ = (0, 1, 2 | 0). Step 2 (m_{32} = −1): $R_3 + R_2$ = (0, 0, 2 | −2). Back substitution: z = −1, y = 2, x = 1. Thus, x = [1; 2; −1]. Verification: A·x = [6; 10; 6] = b.

Exercise 30. 3×3 Elimination. Solve Ax = b by Gaussian elimination for A = [1 1 1; 2 3 2; 1 2 4], b = [2; 6; 1].

Solution. The augmented matrix is [1 1 1 2; 2 3 2 6; 1 2 4 1]. Step 1 (m_{21} = 2, m_{31} = 1): $R_2 - 2R_1$ = (0, 1, 0 | 2); $R_3 - R_1$ = (0, 1, 3 | −1). Step 2 (m_{32} = 1): $R_3 - R_2$ = (0, 0, 3 | −3). Back substitution: z = −1, y = 2, x = 1. Thus, x = [1; 2; −1]. Verification: A·x = [2; 6; 1] = b.

Exercise 31. 3×3 Elimination (infinitely many solutions). Solve Ax = b by Gaussian elimination for A = [1 2 1; 2 4 2; 1 3 5], b = [4; 8; 2].

Solution. The augmented matrix is [1 2 1 4; 2 4 2 8; 1 3 5 2]. Step 1 (m_{21} = 2, m_{31} = 1): $R_2 - 2R_1$ = (0, 0, 0 | 0) — the second equation is twice the first; $R_3 - R_1$ = (0, 1, 4 | −2). After swapping rows 2 and 3: [1 2 1 4; 0 1 4 −2; 0 0 0 0]. Free variable z = t: y = −2 − 4t, x = 4 − 2y − z = 8 + 7t. Infinitely many solutions: x = [8 + 7t; −2 − 4t; t], t ∈ R; particular solution x = [1; 2; −1] (t = −1). Verification: A·[1; 2; −1] = [4; 8; 2] = b.

Exercise 32. 3×3 Elimination. Solve Ax = b by Gaussian elimination for A = [1 3 1; 2 5 2; 1 4 4], b = [6; 10; 5].

Solution. The augmented matrix is [1 3 1 6; 2 5 2 10; 1 4 4 5]. Step 1 (m_{21} = 2, m_{31} = 1): $R_2 - 2R_1$ = (0, −1, 0 | −2); $R_3 - R_1$ = (0, 1, 3 | −1). Step 2 (m_{32} = −1): $R_3 + R_2$ = (0, 0, 3 | −3). Back substitution: z = −1, y = 2, x = 1. Thus, x = [1; 2; −1]. Verification: A·x = [6; 10; 5] = b.

Exercise 33. LU and LDU 3×3. Compute LU and LDU for A = [2 1 1; 2 3 2; −2 1 1].

Solution. m_{21} = 1, m_{31} = −1, m_{32} = 1: $R_2 - R_1$ = (0, 2, 1); $R_3 + R_1$ = (0, 2, 2); $R_3 - R_2$ = (0, 0, 1). L = [1 0 0; 1 1 0; −1 1 1], U = [2 1 1; 0 2 1; 0 0 1]; verification: LU = A. D = [2 0 0; 0 2 0; 0 0 1], $D^{-1}U$ = [1 1/2 1/2; 0 1 1/2; 0 0 1]; A = LU = L·D·D^{-1}U.

Exercise 34. LU and LDU 3×3. Compute LU and LDU for A = [2 1 1; 4 5 3; −2 5 3].

Solution. m_{21} = 2, m_{31} = −1, m_{32} = 2: $R_2 - 2R_1$ = (0, 3, 1); $R_3 + R_1$ = (0, 6, 4); $R_3 - 2R_2$ = (0, 0, 2). L = [1 0 0; 2 1 0; −1 2 1], U = [2 1 1; 0 3 1; 0 0 2]; verification: LU = A (third row: −[2 1 1] + 2·[0 3 1] + [0 0 2] = [−2 5 3]). D = [2 0 0; 0 3 0; 0 0 2], D^{-1}U = [1 1/2 1/2; 0 1 1/3; 0 0 1]; A = LU = L·D·D^{-1}U.

Exercise 35. LU and LDU 3×3. Compute LU and LDU for A = [2 1 1; 2 5 2; −2 11 5].

Solution. m_{21} = 1, m_{31} = −1, m_{32} = 3: $R_2 - R_1$ = (0, 4, 1); $R_3 + R_1$ = (0, 12, 6); $R_3 - 3R_2$ = (0, 0, 3). L = [1 0 0; 1 1 0; −1 3 1], U = [2 1 1; 0 4 1; 0 0 3]; verification: LU = A (third row: −[2 1 1] + 3·[0 4 1] + [0 0 3] = [−2 11 5]). D = [2 0 0; 0 4 0; 0 0 3], D^{-1}U = [1 1/2 1/2; 0 1 1/4; 0 0 1]; A = LU = L·D·D^{-1}U.

Exercise 36. LU and LDU 3×3. Compute LU and LDU for A = [2 1 1; 4 4 3; −2 1 1].

Solution. $m_{21} = 2$, $m_{31} = -1$, $m_{32} = 1$: $R_2 - 2R_1 = (0, 2, 1)$; $R_3 + R_1 = (0, 2, 2)$; $R_3 - R_2 = (0, 0, 1)$. L = [1 0 0; 2 1 0; −1 1 1], U = [2 1 1; 0 2 1; 0 0 1]; verification: LU = A. D = [2 0 0; 0 2 0; 0 0 1], $D^{-1}U$ = [1 1/2 1/2; 0 1 1/2; 0 0 1]; A = LU = $L \cdot D \cdot D^{-1}U$.

Exercise 37. Row Permutation and PLU. Find a row permutation P and factors satisfying PA = LU, for A = [0 2 1; 1 1 0; 2 3 2].

Solution. Swap rows 1 and 2 (zero first pivot): P = [0 1 0; 1 0 0; 0 0 1], PA = [1 1 0; 0 2 1; 2 3 2]. Elimination: $m_{31} = 2$ ($R_3 - 2R_1 = (0, 1, 2)$); $m_{32} = 1/2$ ($R_3 - (1/2)R_2 = (0, 0, 3/2)$). L = [1 0 0; 0 1 0; 2 1/2 1], U = [1 1 0; 0 2 1; 0 0 3/2]; verification: LU = PA. Thus, PA = LU.

Exercise 38. Transpose and Inverse. For A = [1 2 0; 0 1 1; 2 0 1] compute A^T and verify $(A^T)^{-1} = (A^{-1})^T$.

Solution. A^T = [1 0 2; 2 1 0; 0 1 1]; det A = 5. $(A^T)^{-1} = (1/5) \cdot$[1 2 −2; −2 1 4; 2 −1 1]; on the other hand, $A^{-1} = (1/5) \cdot$[1 −2 2; 2 1 −1; −2 4 1], and $(A^{-1})^T = (1/5) \cdot$[1 2 −2; −2 1 4; 2 −1 1] = $(A^T)^{-1}$. The equality holds; verification: $A^T \cdot (A^T)^{-1} = I$.

Exercise 39. Symmetric LDL^T. Factor the symmetric A = [2 2 0; 2 5 1; 0 1 3] as LDL^T.

Solution. Multipliers $m_{21} = 1$, $m_{31} = 0$, $m_{32} = 1/3$. L = [1 0 0; 1 1 0; 0 1/3 1], D = [2 0 0; 0 3 0; 0 0 8/3], and $A = LDL^T$. Verification: $L \cdot D \cdot L^T$ = [2 2 0; 2 5 1; 0 1 3] = A.

Exercise 40. Challenge problem: tridiagonal model. Solve Ax = b using LU for A = [2 −1 0; −1 2 −1; 0 −1 2], b = [2; 0; 2]. Interpret the nearest neighbor structure.

Solution. LU: L = [1 0 0; −1/2 1 0; 0 −2/3 1], U = [2 −1 0; 0 3/2 −1; 0 0 4/3] (the multipliers $m_{21} = -1/2$, $m_{32} = -2/3$ are stored in L with their original signs). Forward substitution Ly = b gives y = [2; 1; 8/3]; backward substitution Ux = y gives $x_3 = 2$, $x_2 = 2$, $x_1 = 2$. Thus, x = [2; 2; 2]. Verification: A·x = [2; 0; 2] = b. The matrix is tridiagonal — each unknown is connected only to its nearest neighbors; the factors L and U are bidiagonal, and the narrow structure makes the solution inexpensive.

3.9 Linear Algebra I, week 6: review material with examples

[Source: Week 6 Extra material — Review with examples: I) a brief lesson on LU, LDU, and normalized U; II) a brief lesson on Gaussian elimination for 3×3 systems; III) a detailed example of finding the inverse by the Gauss-Jordan method]

Key points of lesson II (review of week 5). A system can be viewed in three pictures: row (equations are planes), matrix (Ax = b), and column (combination of columns of A gives b). Three reversible row operations are allowed: swapping two equations ($R_i \leftrightarrow R_j$), multiplying an equation by a nonzero number ($R_i \leftarrow c \cdot R_i$, $c \neq 0$), and adding a multiple of another equation ($R_i \leftarrow R_i + c \cdot R_j$); each is applied to the entire equation, including the right-hand side. Procedure for a 3×3 system: write the augmented matrix; select a nonzero first pivot element (swapping rows if necessary); eliminate elements below it; select the second pivot element in the updated matrix; eliminate the element below it; check for contradictions or free variables; with three pivot elements, perform backward substitution; verify the solution in the original system. Backward substitution formulas: $z = c_3/u_{33}$, $y = (c_2 - u_{23} \cdot z)/u_{22}$, $x = (c_1 - u_{12} \cdot y - u_{13} \cdot z)/u_{11}$ (pivot elements are nonzero). Typical errors: forgetting to update the right-hand side; using the old row after it has been changed; confusing the direction of the multiplier relationship; considering

a zero proposed pivot element as proof of degeneracy.

Example 1. LU, normalization, and LDU (lesson I). Find the LU and LDU decompositions of the matrix A = [2 4 2; 4 11 7; 2 7 8].

Solution. Eliminate under the first pivot element 2: $R_2 \leftarrow R_2 - 2R_1$ and $R_3 \leftarrow R_3 - R_1$ give [2 4 2; 0 3 3; 0 3 6]; then under the second pivot element 3: $R_3 \leftarrow R_3 - R_2$ gives [2 4 2; 0 3 3; 0 0 3]. The multipliers are $m_{21} = 2$, $m_{31} = 1$, $m_{32} = 1$, so L = [1 0 0; 2 1 0; 1 1 1] and A = LU with U = [2 4 2; 0 3 3; 0 0 3]; L stores the multipliers with their original signs (the operation subtracts $2R_1$, but $L_{21} = +2$). Normalization: divide each row of U by its diagonal element — the pivot elements 2, 3, 3 are factored out into D = [2 0 0; 0 3 0; 0 0 3], the normalized upper factor is $D^{-1}U$ = [1 2 1; 0 1 1; 0 0 1], and A = L·D·(D^{-1}U); multiplication by D^{-1} on the left scales the rows. Note on conventions: in A = LU, the pivot elements sit on the diagonal of U (D is not separated, L has unit diagonal); in A = LDU, the pivot elements are in D, and both L and the third factor have unit diagonals; many textbooks rename $D^{-1}U$ as U.

Example 2. Solving a system using factors (lesson I). Using the decomposition from example 1, solve Ax = [8; 22; 17].

Solution. Forward substitution Ly = b: $y_1 = 8$; $2y_1 + y_2 = 22$, from which $y_2 = 6$; $y_1 + y_2 + y_3 = 17$, from which $y_3 = 3$, that is, y = [8; 6; 3]. Backward substitution Ux = y: $3x_3 = 3$, $x_3 = 1$; $3x_2 + 3x_3 = 6$, $x_2 = 1$; $2x_1 + 4x_2 + 2x_3 = 8$, $x_1 = 1$. Thus, x = [1; 1; 1]. With LDU, the computation breaks into three steps: Ly = b, Dz = y (division by pivot elements), then (D^{-1}U)x = z. When row exchanges are needed, the standard form is PA = LU or PA = LDU, where P records the exchanges; extraction and division by pivot elements requires them to be nonzero.

Example 3. Complete Gaussian elimination example (lesson II). Solve the system 2x + y − z = 1, 4x + 3y + z = 13, −2x + y + 2z = 6.

Solution. Here A = [2 1 −1; 4 3 1; −2 1 2], b = [1; 13; 6]. Step 1: the first pivot element is 2, multiplier $m_{21} = 4/2 = 2$; $R_2 \leftarrow R_2 - 2R_1$: (4x + 3y + z) − 2(2x + y − z) = 13 − 2·1, that is, y + 3z = 11. Step 2: $m_{31} = -2/2 = -1$; $R_3 \leftarrow R_3 + R_1$: (−2x + y + 2z) + (2x + y − z) = 6 + 1, that is, 2y + z = 7. Step 3: the second pivot element is 1 (coefficient of y in the updated equation 2), $m_{32} = 2$; $R_3 \leftarrow R_3 - 2R_2$: (2y + z) − 2(y + 3z) = 7 − 22, that is, −5z = −15. The pivot elements 2, 1, −5 — the second and third are taken from the updated equations, not from the original diagonal. Back substitution: z = 3, y = 2, x = 1, that is, (x, y, z) = (1, 2, 3). Verification: 2·1 + 2 − 3 = 1; 4·1 + 6 + 3 = 13; −2·1 + 2 + 6 = 6. With the augmented matrix: [2 1 −1 1; 4 3 1 13; −2 1 2 6] → [2 1 −1 1; 0 1 3 11; 0 2 1 7] → [2 1 −1 1; 0 1 3 11; 0 0 −5 −15].

Example 4. Elimination as matrix multiplication (Lesson II). Write the operations of Example 3 in terms of elimination matrices.

Solution. For $R_2 \leftarrow R_2 - 2R_1$: E_{21} = [1 0 0; −2 1 0; 0 0 1] (the second row of the product E_{21}·A equals −2·(row 1 of A) + (row 2 of A)); for adding row 1 to row 3: E_{31} = [1 0 0; 0 1 0; 1 0 1]; for $R_3 \leftarrow R_3 - 2R_2$: E_{32} = [1 0 0; 0 1 0; 0 −2 1]. Complete elimination: E_{32}·E_{31}·E_{21}·A = U and E_{32}·E_{31}·E_{21}·b = c, where U = [2 1 −1; 0 1 3; 0 0 −5], c = [1; 11; −15]; the rightmost operation acts first. Rules: $(BC)_{ij} = \sum_k b_{ik} \cdot c_{kj}$; associativity (BC)D = B(CD); non-commutativity BC ≠ CB — therefore the order of row operations is important.

Example 5. Zero proposed pivot element (Lesson II). Solve the system y + z = 5, x + y + z = 6, 2x + 3y + 4z = 20.

Solution. A = [0 1 1; 1 1 1; 2 3 4], b = [5; 6; 20]; the first proposed pivot element is zero — we swap rows 1 and 2 (permutation matrix P = [0 1 0; 1 0 0; 0 0 1], reordered system PAx = Pb): [1 1 1 6; 0 1 1 5; 2 3 4 20]. Then $R_3 \leftarrow R_3 - 2R_1$: (0, 1, 2 | 8); then $R_3 \leftarrow R_3 - R_2$: (0, 0, 1 | 3). Back substitution: z = 3, y = 2, x = 1. A zero in the proposed pivot position does not automatically mean no unique solution; for numerical calculations, partial pivoting takes the largest available element in absolute value in the current column, controlling the elimination multipliers.

Example 6. How many solutions does the system have (Lesson II). Consider the family x + y + z = 6, 2x + 3y + 4z = 20, 3x + 4y + 5z = k. For which k are there no solutions, for which are there infinitely many?

Solution. Apply $R_2 \leftarrow R_2 - 2R_1$ and $R_3 \leftarrow R_3 - 3R_1$: we get y + 2z = 8 and y + 2z = k − 18. Subtracting equation 2 from equation 3: 0 = k − 26. If k ≠ 26 — no solutions (when k = 27 we get 0 = 1 — the equations are inconsistent). If k = 26 — the last equation is 0 = 0 and adds no constraints: with free variable z = t we have y = 8 − 2t and x = 6 − y − z = t − 2, that is, (x, y, z) = (t − 2, 8 − 2t, t), t ∈ R — infinitely many solutions. Outcome table for 3×3: three pivot elements — unique solution; a row [0 0 0 | d], d ≠ 0 — no solutions; fewer than three pivot elements without a contradictory row — infinitely many solutions (one zero row by itself does not guarantee consistency — need to check the other rows).

Example 7. LU-factorization and reuse (Lesson II). Write A = LU for the system of Example 3 and solve Lc = b, Ux = c.

Solution. The multipliers of Example 3: $m_{21} = 2$, $m_{31} = -1$, $m_{32} = 2$; we place them under the diagonal: L = [1 0 0; 2 1 0; −1 2 1], and A = LU with U = [2 1 −1; 0 1 3; 0 0 −5]; the signs in L are the multipliers themselves, while the elimination operations use their negatives. Solve in two steps: Lc = b gives $c_1 = 1$, $c_2 = 11$, $c_3 = -15$ (forward substitution); then Ux = c by back substitution gives (1, 2, 3). The advantage appears when several systems have the same matrix A but different right-hand sides: A is factored once, and L and U are reused. If row exchanges are needed, the standard factorization is PA = LU.

Example 8. Exercises A and B (Lesson II). Solve the systems: (A) x + y + z = 6, 2x + 3y + z = 11, x + 2y + 3z = 14; (B) y + 2z = 8, x + y + z = 6, 2x + 3y + z = 11.

Solution. (A) Eliminate x: $R_2 - 2R_1$ gives y − z = −1; $R_3 - R_1$ gives y + 2z = 8. Subtracting: 3z = 9, z = 3, y = z − 1 = 2, x = 6 − 2 − 3 = 1. Answer (1, 2, 3); check: 1 + 2 + 3 = 6, 2 + 6 + 3 = 11, 1 + 4 + 9 = 14. (B) There is no x in the first equation (zero proposed pivot element) — before eliminating x, swap the first two rows: x + y + z = 6, y + 2z = 8, 2x + 3y + z = 11. Eliminate x: $R_3 - 2R_1$ gives y − z = −1; together with y + 2z = 8, subtraction gives 3z = 9, z = 3, y = 2, x = 1. Answer (1, 2, 3); check: 2 + 2·3 = 8, 1 + 2 + 3 = 6, 2 + 6 + 3 = 11.

Example 9. Exercises C and D (Lesson II). (C) Determine for which k the system x + y + z = 6, 2x + 3y + 4z = 20, 3x + 4y + 5z = k has no solutions, and for which it has infinitely many. (D) Find the matrix of the operation $R_3 \leftarrow R_3 - 4R_1$.

Solution. (C) This is a family of example 6: after elimination, the third equation reduces to 0 = k − 26, so when k ≠ 26 there are no solutions, when k = 26 there are infinitely many: (x, y, z) = (t − 2, 8 − 2t, t), t ∈ R; there is no value of k that gives a unique solution (the coefficient matrix has only two pivots). (D) The operation changes only row 3, subtracting 4·(row 1): E = [1 0 0; 0 1 0; −4 0 1];

the third row of the product E·A is (row 3 of A) − 4·(row 1 of A), the other rows do not change.

Example 10. Inverse matrix by the Gauss-Jordan method (Lesson III). Find the inverse matrix of A = [1 1 0; 1 2 1; 0 1 2], by adjoining the identity matrix and reducing the left part to the identity.

Solution. Theory: the inverse of an n×n matrix A is the matrix A^{-1} with $AA^{-1} = A^{-1}A = I$; each elementary row operation is equivalent to left multiplication by an elementary matrix, and if $E_k \cdots E_2 E_1 A = I$, then $A^{-1} = E_k \cdots E_2 E_1$ — therefore the same operations, applied to I, give A^{-1}: $[A \mid I] \to [I \mid A^{-1}]$; each operation acts on the entire augmented row. Step 1: [A | I] = [1 1 0 1 0 0; 1 2 1 0 1 0; 0 1 2 0 0 1]. Step 2 (eliminate under the first pivot): $R_2 \leftarrow R_2 - R_1$: [1 1 0 1 0 0; 0 1 1 −1 1 0; 0 1 2 0 0 1]. Step 3 (under the second): $R_3 \leftarrow R_3 - R_2$: [1 1 0 1 0 0; 0 1 1 −1 1 0; 0 0 1 1 −1 1]. Step 4 (above the third): $R_2 \leftarrow R_2 - R_3$: [1 1 0 1 0 0; 0 1 0 −2 2 −1; 0 0 1 1 −1 1]. Step 5 (above the second): $R_1 \leftarrow R_1 - R_2$: [1 0 0 3 −2 1; 0 1 0 −2 2 −1; 0 0 1 1 −1 1]. The left part became the identity, so A^{-1} = [3 −2 1; −2 2 −1; 1 −1 1]. Check by multiplication: $A \cdot A^{-1} = I$ (for example, element (2, 1): 1·3 + 2·(−2) + 1·1 = 0).

Example 11. When the method doesn't work (Lesson III). Show that the matrix [1 2; 2 4] has no inverse.

Solution. A square matrix is invertible only if the left block in [A | I] reduces to the identity matrix. Apply $R_2 \leftarrow R_2 - 2R_1$: [1 2 1 0; 2 4 0 1] → [1 2 1 0; 0 0 −2 1]. A row of zeros appeared in the left block — there are too few pivots to form I; the matrix is degenerate (singular), no inverse exists. Note: Gaussian elimination stops at an upper triangular matrix, while Gauss-Jordan continues until each pivot is 1 and all elements above and below each pivot are zero.

Week 7 — Related rates and review of chapter 3 (Math I); Transpositions, permutations and vector spaces (Linear Algebra I)

Week 7 of the **Math I** course (Fall 2026, lecturer G. Neofotistos) concludes the "Derivatives" block before the midterm. Two parts of the week: **related rates** — a practical session on applying the derivative with respect to time to changing quantities — and **derivatives of elementary functions** (continuation of week 6): sections **3.5** (trigonometric functions) and **3.9** (exponential and logarithmic functions) of the OpenStax Calculus textbook, vol. 1. Objectives (Learning Objectives 3.5.1–3.5.3, 3.9.1–3.9.2): derivatives of sin x, cos x and other trigonometric functions; higher-order derivatives of sine and cosine; derivatives of exponential and logarithmic functions.

4.1 Math I: Formula catalog for week 7

[Source: Week 7 lecture "Derivatives of elementary functions" (§3.5, §3.9 of OpenStax Calculus, vol. 1), files $m_{w7lecg01}$–$g10$; related rates worksheet, file $m_{w7pracg01}$]

4.1.1 Related rates: idea and solution strategy

In **related rates** problems, several quantities depend on time t and are related by an equation (volume or area formula, Pythagorean theorem, similarity of triangles). The rate of change of the desired quantity is found by differentiating

the relationship with respect to t: the equation relates the quantities, and the differentiated equation relates their rates. An example is inflating a balloon:

$$V = (4/3)\pi r^3 \to dV/dt = 4\pi r^2 \cdot dr/dt$$

- Step 1. Read the problem, make a diagram, label all changing (dependent on t) quantities and constants.
- Step 2. Write down the known rates and the desired rate; note the specific moment ("when r = 30") — this is an instantaneous snapshot, numbers are substituted at the end.
- Step 3. Write the relationship equation (geometric formula, Pythagorean theorem, similarity); eliminate extra variables, for example r = h/2 from similarity.
- Step 4. Differentiate both sides with respect to t (chain rule): each variable dependent on t gives a factor d(·)/dt; numbers — only AFTER differentiation.
- Step 5. Substitute known values, solve for the desired rate; check units and sign: minus indicates decreasing.

A typical mistake is to substitute numbers before differentiation: the equation becomes constant, and rates will be lost. Decreasing quantities have negative speed (a rope being pulled at 1 m/s: dl/dt = −1 m/s); the answer is the instantaneous speed at the specified moment, not the average.

4.1.2 Typical geometric formulas

Quantity / figure	Formula
Sphere: volume	$V = (4/3)\pi r^3$
Sphere: surface area	$S = 4\pi r^2$
Circle: area	$A = \pi r^2$
Circle: circumference	$C = 2\pi r$
Cylinder: volume	$V = \pi r^{2h}$
Cone: volume	$V = (1/3)\pi r^{2h}$
Triangle: area	A = (1/2) · base · height
Right triangle	$a^2 + b^2 = c^2$ (Pythagorean theorem)
Similar triangles	corresponding sides are proportional, for example 15/(x + s) = 6/s

A useful consequence: if a cone has base diameter equal to height (2r = h), then $V = (1/3)\pi(h/2)^2 \cdot h = (\pi/12)h^3$ and $dV/dt = (\pi h^2/4) \cdot (dh/dt)$ — this formula appears in two exercises. Water in a conical tank forms a cone similar to the tank, so r/h = R/H; the shadow problem is also solved by similarity of triangles.

4.1.3 Review: differentiation rules (from lecture)

$$d/dx\,(x^n) = n \cdot x^{(n-1)},\ (f+g)' = f' + g',\ (f-g)' = f' - g',\ (k \cdot f)' = k \cdot f' \text{ (theorems 3.3–3.4)}$$

$$(f \cdot g)' = f' \cdot g + g' \cdot f \text{ (product rule, theorem 3.5)}$$

$$(f/g)' = (f' \cdot g - g' \cdot f)/g^2 \text{ (quotient rule, theorem 3.6)}$$

Example (lecture, In-Class activity): for $j(x) = (x^2 + 2)(3x^3 - 5x)$ by the product rule $j'(x) = 2x \cdot (3x^3 - 5x) + (9x^2 - 5) \cdot (x^2 + 2) = 15x^4 + 3x^2 - 10$. Example 3.25: for $k(x) = 5x^2/(4x + 3)$ by the quotient rule $k'(x) = (10x(4x + 3) - 20x^2)/(4x + 3)^2$

$= (20x^2 + 30x)/(4x + 3)^2$.

4.1.4 Derivatives of trigonometric functions (§3.5)

$$d/dx\ (\sin x) = \cos x,\ d/dx\ (\cos x) = -\sin x\ \text{(theorem 3.8)}$$

Graphically (Fig. 3.27): at maximum and minimum points of sin x, the tangent is horizontal and cos x = 0; where sin x is increasing — cos x > 0, where decreasing — cos x < 0. The proof for sine relies on the definition of derivative, the sine addition formula, and two remarkable limits:

$$\lim h\to 0\ (\sin h)/h = 1,\ \lim h\to 0\ (\cos h - 1)/h = 0,\ \sin(x + h) = \sin x\cdot\cos h + \cos x\cdot\sin h$$

$$d/dx\ (\tan x) = \sec^2 x,\ d/dx\ (\cot x) = -\csc^2 x,\ d/dx\ (\sec x) = \sec x\cdot\tan x,\ d/dx\ (\csc x) = -\csc x\cdot\cot x\ \text{(theorem 3.9)}$$

Lecturer's advice: "Do not memorize them!" — derivatives of tan, cot, sec, csc are not memorized: each is derived from sin and cos using the quotient rule ($\tan x = \sin x/\cos x \to \sec^2 x$). Harmonic oscillation: s′ — velocity, s″ — acceleration. Higher-order derivatives of sine are cyclic with period 4:

$$\sin x \to \cos x \to -\sin x \to -\cos x \to \sin x \to \ldots$$

Example (lecture; textbook examples 3.45–3.46): d^{74}/dx^{74} (sin x). Since 74 = 4·18 + 2, the 74th derivative in the cycle is in the position of the 2nd: −sin x.

4.1.5 Derivatives of exponential and logarithmic functions (§3.9)

$$d/dx\ (e^x) = e^x,\ d/dx\ (e\text{^}(g(x))) = e\text{^}(g(x))\cdot g'(x)\ \text{(theorem 3.14)}$$

$$d/dx\ (b^x) = b^x\cdot\ln b,\ d/dx\ (\ln x) = 1/x,\ d/dx\ (\ln(g(x))) = g'(x)/g(x),\ d/dx\ (\log_b x) = 1/(x\cdot\ln b)\ \text{(theorems 3.15–3.16)}$$

The number e is the unique base b > 0 for which the function $B(x) = b^x$ satisfies B′(0) = 1; 2.7182 < e < 2.7183 (e ≈ 2.718282), and the tangent to e^x at zero is the line y = x + 1. For another base: $2^x = e^{(x\cdot\ln 2)}$, from which $(2^x)' = 2^x\cdot\ln 2$. Example (lecture, In-Class activity): $(e^{(2x)})' = 2\cdot e^{(2x)}$; $(e^{(3x^2)})' = 6x\cdot e^{(3x^2)}$; $(e^{(3x^2+5x^3)})' = (6x + 15x^2)\cdot e^{(3x^2+5x^3)}$.

- Exponent laws (a, b > 0): $b^x\cdot b^y = b^{(x+y)}$; $b^x/b^y = b^{(x-y)}$; $(b^x)^y = b^{(xy)}$; $(ab)^x = a^x\cdot b^x$; $(a/b)^x = a^x/b^x$.
- Logarithm properties (a, b, c > 0, b ≠ 1): $\log_b(a\cdot c) = \log_b a + \log_b c$; $\log_b(a/c) = \log_b a - \log_b c$; $\log_b(a^r) = r\cdot\log_b a$.
- Change of base: $a^x = e^{(x\cdot\ln a)}$ and $\log_a x = (\ln x)/(\ln a)$; since ln a is constant, $(\log_a x)' = 1/(x\cdot\ln a)$.
- Logarithmic differentiation (3.9.3): take the logarithm of y = h(x), expand the logarithm using its properties, differentiate (the left side will yield (1/y)·y′), multiply by y, and replace y with h(x). This method is needed for functions of the form g(x)^(f(x)) (for example, x^x); according to the lecturer's decision, this section is postponed until the midterm.

4.2 Math I: Week 7 Practical Session — questions and complete solutions

[Source: Week 7 Practical Session exercises with solutions, MATH I (Fall 2026), prepared by G. Neofotistos — files $m_{w7pracg01}$–g23; sheet 1 "related rates" (exercises 1–10) and sheet 2 "derivatives of elementary functions" (exercises 1–11), numbering of each sheet is preserved]

The session consists of two sheets. Sheet 1 contains ten classic problems on **related rates**; all velocities are instantaneous, units are mandatory. Sheet 2 contains eleven exercises on **derivatives of elementary functions** (the last two are marked REVIEW — review of week 6 on electrical circuits); note on the sheet: "differentiate" means "find the first derivative".

Exercise 1 (related rates). Inflating a balloon. A spherical balloon is being inflated with gas at a rate of 800 cm^3/min. How fast is the radius of the balloon increasing at the moment when the radius is 30 cm?

Solution. Volume of a sphere $V = (4/3)\pi r^3$. Differentiate with respect to t: $dV/dt = 4\pi r^2 \cdot (dr/dt)$. Substitute dV/dt = 800 and r = 30: $dr/dt = 800/(4\pi \cdot 30^2) = 800/(3600\pi) = 2/(9\pi) \approx 0.0707$ cm/min. The radius is increasing at a rate of $2/(9\pi)$ cm/min (approximately 0.071 cm/min).

Exercise 2 (related rates). Ladder against a wall. A 10-foot ladder leans against a vertical wall. If the bottom end of the ladder slides away from the wall at a rate of 1 ft/s, how fast is the top end sliding down the wall when the bottom end is 6 feet from the wall?

Solution. Let x be the distance from the bottom end to the wall, y be the height of the top end. By Pythagoras' theorem $x^2 + y^2 = 10^2 = 100$; given dx/dt = 1 ft/s, we find dy/dt when x = 6. Then y = 8 ($36 + y^2 = 100 \rightarrow y^2 = 64$). Differentiate: $2x \cdot (dx/dt) + 2y \cdot (dy/dt) = 0$, so $dy/dt = -(x/y) \cdot (dx/dt) = -(6/8) \cdot 1 = -3/4$ ft/s. The "minus" sign indicates the height is decreasing: the top end slides down at a rate of 3/4 ft/s (0.75 ft/s).

Exercise 3 (related rates). Shadow from a lamp. A 6-foot tall person walks away from a lamp post that is 15 feet tall at a rate of 5 ft/s. How fast is the tip of their shadow moving when they are 40 feet from the post?

Solution. Let x be the distance from the person to the post, s be the length of the shadow, L = x + s be the distance from the post to the tip of the shadow; dx/dt = 5 ft/s. From similar triangles: $15/(x + s) = 6/s \rightarrow 15s = 6(x + s) \rightarrow 9s = 6x \rightarrow s = (2/3)x$, so $L = (5/3)x$. Differentiate: $dL/dt = (5/3) \cdot (dx/dt) = (5/3) \cdot 5 = 25/3 \approx 8.33$ ft/s. The speed of the shadow tip is constant and does not depend on the distance to the post (therefore x = 40 is not used in the calculation).

Exercise 4 (related rates). Conical tank with leak. Water is supplied to an inverted conical tank at a constant rate of 9 m^3/min. The tank has a height of 12 m and a radius of 6 m at the top edge. If the water level is rising at a rate of 0.2 m/min at the moment when the water depth is 4 m, find the rate at which water is leaking from the tank.

Solution. Let V be the volume of water, h be the depth, r be the radius of the surface. From similarity $r/h = 6/12 = 1/2$, so $r = h/2$. Then $V = (1/3)\pi(h/2)^2 \cdot h = (\pi/12)h^3$, and after differentiating with respect to t: $dV/dt = (\pi h^2/4) \cdot (dh/dt)$ — the net rate of volume change (inflow minus outflow). When h = 4 and dh/dt = 0.2: $dV_{net}/dt = (\pi \cdot 16/4) \cdot 0.2 = 0.8\pi \approx 2.513$ m^3/min. From $dV_{net}/dt = Q_{in} - Q_{out}$: $Q_{out} = 9 - 0.8\pi \approx 6.487$ m^3/min (approximately 6.49 m^3/min).

Exercise 5 (related rates). Two cars. Two cars start from the same point. One travels south at 60 km/h, the other west at 50 km/h. At what rate is the distance between the cars increasing two hours later?

Solution. Let x be the distance traveled by the westbound car (dx/dt = 50 km/h), y by the southbound car (dy/dt = 60 km/h), z be the distance: $z^2 = x^2 + y^2$. After 2 hours: x = 100 km, y = 120 km, z = sqrt($100^2 + 120^2$) = sqrt(24400) =

20·sqrt(61) ≈ 156.2 km. Differentiate: 2z·(dz/dt) = 2x·(dx/dt) + 2y·(dy/dt), so dz/dt = (100·50 + 120·60)/(20·sqrt(61)) = 12200/(20·sqrt(61)) = 10·sqrt(61) ≈ 78.1 km/h.

Exercise 6 (related rates). Plane over radar. A plane flies horizontally at an altitude of 2 km with a speed of 800 km/h and passes directly over a radar station. Find the rate at which the distance from the plane to the station is increasing when it is equal to 3 km.

Solution. Let x be the horizontal distance from the station to the point under the plane, s be the direct distance to the plane; the height h = 2 km is constant, so $s^2 = x^2 + 4$. When s = 3: $x^2 = 5$, x = sqrt(5) km. Differentiating: 2s·(ds/dt) = 2x·(dx/dt), so ds/dt = (x/s)·(dx/dt) = (sqrt(5)/3)·800 = 800·sqrt(5)/3 ≈ 596.27 km/h (approximately 596 km/h).

Exercise 7 (related rates). Gravel pile. Gravel falls from a conveyor belt at a rate of 30 ft^3/min, and its granularity is such that it forms a pile in the shape of a cone whose base diameter and height are always equal. How fast is the height of the pile growing when it is 10 feet high?

Solution. Diameter equals height: 2r = h → r = h/2. As in exercise 4: $V = (\pi/12)h^3$ and $dV/dt = (\pi h^2/4)\cdot(dh/dt)$, so $dh/dt = 4\cdot(dV/dt)/(\pi h^2)$. Substituting dV/dt = 30 and h = 10: dh/dt = 4·30/(π·100) = 120/(100π) = 6/(5π) = 1.2/π ≈ 0.382 ft/min.

Exercise 8 (related rates). Boat and pulley. A boat is pulled toward a dock by a rope attached to the bow of the boat and passing through a pulley on the dock located 1 m above the bow of the boat. If the rope is pulled at a rate of 1 m/s, how fast is the boat approaching the dock when it is 8 m from it?

Solution. Let x be the horizontal distance from the boat to the dock, l be the length of the rope from the pulley to the boat; the pulley is 1 m above the bow, so $l^2 = x^2 + 1$. The rope is shortening: dl/dt = −1 m/s. When x = 8: l = sqrt(65). Differentiating: 2l·(dl/dt) = 2x·(dx/dt), so dx/dt = (l/x)·(dl/dt) = (sqrt(65)/8)·(−1) ≈ −1.008 m/s. The distance is decreasing — the boat is approaching the dock at a rate of sqrt(65)/8 ≈ 1.01 m/s.

Exercise 9 (related rates). Particle on a curve. A particle moves along the curve y = sqrt(x) such that its x-coordinate increases at a rate of 2 units/s. How fast is the y-coordinate of the particle changing when the particle is at the point (4, 2)?

Solution. $y = sqrt(x) = x^{(1/2)}$. Differentiate with respect to t (chain rule): $dy/dt = (1/2)\cdot x^{(-1/2)}\cdot(dx/dt) = (1/(2\cdot sqrt(x)))\cdot(dx/dt)$. Substituting x = 4 and dx/dt = 2: dy/dt = (1/4)·2 = 1/2 units/s — the y-coordinate is increasing (0.5 units/s).

Exercise 10 (related rates). Oil slick. The radius of a circular oil slick is expanding at a constant rate of 2 m/min. At what rate is the area of the slick growing when the radius is 30 m?

Solution. Area of a circle $A = \pi r^2$; differentiate with respect to t: $dA/dt = 2\pi r\cdot(dr/dt)$. Substituting r = 30 and dr/dt = 2: dA/dt = 2π·30·2 = 120π ≈ 376.99 m^2/min (approximately 377 m^2/min).

[Source: Week 7 Practical Session exercises with solutions — sheet 2, exercises on derivatives of elementary functions (1–9) and REVIEW (10–11); files $m_{w7pracg02}$–g23]

Exercise 1 (derivatives). Trigonometric functions. "Differentiate" means "find the first derivative". Differentiate: a) 3·sin x − 2·cos x; b) tan x + cot x; c) sec x − csc x.

Solution. a) d/dx(3 sin x − 2 cos x) = 3 cos x − 2·(−sin x) = 3 cos x + 2 sin x. b) d/dx(tan x + cot x) = $\sec^2 x - \csc^2 x$. c) d/dx(sec x − csc x) = sec x·tan x − (−csc x·cot x) = sec x·tan x + csc x·cot x.

Exercise 2 (derivatives). Derivative of tangent. Starting from the representation tan x = sin x/cos x, derive the derivative of tan x using the quotient rule.

Solution. f(x) = tan x = sin x/cos x. By the quotient rule: f′(x) = (cos x·cos x − (−sin x)·sin x)/$\cos^2 x$ = ($\cos^2 x + \sin^2 x$)/$\cos^2 x$. By the Pythagorean identity $\cos^2 x + \sin^2 x = 1$, so f′(x) = 1/$\cos^2 x = \sec^2 x$ (for cos x ≠ 0; used sec x = 1/cos x).

Exercise 3 (derivatives). Quotient with cosine. Differentiate f(x) = sin x/(1 + cos x), where 1 + cos x ≠ 0.

Solution. By the quotient rule: f′(x) = (cos x·(1 + cos x) − sin x·(−sin x))/$(1 + \cos x)^2$ = ($\cos x + \cos^2 x + \sin^2 x$)/$(1 + \cos x)^2$. Since $\cos^2 x + \sin^2 x = 1$, the numerator equals 1 + cos x, and f′(x) = (1 + cos x)/$(1 + \cos x)^2$ = 1/(1 + cos x) — on the original domain.

Exercise 4 (derivatives). Vibrating component. A vibrating component has displacement s(t) = 0.02·cos t m at angular frequency 1 rad/s. Find the velocity and acceleration, calculate them at t = π/2 s, and also find $d^{4}s/dt^{4}$.

Solution. Velocity v(t) = s′(t) = −0.02·sin t m/s; acceleration a(t) = v′(t) = −0.02·cos t m/s^2. At t = π/2: v = −0.02·sin(π/2) = −0.02 m/s, a = −0.02·cos(π/2) = 0. Differentiate twice more: s‴(t) = 0.02·sin t, $s^{(4)}(t)$ = 0.02·cos t m/s^4 — the fourth derivative repeats the original form (the cosine cycle has period 4).

Exercise 5 (derivatives). Different bases. (Note: "Beware of the different bases".) Differentiate $f(x) = 3e^x + 2^x - 5$ and find f′(0).

Solution. $(e^x)' = e^x$. For 2^x: $2^x = (e^{(\ln 2)})^x = e^{(x\cdot\ln 2)}$, therefore $(2^x)' = e^{(x\cdot\ln 2)}\cdot\ln 2 = 2^x\cdot\ln 2$. Thus f′(x) = $3e^x + 2^x\cdot\ln 2$. At x = 0: f′(0) = $3\cdot e^0 + 2^0\cdot\ln 2$ = 3 + ln 2 ≈ 3.693.

Exercise 6 (derivatives). Logarithms and tangent. Differentiate f(x) = 2·ln x − $\log_{10}$ x, x > 0, and find the equation of the tangent at x = 1.

Solution. (ln x)′ = 1/x; by the change of base formula $\log_{10} x$ = ln x/ln 10, therefore ($\log_{10} x$)′ = 1/(x·ln 10). Hence f′(x) = 2/x − 1/(x·ln 10). At the point of tangency f(1) = 2·ln 1 − $\log_{10} 1$ = 0, slope f′(1) = 2 − 1/ln 10. Tangent: y = (2 − 1/ln 10)·(x − 1).

Exercise 7 (derivatives). Product and quotient. Differentiate: a) f(x) = x·ln x, x > 0; b) g(x) = e^x/x, x ≠ 0.

Solution. a) Product rule: f′(x) = 1·ln x + x·(1/x) = ln x + 1. b) Quotient rule: g′(x) = ($x\cdot e^x - e^x\cdot 1$)/x^2 = $e^x\cdot(x - 1)/x^2$.

Exercise 8 (derivatives). Product with exponential. Find the first derivative of the function $f(x) = x^2\cdot e^x$.

Solution. Product of $u(x) = x^2$ and $v(x) = e^x$; by the rule (u·v)′ = u′·v + u·v′: f′(x) = $2x\cdot e^x + x^2\cdot e^x = e^x\cdot(x^2 + 2x) = x\cdot e^x\cdot(x + 2)$, x ∈ R.

Exercise 9 (derivatives). Discharging capacitor. A discharging capacitor has voltage $V(t) = 12\cdot e^{(-t/2)}$ V, t ≥ 0 (time in seconds). Find V′(t), initial rate of voltage change, and V′(2).

Solution. Derivative of the exponent $(-t/2)' = -1/2$, therefore $V'(t) = 12 \cdot e^{(-t/2)} \cdot (-1/2) = -6 \cdot e^{(-t/2)}$ V/s. Initial rate: $V'(0) = -6$ V/s. At t = 2: $V'(2) = -6 \cdot e^{(-1)} = -6/e \approx -2.21$ V/s. The "minus" sign indicates the voltage is decreasing, and the rate of decrease decreases in magnitude over time.

Exercise 10 (REVIEW). Charge and electric current. Charge passing through a point in a wire is $q(t) = t^3 + 2t^2 + 3t$ C. Find the electric current at t = 2, average current on [0, 2], and rate of change of current at t = 2. (Electric current is $I(t) = q'(t)$.)

Solution. Current is the derivative of charge: $I(t) = q'(t) = 3t^2 + 4t + 3$, therefore $I(2) = 12 + 8 + 3 = 23$ A. Average current is the average rate of charge change: $q(2) = 8 + 8 + 6 = 22$ C, $q(0) = 0$, therefore $I_{avg} = 22/2 = 11$ A. Rate of change of current: $I'(t) = 6t + 4$, hence $I'(2) = 16$ A/s.

Exercise 11 (REVIEW). Electric power. For $0 \le t \le 4$ a device has voltage $V(t) = 12 - t$ V and current $I(t) = 1 + t$ A. Find the electric power P(t), its derivative P′(t), and values of both quantities at t = 2. (Electric power is $P(t) = V(t) \cdot I(t)$.)

Solution. $P(t) = V(t) \cdot I(t) = (12 - t)(1 + t) = 12 + 11t - t^2$ W. By the product rule $P' = V' \cdot I + V \cdot I' = (-1) \cdot (1 + t) + (12 - t) \cdot 1 = 11 - 2t$ W/s. At t = 2: $P(2) = 12 + 22 - 4 = 30$ W, $P'(2) = 11 - 4 = 7$ W/s — the power being consumed is increasing.

Main ideas of the lesson: related rates — follow the "picture → relation → d/dt → plug in numbers" scheme; derivatives of elementary functions are combined using sum, product, and quotient rules; physical quantities are related by derivatives: velocity s′, acceleration s″, current I = q′, power P = V·I.

4.3 Math I: Week 7 Independent Study Assignments — questions and full solutions

[Source: Week 7 Independent Study Assignments (Homework), MATH I (Fall 2026), file Week_7_-_IndependentStudyAssignments.txt; exercises 178, 181, 191, 192, 193 and 331 — from Chapter 3 of OpenStax Calculus, vol. 1]

ISA — homework assignments that are not submitted or graded: their purpose is to master the material more deeply before the midterm by solving problems independently, with classmates, or with an AI assistant (to clarify concepts, not to get ready-made solutions). Assignment 3 — solve exercises 178, 181, 191, 192, 193 and 331 from chapter 3 of the textbook; full solutions are below.

Assignment 1 (optional). Interact with an AI assistant (ChatGPT, Claude, DeepSeek, Gemini, etc.) in a structured conversation: formulate the most important points of the lecture content and ask the assistant to evaluate your answer (for content and clarity; you can choose other criteria). If you find erroneous conclusions, discrepancies, or contradictions — ask for explanations; if the contradiction is not resolved, consult the instructor.

How to complete. This is an instructional assignment for working with AI; it has no single "solution." Create a list of key ideas for week 7 (rules of differentiation; derivatives of sin, cos, tan, cot, sec, csc; derivatives of e^x, b^x, ln x, $\log_b x$; related rates strategy), ask the assistant to evaluate the completeness and accuracy of the formulations, and analyze the feedback.

Assignment 2. Thoroughly study all problems, exercises, and examples covered in the lecture and practical session of week 7.

How to complete. All 21 exercises of the practical session (10 problems on related rates and 11 on derivatives) — with solutions in the previous section;

examples from the lecture and required reading (3.39–3.47, 3.74–3.80, problem 204 on precipitation, the flu problem) — in the Required reading section below.

Assignment 3, exercise 178. Find dy/dx for the function $y = x - x^3 \cdot \sin x$.
Solution. $y' = 1 - (x^3 \cdot \sin x)'$. By the product rule $(x^3 \cdot \sin x)' = 3x^2 \cdot \sin x + \cos x \cdot x^3$. Thus $y' = 1 - 3x^2 \cdot \sin x - x^3 \cdot \cos x$.

Assignment 3, exercise 181. Find dy/dx for the function $y = (x + \cos x)(1 - \sin x)$.
Solution. Product rule with $u = x + \cos x$ ($u' = 1 - \sin x$) and $v = 1 - \sin x$ ($v' = -\cos x$): $y' = (1 - \sin x)^2 + (x + \cos x) \cdot (-\cos x) = (1 - \sin x)^2 - x \cdot \cos x - \cos^2 x$. After expanding the square and replacing $\sin^2 x - \cos^2 x = -\cos 2x$: $y' = 1 - 2 \cdot \sin x - \cos 2x - x \cdot \cos x$.

Assignment 3, exercise 191. Find d^{2y}/dx^2 for the function $y = x \cdot \sin x - \cos x$.
Solution. $y' = \sin x + x \cdot \cos x + \sin x = x \cdot \cos x + 2 \cdot \sin x$. Then $y'' = \cos x - x \cdot \sin x + 2 \cdot \cos x = 3 \cdot \cos x - x \cdot \sin x$.

Assignment 3, exercise 192. Find d^{2y}/dx^2 for the function $y = \sin x \cdot \cos x$.
Solution. $y' = \cos^2 x - \sin^2 x = \cos 2x$. Then $y'' = -2 \cdot \sin 2x$.

Assignment 3, exercise 193. Find d^{2y}/dx^2 for the function $y = x - (1/2) \cdot \sin x$.
Solution. $y' = 1 - (1/2) \cdot \cos x$; $y'' = (1/2) \cdot \sin x$.

Assignment 3, exercise 331. Find the derivative of the function $f(x) = e^{(x^3)}$. (Exercise 331 is the first exercise of section 3.9 of the textbook; its text is missing in the course transcripts, provided from the textbook.)
Solution. By Theorem 3.14 in general form d/dx e^(g(x)) = e^(g(x))·g′(x) with $g(x) = x^3$, $g'(x) = 3x^2$: $f'(x) = 3x^2 \cdot e^{(x^3)}$. The exponential "stays as is," and the chain rule adds the derivative of the exponent.

Assignment 4. Read the instructions and important points presented in the "Guide to Midterm Exam" document (posted on Moodle, week 7 area).
Assignment 5. Practice on problems and examples from lectures, practical sessions, and Independent Study Assignments for sections 2.1–2.4, 3.1–3.5 and 3.9, on which the midterm problems will be mainly based.
How to complete. The guide sets the organization of the midterm (written "pen and paper" exam, 2 hours, no books, notes, or devices) and covers the material: chapters 2.1–2.4, 3.1–3.5 and 3.9 (except 3.9.3) and the NOTES handouts. Sections 3.5 and 3.9 are covered by exercises of this chapter; material 2.1–2.4 (limits, continuity) and 3.1–3.4 (definition of derivative, rules) — weeks 5–6, review them in their chapters. The exam values concepts, solution procedure, and clear reasoning, not heavy algebra.

4.4 Math I: Required reading for week 7 — handout, examples, and solutions

[Source: Week 7 Required Reading — section 3.5 of the textbook (Derivatives of Trigonometric Functions) and the beginning of section 3.6 (The Chain Rule), files $m_{w7rrg01}$–g05; section 3.9 of the textbook (Derivatives of Exponential and Logarithmic Functions), files $m_{w7rr39g01}$–g04; week 7 lecture examples]

Section **3.5** derives the derivatives of trigonometric functions; the starting point is a numerical experiment: the graph of D(x) = (sin(x + 0.01) − sin x)/0.01 almost coincides with a cosine curve. Key points:

- Theorem 3.8: d/dx(sin x) = cos x, d/dx(cos x) = −sin x; the proof for sine uses the definition of the derivative, the sine addition formula, and the limits lim h→0 (sin h)/h = 1, lim h→0 (cos h − 1)/h = 0.
- Connection between graphs (Fig. 3.27): at maximum/minimum points of sin x, the tangent is horizontal and cos x = 0; where sin x is increasing, cos x > 0, where it is decreasing, cos x < 0.
- Derivatives of tan, cot, sec, csc (Theorem 3.9) are derived using the quotient rule from tan x = sin x/cos x, cot x = cos x/sin x, sec x = 1/cos x, csc x = 1/sin x — no need to memorize. Example 3.42 (deriving $\sec^2 x$) — Exercise 2 on sheet 2 above; Examples 3.45–3.46 (cycle of derivatives of sin x, d^{74}/dx^{74}) — formula catalog; Examples 3.39–3.40 (product and quotient with sin/cos, In-Class activities of the lecture): $(5x^3\cdot\sin x)' = 15x^2\cdot\sin x + 5x^3\cdot\cos x$, $(\cos x/(4x^2))' = (-x\cdot\sin x - 2\cdot\cos x)/(4x^3)$.
- Harmonic motion (mass on a spring, pendulum) is described by sine or cosine: s′ is velocity, s″ is acceleration; higher-order derivatives of sin x are cyclic with period 4.
- The required reading also includes the beginning of section 3.6 (chain rule): the idea of a "chain reaction" (x changes x^3, and x^3 changes $\sin(x^3)$) and deriving the derivative of $\sin(x^3)$ by multiplying the difference quotient by $(x^3 - a^3)/(x^3 - a^3)$.
- Textbook Exercises 174–213: 174 [T] — fitting a Holling function to data on snowshoe hare and lynx populations; 175–196 — derivatives, tangents, second derivatives; 197–213 — applications. The ISA-assigned exercises 178, 181, 191, 192, 193 are solved above.

Example 3.41 (section 3.5). Application to velocity. A particle moves along the coordinate axis, its position at time t: s(t) = 2·sin t − t for 0 ≤ t ≤ 2π. At what moments is the particle at rest?

Solution. The particle is at rest when v(t) = s′(t) = 2·cos t − 1 = 0, that is, cos t = 1/2. On [0, 2π], the solutions are: t = π/3 and t = 5π/3 — at these moments the particle is at rest.

Example 3.43 (section 3.5). Tangent to cotangent. Find the equation of the line tangent to the graph of f(x) = cot x at the point x = π/4.

Solution. Point of tangency: f(π/4) = cot(π/4) = 1, that is (π/4, 1). Slope: f′(x) = $-\csc^2 x$, so f′(π/4) = $-\csc^2(\pi/4) = -(\text{sqrt}(2))^2 = -2$. Equation: y − 1 = −2·(x − π/4), that is y = −2x + 1 + π/2.

Example 3.44 (section 3.5). Sum and product. Find the derivative f(x) = csc x + x·tan x.

Solution. By the sum rule f′(x) = d/dx(csc x) + d/dx(x·tan x) = −csc x·cot x + [1·tan x + $\sec^2$ x·x]. Thus f′(x) = −csc x·cot x + tan x + x·$\sec^2$ x.

Example 3.47 (section 3.5). Application to acceleration. A particle moves such that s(t) = 2 − sin t. Find v(π/4) and a(π/4); compare these values and determine whether the particle is speeding up or slowing down.

Solution. v(t) = s′(t) = −cos t, so v(π/4) = −1/sqrt(2). a(t) = v′(t) = sin t, so a(π/4) = 1/sqrt(2). Velocity is negative, acceleration is positive — they act in opposite directions: the object is accelerating against the direction of motion, meaning the

particle is slowing down.

Section **3.9** derives the derivatives of exponential and logarithmic functions. Key points:

- Assumptions: B(x) = b^x (b > 0) is defined everywhere, continuous, and differentiable at zero; then B′(x) = b^x·B′(0) (the proof uses $b^{(x+h)} = b^x \cdot b^h$). The number e is the unique base with B′(0) = 1; 4^π ≈ 77.88 illustrates the definition of b^r for irrational r as a limit over rational approximations.
- Theorem 3.14: E′(x) = e^x and in general d/dx e^(g(x)) = e^(g(x))·g′(x); the graph of e^x lies between 2^x and 3^x, the tangent at zero is y = x + 1 (Fig. 3.34).
- Theorem 3.15: (ln x)′ = 1/x for x > 0; for ln(g(x)) — g′(x)/g(x) for g(x) > 0. The proof uses implicit differentiation of $e^y = x$: $e^y \cdot (dy/dx) = 1$, from which dy/dx = 1/x.
- Theorem 3.16: $(b^x)' = b^x \cdot \ln b$ and $(\log_b x)' = 1/(x \cdot \ln b)$; general forms with g(x) follow from the chain rule (using $b^x = e^{(x \cdot \ln b)}$, $\log_b x = \ln x / \ln b$).
- Logarithmic differentiation (3.9.3, a five-step strategy — at the end of the section) differentiates functions of the form g(x)^(f(x)) (e.g., x^x) and simplifies cumbersome products; by the lecturer's decision, this subsection is postponed until the midterm.

Example 3.74 (section 3.9). Composition with an exponential. Find the derivative f(x) = e^(tan(2x)).

Solution. f′(x) = e^(tan(2x))·d/dx(tan(2x)) = e^(tan(2x))·$\sec^2(2x)$·2.

Example 3.75 (section 3.9). Combination of rules. Find the derivative y = $e^{(x^2)}/x$.

Solution. The quotient rule together with the derivative of the exponential and the chain rule: $y' = ((e^{(x^2)} \cdot 2x) \cdot x - e^{(x^2)} \cdot 1)/x^2 = e^{(x^2)} \cdot (2x^2 - 1)/x^2$.

Example 3.76 (section 3.9). Population growth. A mosquito colony has an initial population of 1000; after t days the population is A(t) = $1000 \cdot e^{(0.3t)}$. Show that the ratio of the rate of change of the population A′(t) to the population A(t) is constant.

Solution. By the chain rule $A'(t) = 1000 \cdot e^{(0.3t)} \cdot 0.3 = 300 \cdot e^{(0.3t)}$. The ratio A′(t)/A(t) $= 300 \cdot e^{(0.3t)}/(1000 \cdot e^{(0.3t)}) = 0.3$ — a constant: the growth rate of an exponential population is proportional to the population itself.

Example 3.77 (section 3.9). Derivative of the natural logarithm. Find the derivative f(x) = ln(x^3 + 3x − 4).

Solution. Using the formula h′(x) = g′(x)/g(x) with $g(x) = x^3 + 3x - 4$: $f'(x) = (3x^2 + 3)/(x^3 + 3x - 4)$.

Example 3.78 (section 3.9). Logarithmic properties in the derivative. Find the derivative f(x) = ln(x^2·sin x/(2x + 1)).

Solution. First, expand the logarithm using properties: f(x) = 2·ln x + ln(sin x) − ln(2x + 1). Differentiate term by term: f′(x) = 2/x + cot x − 2/(2x + 1).

Example 3.79 (section 3.9). Common base. Find the derivative h(x) = $3^x/(3^x + 2)$.

Solution. The quotient rule and the formula $(3^x)' = 3^x \cdot \ln 3$: $h'(x) = (3^x \cdot \ln 3 \cdot (3^x + 2) - 3^x \cdot \ln 3 \cdot 3^x)/(3^x + 2)^2$. In the numerator, $3^x \cdot \ln 3 \cdot ((3^x + 2) - 3^x) = 2 \cdot 3^x \cdot \ln 3$ remains, so $h'(x) = 2 \cdot 3^x \cdot \ln 3/(3^x + 2)^2$.

Example 3.80 (section 3.9). Slope of the tangent. Find the slope of the line tangent to the graph $y = \log_2(3x + 1)$ at the point $x = 1$.
Solution. Using the formula $(\log_b(g(x)))' = g'(x)/(g(x)\cdot\ln b)$: $dy/dx = 3/((3x + 1)\cdot\ln 2)$. At $x = 1$ the slope is $3/(4\cdot\ln 2) = 3/\ln 16$, since $4\cdot\ln 2 = \ln 2^4 = \ln 16$.

Example (lecture, problem 204 [T]). Rainfall in Phoenix. The amount of rainfall in a month in Phoenix (Arizona) is approximately described by the function $y(t) = 0.5 + 0.3\cdot\cos t$, where t is the number of months since January. Find y' and determine the intervals on which the amount of rainfall is decreasing.
Solution. $y'(t) = -0.3\cdot\sin t$ (radians!). Rainfall decreases where $y' < 0$, i.e., where $\sin t > 0$: on the intervals $(0, \pi)$ and $(2\pi, 3\pi)$. On $(\pi, 2\pi)$ and $(3\pi, 12)$ (note: $12 < 4\pi$) $\sin t < 0$ and rainfall increases.

Example (lecture). Flu in New York. The number of flu cases in New York from the beginning of 1960 to the beginning of 1964 is modeled by the function $N(t) = 5.3\cdot e^{(0.093t^2 - 0.87t)}$, $0 \le t \le 4$, where N(t) is the number of cases (in thousands), t is time in years, t = 0 is the beginning of 1960. a) Calculate N(0) and N(4) and describe their meaning. b) Calculate $N'(0)$ and $N'(3)$ and describe their meaning.
Solution. a) $N(0) = 5.3\cdot e^0 = 5.3$ — approximately 5300 cases at the beginning of 1960. At $t = 4$: $0.093\cdot 16 - 0.87\cdot 4 = -1.992$, so $N(4) = 5.3\cdot e^{(-1.992)} \approx 0.723$ — approximately 723 cases at the beginning of 1964: the number of cases decreased significantly over four years. b) Chain rule: $N'(t) = 5.3\cdot e^{(0.093t^2 - 0.87t)}\cdot(0.186t - 0.87) = N(t)\cdot(0.186t - 0.87)$ (thousands of cases per year). $N'(0) = 5.3\cdot(-0.87) = -4.611$ — at the beginning of 1960, a decrease of about 4611 cases per year. At $t = 3$: $0.093\cdot 9 - 0.87\cdot 3 = -1.773$ and $0.186\cdot 3 - 0.87 = -0.312$, so $N'(3) = 5.3\cdot e^{(-1.773)}\cdot(-0.312) \approx -0.2808$ — about 281 cases per year. Both derivatives are negative (the incidence is decreasing), but their absolute values are decreasing — the decline is slowing down; these are instantaneous rates, not losses over a year.

Visual support for the week: fig. 3.27 (connection between the graphs of sin x and cos x — the derivative of sine is cosine), fig. 3.33 (graphs of 2^x, 2.7^x, 2.8^x, 3^x and the position of e between 2.7 and 2.8), fig. 3.34 (the tangent $y = x + 1$ to e^x at zero). The week concludes with preparation for the midterm (problems 4–5 ISA).

Week 7 of **Linear Algebra I** (Fall 2026) connects matrix algebra with the geometry of spaces: section **2.7** of G. Strang's textbook (Introduction to Linear Algebra, 4th ed., pp. 108–120) — on transposes and permutations — and section **3.1** (pp. 121–132) — on vector spaces and subspaces. Key results: the identities $(A^T)^T = A$, $(A + B)^T = A^T + B^T$, $(AB)^T = B^T A^T$, $(A^{-1})^T = (A^T)^{-1}$; symmetric matrices $A^T = A$ and always symmetric products of the form $R^T R$; permutation matrices, for which $P^T = P^{-1}$, and the PA = LU decomposition; vector spaces R^n, their subspaces, the column space C(A) and the null space N(A) — the language of solvability of the system Ax = b. Week's objectives: use transpose identities, compute row and column permutations, understand PA = LU, recognize and construct subspaces. Notation: matrices are written row-wise — [a b; c d] means "first row (a b), second (c d)"; column vector — [x; y], row vector — [x y].

4.5 Linear Algebra I, week 7: formula catalog

[Source: Week 7 lecture "Transposes, permutations, vector spaces and subspaces" (Strang, Introduction to Linear Algebra, 4th ed., §2.7 pp. 108–120, §3.1 pp. 121–132)]

4.5.1 Review of Week 6: A = LU, LDU, PA = LU

Gaussian elimination gives $E_k \cdots E_1 A = U$. Without row exchanges, the elimination factors combine into a lower triangular L (with ones on the diagonal), and A = LU: for example, A = [2 1; 4 5], the step R2 ← R2 − 2R1 gives U = [2 1; 0 3], L = [1 0; 2 1]. Factoring out the pivots gives A = LDŪ: D = [2 0; 0 3] is the diagonal of pivots, and Ū (U with a bar) = [1 1/2; 0 1] is upper triangular with ones on the diagonal.

$$A = LU,\ A = LD\bar{U};\ \text{when exchanging rows} — PA = LU$$

If the leading element is zero (A = [0 2; 3 4]), the rows are swapped: P = [0 1; 1 0] gives PA = [3 4; 0 2]; all row swaps are recorded in P.

4.5.2 Transposition: definition and basic identities

If A is an m × n matrix, then A^T has size n × m and elements $(A^T)_{ij} = a_{ji}$: columns become rows. Example: A = [1 2 3; 4 5 6] gives A^T = [1 4; 2 5; 3 6].

$$(A^T)^T = A,\ (A + B)^T = A^T + B^T,\ (cA)^T = cA^T$$

Element-wise proof (for the sum): $((A + B)^T)_{ij} = (A + B)_{ji} = a_{ji} + b_{ji} = (A^T + B^T)_{ij}$.

$$(AB)^T = B^T A^T$$

The order of factors is reversed ($A^T B^T$ may not even be defined). Proof: $((AB)^T)_{ij} = (AB)_{ji} = \Sigma_k a_{jk} b_{ki} = \Sigma_k (B^T)_{ik} (A^T)_{kj} = (B^T A^T)_{ij}$. Lecture check: A = [1 2; 0 1], B = [2 0; 3 1]: $(AB)^T$ = [8 3; 2 1] = $B^T A^T$.

$$(A^{-1})^T = (A^T)^{-1}$$

For an invertible A, we transpose $AA^{-1} = I$: we get $(A^{-1})^T A^T = I$; similarly $A^T (A^{-1})^T = I$. Thus, transposition and inversion commute.

4.5.3 Symmetric and skew-symmetric matrices; the product $A^T A$

A matrix S is **symmetric** if $S^T = S$ ($s_{ij} = s_{ji}$); example: S = [2 −1; −1 3]. A matrix K is **skew-symmetric** if $K^T = -K$; example: K = [0 4; −4 0]; the diagonal is zero: $k_{ii} = -k_{ii} \Rightarrow k_{ii} = 0$.

$$A = (A + A^T)/2 + (A - A^T)/2 = S + K,\ S^T = S,\ K^T = -K$$

Any square matrix decomposes this way; lecture example: A = [1 4; 2 3] → S = [1 3; 3 3], K = [0 1; −1 0]; verification: S + K = A.

$$(A^T A)^T = A^T A,\ x^T A^T A\, x = (Ax)^T (Ax) = ||Ax||^2 \geq 0$$

The product $A^T A$ is always symmetric (a special case of $(AB)^T = B^T A^T$), and $x^T A^T A\, x = ||Ax||^2 \geq 0$ for any x. Example: A = [1 2; 0 1] gives $A^T A$ = [1 2; 2 5]; matrices $R^T R$ are fundamental to least squares.

Scalar and outer product: for columns u, v ∈ R^n, the number $u^T v$ is the scalar product, while uv^T is an n × n matrix. Example: u = [1; 2], v = [3; 4]: $u^T v = 11$, uv^T = [3 4; 6 8].

4.5.4 Permutations: $P^T = P^{-1}$, six 3×3 matrices, partial pivoting

A permutation matrix is obtained by permuting the rows of the identity matrix: exactly one 1 in each row and each column. Example: P = [0 1 0; 0 0 1; 1 0 0] maps [x_1; x_2; x_3] to [x_2; x_3; x_1].

$$PA — \text{matrix } A \text{ with permuted rows; } AQ — \text{with permuted columns}$$

Left multiplication permutes rows: A = [1 2; 3 4; 5 6], P = [0 1 0; 0 0 1; 1 0 0] → PA = [3 4; 5 6; 1 2]. Right multiplication permutes columns: A = [1 2 3; 4 5 6], Q = [0 0 1; 1 0 0; 0 1 0] → AQ = [2 3 1; 5 6 4]. In general, PA ≠ AP.

$$P^T P = I, \text{ i.e. } P^{-1} = P^T; \text{ consequences: } ||Px|| = ||x||, \det P = \pm 1$$

The columns of P are different coordinate unit vectors, so their scalar products form I: $P^T P = I$, hence $P^{-1} = P^T$. Consequences: ||Px|| = ||x|| and det P = ±1. One row exchange changes the sign of the determinant: det [0 1; 1 0] = −1, so det(PA) = −det A; two exchanges restore the sign.

There are exactly n! permutations of order n; for n = 3 there are six (Strang, §2.7) — three even (det P = +1) and three odd (det P = −1); $P^{-1} = P^T$, therefore $(P_{32} P_{21})^{-1} = P_{21} P_{32}$:

Permutation	**Matrix (row-wise)**	**det P**	**$P^{-1} = P^T$**
I	[1 0 0; 0 1 0; 0 0 1]	+1	I
P_{21} (swap rows 1, 2)	[0 1 0; 1 0 0; 0 0 1]	−1	P_{21}
P_{32} (swap rows 2, 3)	[1 0 0; 0 0 1; 0 1 0]	−1	P_{32}
P_{31} (swap rows 1, 3)	[0 0 1; 0 1 0; 1 0 0]	−1	P_{31}
$P_{32} P_{21}$ (cyclic)	[0 1 0; 0 0 1; 1 0 0]	+1	$P_{21} P_{32}$
$P_{21} P_{32}$ (cyclic)	[0 0 1; 1 0 0; 0 1 0]	+1	$P_{32} P_{21}$

Partial pivoting: if the pivot element is zero, rows are swapped, taking in the current column the candidate with the largest absolute value. Example: A = [10^{-5} 1; 1 1]: without swap the multiplier is $1/10^{-5} = 10^5$, after swap — $10^{-5}/1 = 10^{-5}$. Swaps are recorded in P: PA = LU.

4.5.5 Vector spaces: definition and examples

Vector space over R — a set of vectors with addition and scalar multiplication operations, satisfying vector space axioms: zero vector, opposite elements, associativity, distributivity, etc. Lecture examples: R^n, real matrices, polynomials, continuous functions — any object can be a "vector".

4.5.6 Subspaces and closure testing

$$W \subseteq V \text{ — subspace} \Leftrightarrow 0 \in W;\ u + v \in W \text{ for all } u, v \in W;\ cu \in W \text{ for all } u \in W, c \in R$$

Equivalently: non-empty W is closed under combinations au + bv. Test: (1) zero vector; (2) sum; (3) scalar multiplication — violation of any point refutes.

- **Line through origin:** W = {(t, 2t)} = span{(1, 2)}: for u = (s, 2s), v = (t, 2t) the combination au + bv = (as + bt, 2(as + bt)) is again in W — subspace.
- **Shifted line (counterexample):** S = {(x, y): y = 2x + 1} — zero vector does not satisfy the equation; a subspace must pass through the origin.
- **Plane through origin:** W = {(x, y, z): x + y + z = 0} = span{(1, 0, −1), (0, 1, −1)} (from z = −x − y: (x, y, z) = x(1, 0, −1) + y(0, 1, −1)); plane x + y + z = 1 is not a subspace (no zero).
- **Geometry of R^3:** subspaces are only {0}, lines through 0, planes through 0 and all R^3.

- **Intersection and union:** U ∩ W — subspace; union — not necessarily: x and y axes are subspaces of R^2, but (1, 0) + (0, 1) = (1, 1) lies on neither axis.
- **Matrices and polynomials:** symmetric 2×2 — subspace of all 2×2 (sums and multiples are symmetric); polynomials with p(0) = 0 — subspace, with p(0) = 1 — no.

4.5.7 Linear combinations and linear spans

$$span\{v_1, ..., v_k\} = \{c_1 v_1 + ... + c_k v_k : c_i \in R\}$$

Linear span — the set of all combinations of given vectors; it is always a subspace: if $u = \Sigma a_i v_i$ and $v = \Sigma b_i v_i$, then $su + tv = \Sigma (s a_i + t b_i) v_i$ — again a combination. Lecture example: is (3, 5) in span{(1, 1), (1, −1)}? Solve $c_1 + c_2 = 3$, $c_1 - c_2 = 5 \to c_1 = 4$, $c_2 = -1$ — yes.

4.5.8 Column space C(A)

$$A = [a_1 a_2 ... a_n] \to Ax = x_1 a_1 + ... + x_n a_n; C(A) = Col(A) = span\{a_1, ..., a_n\}$$

The product Ax is a linear combination of columns of A. **Column space** C(A) = Col(A) — the linear span of columns, a subspace of R^m for an m × n matrix A. Example: A = [1 2; 2 4; 0 0] — columns are proportional, Col(A) = span{(1, 2, 0)} — a line in R^3.

$$system\ Ax = b\ is\ solvable \Leftrightarrow b \in C(A)$$

The system Ax = b has a solution if and only if b lies in the column space: the solution x picks the coefficients of the combination of columns that gives b. The main criterion for solvability is weak.

4.5.9 Null space N(A)

$$N(A) = Null(A) = \{x : Ax = 0\}$$

Null space — the set of solutions to the homogeneous system; it is a subspace of R^n: if Au = Av = 0, then A(su + tv) = 0. Example: A = [1 2]: $x_1 + 2x_2 = 0$ gives $x_1 = -2x_2$, therefore Null(A) = span{(−2, 1)} — a line.

$$\{x : Ax = 0\} — always\ subspace; \{x : Ax = b\},\ b \neq 0 — not\ subspace\ (doesn't\ contain\ 0)$$

Solutions to the homogeneous system form a subspace; solutions to the non-homogeneous (b ≠ 0) — no: zero vector does not belong to them. If the system Ax = b is solvable, its solutions are a shift of the null space: $x = x_p + x_n$, where x_p is a particular solution, $x_n \in$ Null(A) (affine set). **By week 8 midterm:** be able to compute A^T and $(AB)^T$, use $P^{-1} = P^T$ and PA = LU, test closure, find spans and membership in C(A).

4.6 Linear Algebra I, week 7: in-class problems — questions and full solutions

[Source: Week 7 Class problems (Solved), Linear Algebra I (Fall 2026) — 20 in-class problems; arithmetic verified in python]

Problem 1. Transpose of 2×2. Compute A^T for A = [1 2; 3 4].

Solution. Transpose swaps rows and columns: $(A^T)_{ij} = a_{ji}$. Therefore A^T = [1 3; 2 4]; in particular, $(A^T)_{12} = a_{21} = 3$, diagonal remains unchanged.

Problem 2. Transposition of a sum. Compute $(A + B)^T$ for A = [1 0; 2 3], B = [2 1; 0 −1].

Solution. First, add: A + B = [3 1; 2 2]. Then transpose: $(A + B)^T$ = [3 2; 1 2]. Check: $A^T + B^T$ = [1 2; 0 3] + [2 0; 1 −1] = [3 2; 1 2] — matches.

Problem 3. Product $A^T A$. Find $A^T A$ for A = [1 2; 0 1].

Solution. $A^T A$ = [1 0; 2 1]·[1 2; 0 1] = [1 2; 2 5] — the result is symmetric, as always (exercise 26 ISA).

Problem 4. Transposition of a product. Find $(AB)^T$ for A = [1 1; 0 2], B = [2 0; 1 3].

Solution. AB = [3 3; 2 6], so $(AB)^T$ = [3 2; 3 6]. The identity gives the same: $B^T A^T$ = [2 1; 0 3]·[1 1; 0 2] = [3 2; 3 6] — the order of factors is reversed.

Problem 5. Checking symmetry. Is the matrix A = [2 −3; −3 1] symmetric?

Solution. A^T = [2 −3; −3 1] = A. Yes, it is symmetric: the corresponding off-diagonal elements coincide ($a_{12} = a_{21} = -3$), and the diagonal does not change under transposition.

Problem 6. Symmetric and skew-symmetric parts. Decompose A = [1 4; 2 3] into symmetric and skew-symmetric parts.

Solution. A^T = [1 2; 4 3]; S = $(A + A^T)/2$ = [1 3; 3 3] (off-diagonal elements are averaged), K = $(A - A^T)/2$ = [0 1; −1 0]. Check: S + K = A, $S^T = S$, $K^T = -K$.

Problem 7. Row permutation: PA. Find PA, where P = [0 1; 1 0], A = [1 2; 3 4].

Solution. PA = [0 1; 1 0]·[1 2; 3 4] = [3 4; 1 2]: the first row of the product is the second row of A, and vice versa. Left multiplication swaps rows 1 and 2.

Problem 8. Column permutation: AP. Find AP for the same A and P (A = [1 2; 3 4], P = [0 1; 1 0]).

Solution. AP = [1 2; 3 4]·[0 1; 1 0] = [2 1; 4 3]: the first column of the product is the second column of A, and vice versa. Right multiplication swaps columns 1 and 2; note that AP ≠ PA.

Problem 9. Inverse of a permutation matrix. Find P^{-1} for P = [0 1 0; 0 0 1; 1 0 0].

Solution. For the permutation matrix $P^{-1} = P^T$ = [0 0 1; 1 0 0; 0 1 0]. Verification: $P^T P$ = [0 0 1; 1 0 0; 0 1 0]·[0 1 0; 0 0 1; 1 0 0] = I — the columns of P are orthonormal.

Problem 10. Exchange rows 1 and 3. Find the permutation matrix P that exchanges rows 1 and 3 of a 3×3 matrix.

Solution. Exchange of rows 1 and 3, applied to I_3: P = [0 0 1; 0 1 0; 1 0 0]. For any A, the product PA exchanges its first and third rows; for example, for A = [1 2 3; 4 5 6; 7 8 9] we get [7 8 9; 4 5 6; 1 2 3].

Problem 11. PA = LU with zero pivot. Find P, L, U for A = [0 2; 3 4].

Solution. The pivot element $a_{11} = 0$ — exchange is unavoidable. Exchange the rows: P = [0 1; 1 0], PA = [3 4; 0 2]. Below the pivot 3 is 0 — elimination is complete, no multipliers: L = I_2, U = PA = [3 4; 0 2]; result: PA = LU.

Problem 12. LU without row exchanges. Find L, U for A = [2 1; 6 5] without row exchanges.

Solution. The pivot element 2 ≠ 0, no exchanges needed; the multiplier $m_{21} = 6/2 = 3$. Step R2 ← R2 − 3R1: [6 5] − 3·[2 1] = [0 2], so U = [2 1; 0 2], L = [1 0; 3 1]. Verification: LU = [2 1; 6 5] = A.

Problem 13. Subspace: line through origin. Is W = {(x, y): y = 3x} a subspace of R^2?

Solution. Yes. W = {t(1, 3): t ∈ R}: the zero vector corresponds to t = 0. For u = (s, 3s), v = (t, 3t) and any a, b: au + bv = (as + bt, 3(as + bt)) ∈ W — closure holds; W is a subspace (line through origin).

Problem 14. Counterexample: shifted line. Is S = {(x, y): y = 3x + 2} a subspace?

Solution. No. The zero vector does not satisfy the equation: 3·0 + 2 = 2 ≠ 0. The zero vector test is sufficient for refutation: a subspace must contain the origin.

Problem 15. Subspace: plane. Is W = {(x, y, z): x + y − z = 0} a subspace?

Solution. Yes. For f(x, y, z) = x + y − z and u, v ∈ W: f(au + bv) = a·f(u) + b·f(v) = 0, so au + bv ∈ W; and f(0, 0, 0) = 0. W is a subspace (plane through origin).

Problem 16. Linear combination of vectors. Express (4, 1) as a combination of vectors (1, 1) and (1, −1).

Solution. We seek $c_1(1, 1) + c_2(1, −1) = (4, 1)$: $c_1 + c_2 = 4$, $c_1 − c_2 = 1$. Adding: $2c_1 = 5$, $c_1 = 5/2$, $c_2 = 3/2$. Verification: (5/2 + 3/2, 5/2 − 3/2) = (4, 1).

Problem 17. Span of proportional vectors. Find the linear span of vectors (1, 2, 0) and (2, 4, 0).

Solution. The second vector is twice the first: (2, 4, 0) = 2(1, 2, 0), so any combination is a multiple of (1, 2, 0). Result: span{(1, 2, 0), (2, 4, 0)} = {t(1, 2, 0): t ∈ R} — a line in R^3.

Problem 18. Null space of a 1×3 matrix. Find Null(A) for A = [1 2 −1].

Solution. Equation Ax = 0: $x_1 + 2x_2 − x_3 = 0$. Free variables $x_2 = s$, $x_3 = t$, then $x_1 = −2s + t$ and x = s(−2, 1, 0) + t(1, 0, 1). Result: Null(A) = span{(−2, 1, 0), (1, 0, 1)} — a plane in R^3. Verification: 1·(−2) + 2·1 − 0 = 0 and 1·1 + 2·0 − 1 = 0.

Problem 19. Column space. Find Col(A) for A = [1 2; 2 4; 0 0]; does the vector (3, 6, 0) belong to it?

Solution. Columns $a_1 = (1, 2, 0)$ and $a_2 = (2, 4, 0) = 2a_1$ are proportional, so Col(A) = span{(1, 2, 0)} — a line in R^3. Since (3, 6, 0) = 3(1, 2, 0), the vector lies on this line: yes (for example, A·[3; 0] = (3, 6, 0)).

Problem 20. Transpose and determinant of 3×3. For A = [1 2 0; 0 1 1; 0 0 2] find A^T and det(A).

Solution. A^T = [1 0 0; 2 1 0; 0 1 2]. A is upper triangular, so det A = 1·1·2 = 2 (product of diagonal elements); transposition doesn't change the determinant: $\det(A^T) = 2$.

4.7 Linear Algebra I, week 7: ISA exercises — questions and complete solutions

[Source: Week 7 Independent Study Assignments, Linear Algebra I (Fall 2026) — 50 problems without solutions; solutions created and verified in python]

Independent Study Assignments — ungraded self-practice (instructions: column vectors, intermediate steps, justification of subspace answers). File parts: A (1–12) — transpositions, B (13–20) — permutations and pivoting, C (21–32) — subspace criteria, D (33–42) — linear spans and matrix subspaces, E (43–46) — mixed applications, F (47–50) — advanced problems. Exercises 1–20 exactly match classroom problems 1–20 (below — compact solutions, complete ones

above); the remaining 30 are fully solved.

Exercise 1. Compute A^T for A = [1 2; 3 4].
Solution. A^T = [1 3; 2 4] (rows became columns); in detail — problem 1 above.

Exercise 2. Compute $(A + B)^T$ for A = [1 0; 2 3], B = [2 1; 0 −1].
Solution. A + B = [3 1; 2 2], so $(A + B)^T$ = [3 2; 1 2] = $A^T + B^T$.

Exercise 3. Find $A^T A$ for A = [1 2; 0 1].
Solution. $A^T A$ = [1 0; 2 1]·[1 2; 0 1] = [1 2; 2 5] — the result is symmetric.

Exercise 4. Find $(AB)^T$ for A = [1 1; 0 2], B = [2 0; 1 3].
Solution. AB = [3 3; 2 6], so $(AB)^T$ = [3 2; 3 6] = $B^T A^T$.

Exercise 5. Is the matrix A = [2 −3; −3 1] symmetric?
Solution. Yes: A^T = [2 −3; −3 1] = A — the off-diagonal elements coincide (a_{12} = a_{21} = −3).

Exercise 6. Decompose A = [1 4; 2 3] into symmetric and skew-symmetric parts.
Solution. S = $(A + A^T)/2$ = [1 3; 3 3], K = $(A − A^T)/2$ = [0 1; −1 0]; check: S + K = A.

Exercise 7. Find PA, where P = [0 1; 1 0], A = [1 2; 3 4].
Solution. PA = [3 4; 1 2] — left multiplication swapped rows 1 and 2.

Exercise 8. Find AP for the same A and P (A = [1 2; 3 4], P = [0 1; 1 0]).
Solution. AP = [2 1; 4 3] — right multiplication swapped columns 1 and 2; note that AP ≠ PA.

Exercise 9. Find P^{-1} for P = [0 1 0; 0 0 1; 1 0 0].
Solution. P^{-1} = P^T = [0 0 1; 1 0 0; 0 1 0]; check: $P^T P$ = I.

Exercise 10. Find the permutation matrix P that swaps rows 1 and 3 of a 3×3 matrix.
Solution. P = [0 0 1; 0 1 0; 1 0 0] — swapping rows 1 and 3 in I_3; PA swaps the first and third rows of any matrix.

Exercise 11. Find P, L, U for A = [0 2; 3 4].
Solution. Zero pivot requires exchange: P = [0 1; 1 0], PA = [3 4; 0 2]; elimination is already complete, so L = I_2, U = PA; result: PA = LU.

Exercise 12. Find L, U for A = [2 1; 6 5] without row exchanges.
Solution. m_{21} = 6/2 = 3; step R2 ← R2 − 3R1 gives U = [2 1; 0 2]; L = [1 0; 3 1]; check: LU = A.

Exercise 13. Is W = {(x, y): y = 3x} a subspace of R^2?
Solution. Yes: W = {t(1, 3): t ∈ R}, zero vector when t = 0, and a(s, 3s) + b(t, 3t) = (as + bt, 3(as + bt)) ∈ W.

Exercise 14. Is S = {(x, y): y = 3x + 2} a subspace?
Solution. No: (0, 0) does not satisfy the equation (0 ≠ 3·0 + 2) — no zero vector.

Exercise 15. Is W = {(x, y, z): x + y − z = 0} a subspace?
Solution. Yes: f(x, y, z) = x + y − z is linear, f(0, 0, 0) = 0 and f(au + bv) = a·f(u) + b·f(v) = 0 for u, v ∈ W.

Exercise 16. Express (4, 1) as a combination of vectors (1, 1) and (1, −1).
Solution. $c_1 + c_2 = 4$, $c_1 - c_2 = 1 \to c_1 = 5/2$, $c_2 = 3/2$; verification: $(5/2)(1, 1) + (3/2)(1, -1) = (4, 1)$.

Exercise 17. Find the linear span of vectors (1, 2, 0) and (2, 4, 0).
Solution. $(2, 4, 0) = 2(1, 2, 0)$, so $\operatorname{span}\{(1, 2, 0), (2, 4, 0)\} = \{t(1, 2, 0): t \in R\}$ — a line.

Exercise 18. Find Null(A) for A = [1 2 −1].
Solution. $x_1 + 2x_2 - x_3 = 0 \to x_1 = -2s + t$, $x_2 = s$, $x_3 = t$; $\text{Null}(A) = \operatorname{span}\{(-2, 1, 0), (1, 0, 1)\}$.

Exercise 19. Find Col(A) for A = [1 2; 2 4; 0 0]; does (3, 6, 0) belong to it?
Solution. Columns are proportional: $\text{Col}(A) = \operatorname{span}\{(1, 2, 0)\}$; since $(3, 6, 0) = 3(1, 2, 0)$, the vector belongs to Col(A).

Exercise 20. For A = [1 2 0; 0 1 1; 0 0 2] find A^T and det(A).
Solution. $A^T = [1\ 0\ 0;\ 2\ 1\ 0;\ 0\ 1\ 2]$; A is upper triangular, so $\det A = 1 \cdot 1 \cdot 2 = 2 = \det(A^T)$.

Exercise 21. Dimensions. For A = [1 2 3; 4 5 6] compute A^T and specify dimensions.
Solution. A is 2×3; $A^T = [1\ 4;\ 2\ 5;\ 3\ 6]$ is 3×2: dimensions swap, $(m \times n)^T = n \times m$.

Exercise 22. Proof (sum). Prove $(A + B)^T = A^T + B^T$ element-wise.
Solution. For $A = (a_{ij})$, $B = (b_{ij})$ of size m × n and each i, j: $((A + B)^T)_{ij} = (A + B)_{ji} = a_{ji} + b_{ji} = (A^T)_{ij} + (B^T)_{ij} = (A^T + B^T)_{ij}$. Elements match — matrices are equal.

Exercise 23. Proof (product). Prove $(AB)^T = B^T A^T$ for compatible matrices.
Solution. Let A be m × n, B be n × p. For each i, j: $((AB)^T)_{ij} = (AB)_{ji} = \Sigma_k a_{jk} b_{ki} = \Sigma_k (B^T)_{ik} (A^T)_{kj} = (B^T A^T)_{ij}$. Thus, $(AB)^T = B^T A^T$: the order of factors is reversed.

Exercise 24. Product $A^T A$ for 3×2. Compute $A^T A$ for A = [1 0; 2 1; 0 3] (in the file, the 3×2 matrix is written row-wise).
Solution. $A^T A = [1\ 2\ 0;\ 0\ 1\ 3] \cdot [1\ 0;\ 2\ 1;\ 0\ 3] = [5\ 2;\ 2\ 10]$ — the result is symmetric, as theory predicts (exercise 26).

Exercise 25. Symmetric and skew-symmetric parts. For A = [1 3; −2 4] find the symmetric and skew-symmetric parts.
Solution. $A^T = [1\ -2;\ 3\ 4]$. $S = (A + A^T)/2 = [1\ 1/2;\ 1/2\ 4]$, $K = (A - A^T)/2 = [0\ 5/2;\ -5/2\ 0]$. Verification: $S + K = A$, $S^T = S$, $K^T = -K$.

Exercise 26. Symmetry of $A^T A$. Show that $A^T A$ is symmetric for any real matrix A.
Solution. $(A^T A)^T = A^T (A^T)^T = A^T A$ (rule $(AB)^T = B^T A^T$): the matrix equals its transpose, i.e., is symmetric (for A of size m × n, the product $A^T A$ is square n × n).

Exercise 27. Scalar and outer product. Compute $u^T v$ and uv^T for $u = (2, -1)^T$, $v = (3, 4)^T$.
Solution. $u^T v = 2 \cdot 3 + (-1) \cdot 4 = 2$ — a number (scalar product); $uv^T = [2; -1] \cdot [3\ 4] = [6\ 8;\ -3\ -4]$ — a 2×2 matrix (outer product).

Exercise 28. Permutation of rows 1 and 2. Construct the permutation matrix that exchanges rows 1 and 2 of I_3.
Solution. P = [0 1 0; 1 0 0; 0 0 1] — I_3 with rows 1 and 2 exchanged. For any A with three rows, PA swaps the first two: P·[1 2 3; 4 5 6; 7 8 9] = [4 5 6; 1 2 3; 7 8 9].

Exercise 29. PA vs AP. For P = [0 1; 1 0] compare PA and AP for A = [2 5; 7 1].
Solution. PA = [0 1; 1 0]·[2 5; 7 1] = [7 1; 2 5] — rows are permuted; AP = [2 5; 7 1]·[0 1; 1 0] = [5 2; 1 7] — columns are permuted. PA ≠ AP: left multiplication acts on rows, right on columns.

Exercise 30. Verification $P^T P = I$. For P = [0 1 0; 0 0 1; 1 0 0] verify $P^T P = I$.
Solution. P^T = [0 0 1; 1 0 0; 0 1 0]; $P^T P$ = [0 0 1; 1 0 0; 0 1 0]·[0 1 0; 0 0 1; 1 0 0] = I. Rows of P^T are columns of P (e_2, e_3, e_1), their scalar products give δ_{ij}; consequence: $P^{-1} = P^T$.

Exercise 31. Determinant of permutation. Find the determinant of the permutation matrix that exchanges rows 1 and 3.
Solution. P = [0 0 1; 0 1 0; 1 0 0]; one row exchange changes the sign: det P = −1. Verification by expansion along the first row: $1\cdot(-1)^{1+3}\cdot$det [0 1; 1 0] = 1·1·(−1) = −1.

Exercise 32. PA = LU for 2×2. Find P, L, U for A = [0 1; 4 2].
Solution. The pivot $a_{11} = 0$ — exchange is needed: P = [0 1; 1 0], PA = [4 2; 0 1]. Below the pivot 4 is 0, elimination is complete: L = I_2, U = [4 2; 0 1]; result: PA = LU.

Exercise 33. Partial pivoting. Apply partial pivoting to A = [0.001 1; 2 1]; compare the multipliers before and after the exchange.
Solution. Without pivoting: pivot element 0.001, multiplier $m_{21} = 2/0.001 = 2000$ — huge (subtracting 2000·(first row) would strongly amplify errors). Partial pivoting takes the largest by magnitude candidate in column 1 (number 2); after exchange (P = [0 1; 1 0], PA = [2 1; 0.001 1]) the multiplier $m_{21} = 0.001/2 = 0.0005$. The exchange reduced the multiplier from 2000 to 0.0005 (by a factor of $4\cdot10^6$).

Exercise 34. Subspace? x + 2y = 0. Check if W = {(x, y): x + 2y = 0} is a subspace.
Solution. Yes. (1) 0 + 2·0 = 0 — the zero vector is in W. (2) From x + 2y = 0 and x′ + 2y′ = 0 follows (x + x′) + 2(y + y′) = 0. (3) (cx) + 2(cy) = c(x + 2y) = 0. Conclusion: W = span{(−2, 1)} — a line through the origin.

Exercise 35. Subspace? x + 2y = 3. Check if W = {(x, y): x + 2y = 3} is a subspace.
Solution. No: 0 + 2·0 = 0 ≠ 3 — there is no zero vector in W, but a subspace must contain 0 (the shifted line does not pass through the origin).

Exercise 36. Subspace? x ≥ 0. Check if W = {(x, y): x ≥ 0} is a subspace.
Solution. No. The zero vector is in W, and W is closed under addition (sum of non-negative x-coordinates is non-negative). But closure under scalar multiplication is violated: (1, 0) ∈ W, but (−1)·(1, 0) = (−1, 0) ∉ W. A half-plane is not a subspace.

Exercise 37. Subspace? x − y + z = 0. Check if W = {(x, y, z): x − y + z = 0} is a subspace.

Solution. Yes: f(x, y, z) = x − y + z is linear, f(0, 0, 0) = 0 and f(au + bv) = a·f(u) + b·f(v) = 0 for u, v ∈ W. Explicitly: x = y − z gives (x, y, z) = y(1, 1, 0) + z(−1, 0, 1), that is W = span{(1, 1, 0), (−1, 0, 1)} — a plane through the origin.

Exercise 38. Linear combination. Express (6, 2) as a combination of vectors (1, 1) and (1, −1).

Solution. $c_1 + c_2 = 6$, $c_1 - c_2 = 2 \to c_1 = 4$, $c_2 = 2$. Check: 4(1, 1) + 2(1, −1) = (6, 2).

Exercise 39. Geometry of span. Describe geometrically span{(1, 2, 0), (0, 1, 2)}.

Solution. The vectors are not proportional — the span is two-dimensional: a plane through the origin. Its points: s(1, 2, 0) + t(0, 1, 2) = (s, 2s + t, 2t); eliminating parameters (s = x, t = z/2) we get 4x − 2y + z = 0. Check: 4·1 − 2·2 + 0 = 0 and 4·0 − 2·1 + 2 = 0.

Exercise 40. Null space of 2×3. Find Null(A) for A = [1 1 1; 0 1 1] (2×3 matrix, written row-wise).

Solution. From row 2: $x_2 = -x_3$; from row 1: $x_1 = -x_2 - x_3 = 0$. Free variable $x_3 = t$: x = t(0, −1, 1). Conclusion: Null(A) = span{(0, −1, 1)} — a line in R^3. Check: [1 1 1; 0 1 1]·[0; −1; 1] = [0; 0].

Exercise 41. Column space of 3×2. Find Col(A) for A = [1 2; 2 4; 3 6].

Solution. Columns $a_1 = (1, 2, 3)$ and $a_2 = (2, 4, 6) = 2a_1$ are proportional, so any combination is a multiple of a_1: Col(A) = span{(1, 2, 3)} — a line in R^3 (vectors (t, 2t, 3t)).

Exercise 42. Membership in column space. Does (1, 1, 1) belong to the column space Col(A) for A = [1 0; 0 1; 1 1] (3×2 matrix)?

Solution. We seek $c_1(1, 0, 1) + c_2(0, 1, 1) = (1, 1, 1)$: the first two coordinates give $c_1 = c_2 = 1$, but then the third is 2 ≠ 1 — contradiction. Conclusion: (1, 1, 1) ∉ Col(A) (Col(A) is the plane z = x + y, and 1 ≠ 1 + 1).

Exercise 43. Intersection of subspaces. Prove that the intersection of two subspaces is a subspace.

Solution. For subspaces U, W ⊆ V: (1) 0 ∈ U ∩ W. (2) If u, v ∈ U ∩ W, then u + v ∈ U and u + v ∈ W (closure of each), so u + v ∈ U ∩ W. (3) If u ∈ U ∩ W, then cu ∈ U and cu ∈ W, so cu ∈ U ∩ W. The test is passed.

Exercise 44. Union of subspaces. Give two subspaces whose union is not a subspace.

Solution. In R^2 take the axes U = {(x, 0)} and W = {(0, y)} — both are subspaces. In the union lie (1, 0) and (0, 1), but their sum (1, 1) lies on neither axis. The union is not closed under addition — not a subspace.

Exercise 45. Symmetric 3×3 matrices. Is the set of symmetric 3×3 matrices a subspace of the space of all real 3×3 matrices?

Solution. Yes. (1) The zero matrix is symmetric. (2) $(S + T)^T = S^T + T^T = S + T$ for symmetric S, T. (3) $(cS)^T = c \cdot S^T = cS$. Closure is satisfied — it is a subspace.

Exercise 46. Polynomials with p(0) = 1. Is the set of polynomials satisfying p(0) = 1 a subspace?

Solution. No. The zero polynomial gives $p(0) = 0 \neq 1$ — there is no zero vector. Additionally, for $p(x) = 1$ and $q(x) = 1 + x$: $(p + q)(0) = 2 \neq 1$ — closure under addition is also violated. (Polynomials with $p(0) = 0$, on the other hand, do form a subspace.)

Exercise 47. Advanced problem. Prove that Null($A^T A$) = Null(A) for any real matrix A.

Solution. Two inclusions. (1) If $Ax = 0$, then $A^T A x = A^T (Ax) = 0$, that is, $\text{Null}(A) \subseteq \text{Null}(A^T A)$. (2) Let $A^T A x = 0$; multiply on the left by x^T: $x^T A^T A x = ||Ax||^2 = 0$. The sum of squares of the coordinates of Ax is zero, which means $Ax = 0$. Conclusion: $\text{Null}(A^T A) = \text{Null}(A)$; a consequence is that A and $A^T A$ have the same null space and rank (least squares fact).

Exercise 48. Advanced problem. If P is a permutation matrix, prove that Col(PA) = P Col(A).

Solution. By definition, $\text{Col}(PA) = \{PAx: x \in R^n\}$, $P\,\text{Col}(A) = \{Pb: b \in \text{Col}(A)\}$. (1) If $y = PAx$, then $y = P(Ax)$ with $Ax \in \text{Col}(A)$ — thus $y \in P\,\text{Col}(A)$. (2) If $y = Pb$ with $b = Ax \in \text{Col}(A)$, then $y = PAx \in \text{Col}(PA)$. Conclusion: $\text{Col}(PA) = P\,\text{Col}(A)$ — permuting rows permutes each point of the column space by the same permutation ($b \to Pb$ is bijective, P is invertible).

Exercise 49. Advanced problem. If W is a subspace, prove that span(W) = W.

Solution. span(W) is the set of all finite combinations $c_1 w_1 + ... + c_k w_k$ of elements $w_i \in W$. (1) Each combination lies in W: $c_i w_i \in W$ (scalars), sum of elements of W is in W (addition). (2) Each $w \in W$ is the combination $1 \cdot w$. Conclusion: span(W) = W — a subspace already contains all its linear combinations.

Exercise 50. Advanced problem. For A = [1 2 0; 0 1 1; 0 0 0] describe Col(A) and Null(A).

Solution. Columns $a_1 = (1, 0, 0)$, $a_2 = (2, 1, 0)$, $a_3 = (0, 1, 0) = a_2 - 2a_1$; the third row is zero, so all columns lie in the plane $z = 0$, and a_1, a_2 are independent. Result: $\text{Col}(A) = \text{span}\{(1, 0, 0), (2, 1, 0)\} = \{(s + 2t, t, 0)\}$ — the entire plane $z = 0$. Null space: from row 2 $x_2 = -x_3$, from row 1 $x_1 = -2x_2 = 2x_3$; $x_3 = t$: $x = t(2, -1, 1)$. Result: $\text{Null}(A) = \text{span}\{(2, -1, 1)\}$ — a line. Verification: $A \cdot [2; -1; 1] = [0; 0; 0]$.

5 Midterm Exam Preparation

[Source: MATH I – Fall 2026. GUIDANCE for MIDTERM EXAM Preparation]

The Midterm Exam for Math I will be held on **Monday, November 2, 2026**, during the regular lecture time. Below are the official rules and preparation strategy from the course guide.

5.1 Exam Rules

- Total duration — 2 hours; the exam starts promptly at the beginning of the lecture and ends exactly 2 hours later.
- The exam is pen-paper. Electronic and digital devices, books, and notes are prohibited; phones and all devices must be placed in bags.
- There will be no bathroom breaks during the exam.

- The exam focuses on Chapter 2 (§§2.1–2.4) and Chapter 3 (§§3.1–3.5 and 3.9, except subsection 3.9.3) of the textbook and on the Instructor's NOTES; material from weeks 1–3 is considered known from Test 1.
- All distributed materials (problem sheets, solution sheets, drafts) must be returned.
- Those who finish earlier may leave, but only after submitting all distributed materials.
- The expected responses include: knowledge of concepts and methods, demonstration of mathematical procedures, clear explanations and justifications. Heavy algebra is not required — if you get stuck in algebra, just clearly explain what you intend to do.

5.2 Preparation Strategy

- Study the required reading (textbook chapters and sections) and the Instructor's NOTES.
- Study thoroughly the lecture presentations and the examples and exercises covered in them — they cover the most important concepts and methods.
- Study thoroughly the workshop exercises and Independent Study Assignments (ISA) — all of them are covered in chapters 2–4 of this manual.

For linear algebra, the reference is the same: lecture notes from weeks 1–7 (Strang §§1.2–1.4, 2.1–2.7 and beginning of §3.1) with complete solutions to in-class problems, workshops and ISA are compiled in chapters 2–4; the formula catalogs for weeks provide a condensed summary of theory for review.